
Extended Essay

Subject: Physics HL

Exploring the Relationship Between Magnet Configuration and Magnetic Field Density

Research Question: How does the variation in Halbach array configurations (characterized by the parameter $k=0,1,2,3,4$) and speed of rotation of magnets (2.778, 4.167, 5.556, 6.945, 8.333 RPS) at constant radial separation affect the magnetic field strength in an electromagnetic induction setup?

Word count: 3907

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1. Introduction:

While researching about wind turbines and energy conversion, I came across the Halbach array, an interesting magnetic configuration that intensifies the field from one side while cancelling it at the other. Intrigued by this phenomenon and its versatility, I decided to investigate this concept and develop a method to quantify its magnetic field strength in a real-world context, which inspired this Extended Essay.

Two parameters highly influential on magnetic field strength are:

- 1) The arrangement of magnets in different Halbach arrays
- 2) The rotational speed of magnets around a centrally placed coil

I realized that a dynamo-like setup would effectively translate the theoretical properties of Halbach arrays to measurable outcomes. By placing the arrays on a rotating platform around a stationary copper coil, I could induce an electromotive force (EMF) that is evidence for magnetic field strength. This is used in credit card readers, tablets etc (Ling et al.).

I investigated the effects of altering parameter k of Halbach arrays, which defines number of magnetic dipoles, and rotational speed of magnets on magnetic field strength to identify the strongest magnet configurations in a dynamo system. Technology was used for data collection and analysis, considering experimental uncertainties. Using Gaussmeter readings as real values of field strength and comparing them with theoretical values obtained by a formula, I tried to identify the trend of field strength among different arrays.

2. Research Question:

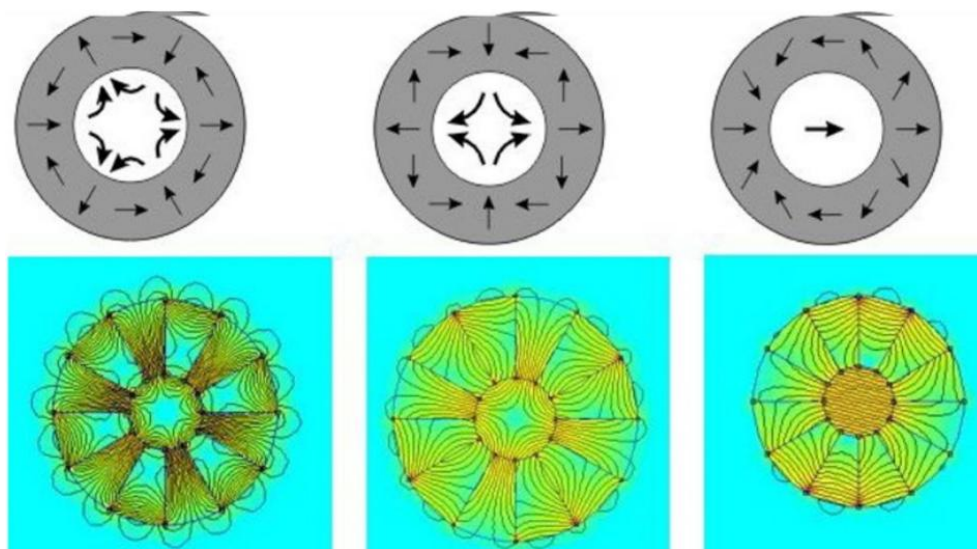
How does the variation in Halbach array configurations (characterized by the parameter $k=0,1,2,3,4$) and speed of rotation of magnets (2.778, 4.167, 5.556, 6.945, 8.333 RPS) at constant radial separation affect the magnetic field strength in an electromagnetic induction setup?

3. Theoretical Background:

3.1. Circular Halbach Arrays

A Circular Halbach array is an arrangement of permanent magnets that greatly enhances the magnitude of the magnetic field on one side but reduce it substantially on the other side. Its disposition maximizes the strength and directionality of magnetic fields, so the Halbach array is very useful in several technologies and industries (Z. Q. Zhu). Halbach arrays are commonly used in energy conversion technologies, such as in renewable energy systems, where spatially localized magnetic fields are needed. Examples of such systems are magnetic levitation, particle accelerators, and advanced imaging technologies (Suman Dwari et al.). An image of Halbach arrays with their magnetic fields is presented in Fig.1:

Figure 1: Circular Halbach arrays with their magnetic fields (“A Quick Introduction to Halbach Array Rings - XF Magnet”)



The arrangement of magnets in a particular configuration is characterized by a parameter k , which denotes $(n+1)$ of complete magnetic cycles of the array, n being the number of desired dipoles. In a circular Halbach array with 12 Ferrite magnets, the phase angle

increment between consecutive magnets depends on the number of magnetic cycles required around the circle (represented by k) (MPCO Magnetics, n.d.).

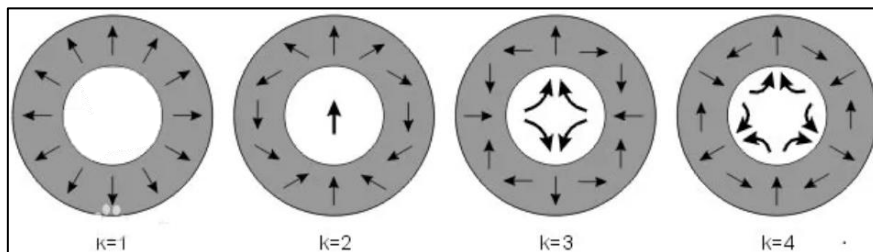
- k : Number of magnetic cycles.
- N : Total number of magnets (12 here).
- $\Delta\theta_n$: Increment in angle for each consecutive magnet according to the previous one.

$$\Delta\theta_n = \frac{2k\pi}{\text{No. of magnets}} = \frac{2k\pi}{12} \quad \text{Eq. 1}$$

(MPCO Magnetics, n.d.)

The magnetic dipoles are illustrated in the Fig.2 below:

Figure 2: Halbach arrays (MPCO Magnetics, n.d.)



As k increases, number of dipoles also increases. This causes hot spots (areas of high field strength) and cold spots (areas of low field strength). This usually influences induced voltage positively since more dipoles cause more change in flux, which is also more rapid due to the greater number of dipoles cutting into the area of the coil at an instant. Therefore, there is usually a positive correlation between the value of k and the induced voltage.

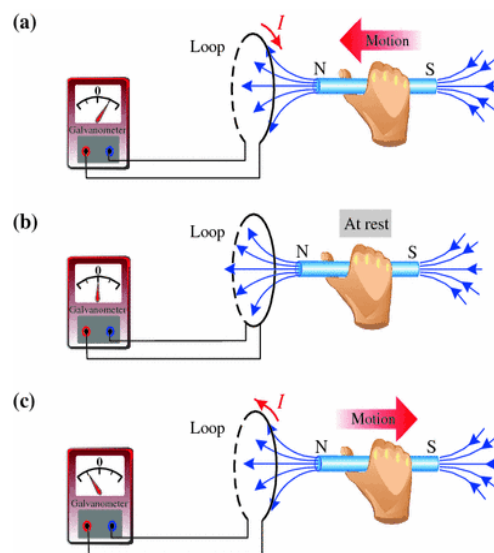
This behaviour is important because the choice of k influences design and operation of devices employing Halbach arrays. For example, in electric motors and generators, optimizing k ensures maximum energy conversion efficiency (Pacurar et al.). Induced voltage, for this

investigation, will thus assign a tangible value to magnetic field strength for each array and help to understand its effectiveness.

3.2- Electromagnetic Induction:

Electromagnetic induction is the process of generation of an electric current in a conductor due to a varying magnetic field. This principle is crucial to many modern technologies, from electric power generators to transformers to induction stoves (Grubb, 2021). This is based off Faraday's Law, which states that EMF induced is directly proportional to change in magnetic flux (Gregersen). This is shown in Fig.3 below:

Figure 3: Representation of Faraday's Law and Lenz's Law (Zhu, 2020)



Magnetic flux is a measure of the total magnetic field that passes through a given area (Khan Academy, 2022). The induced EMF is proportional to the rate of change of magnetic flux passing through the coil, which can be described by the following equation:

$$\varepsilon = -N \frac{d}{dt} \Phi_B$$

- N : Number of turns in the coil
- $\frac{d}{dt} \Phi_B$: Rate of change of magnetic flux linked with the coil

Further derivation of the formula used in this experiment is explained in Section 3.4.

In this experiment, the principle of electromagnetic induction is leveraged to quantify the magnetic field strength of different Halbach array configurations (Zhao, 2022) and hence to investigate different arrays under varying rotational speeds. The rotating magnets around a stationary coil rather than the other-way-round, reflect real-world dynamos like in wind turbines to handheld flashlights (Sweeney & O'Sullivan, 2021).

The induced voltage from this setup is proportional to magnetic field strength (Baker, 2015), allowing the evaluation of Halbach arrays' effectiveness in generating magnetic fields. This study will be able to identify the ideal arrangement and conditions to maximize induced voltage, which is pertinent in improving power output for wind turbine generators (Blaabjerg et al., 2014), MRI machines (J. Brenner, 2017), and in transportation and industrial automation applications (Ustun et al., 2020). It can also predict the outcomes or anomalies in inducing EMF practically.

3.3. Nomenclature:

Table 1: Nomenclature of all symbols used

k	Halbach array parameter
n	Number of desired dipoles
Φ_B	Magnetic Flux
B	Magnetic field vector
A	Area of plane of central coil
θ	Angle between A and B
ω	Angular speed

t	Time
ε	Induced EMF
N	Number of turns in the central coil
ν	Frequency of revolutions
$\Delta\theta_n$	Change in angle (increment magnitude) for each consecutive magnet from the previous
μ_0	Permeability of free space

3.4. Mathematical Modelling:

The magnetic flux (Φ_B) linked to the coil can be related to the magnetic field strength (B) and area of coil (A) as (Pausa, 2006):

$$\Phi_B = BA \cos \theta \quad \text{Eq. 2}$$

Now, as

$$\theta = \omega t$$

Rearranging (Eq. 2),

$$\Phi_B = BA \cos \omega t \quad \text{Eq. 3}$$

According to Faraday's Law of Induction,

$$\varepsilon = -N \frac{d}{dt} \Phi_B$$

Replacing Φ_B from (Eq. 3):

$$\varepsilon = -NBA \frac{d}{dt} \cos (\omega t)$$

Differentiating:

$$\varepsilon = NBA \omega \sin(\omega t)$$

As ω can be written using frequency of revolutions (ν) as:

$$\omega = 2\pi\nu$$

Upon substituting ω :

$$\varepsilon = NBA 2\pi\nu \sin(2\pi\nu t) \quad \text{Eq. 4}$$

Where-

- ε : Induced EMF - volts
- N : number of turns in the coil
- B : Strength of magnetic field – Tesla
- A : Area of coil - m²
- ν : Frequency of rotation – RPS
- t : time – s

3.5. Hypothesis:

I make the following 2 predictions:

- 1) For different Halbach arrays, the value of induced EMF will increase as the Halbach array parameter k -value increases.
- 2) For the same Halbach array, an increase in speed of rotation will increase the induced EMF proportionately.

4. Variables:

4.1. Independent variables-

Table 2:Independent variables

Variable	Range	Increment method	Magnitude of increment
Halbach array configuration	$k = 0,1,2,3,4$	-	+1
Speed of rotation of magnets	2.778, 4.167, 5.556, 6.945, 8.333 rotations per second (RPS)	Varying preset intervals of DC power supply- 4,6,8,10,12V	+2V

4.2. Dependent variables-

Table 3: Dependent variables

Variable	Measurement method	Unit	Uncertainty
Magnetic field strength	Digital Gaussmeter placed in the centre of the coils	milli-Tesla (mT)	± 0.0001 T
Induced electromotive force	Logger Pro 3 digital voltmeter connected to the coils	Volts (V)	± 0.0001 V

4.3. Controlled variables-

Table 4: Controlled Variables

Variable	Rationale for control	Control method
Radial separation between central coil and rotating magnets	Changing radial distance between the magnets and the coils will affect the concentration of the magnetic field as magnetic force is inversely proportional to the square of distance according to Coulomb's Law ("[Solved] in the Coulomb's Law of Magnetism, Force Is Inversely Pr")	Arrange and fix the magnets around the coil in a circle of constant radius.
Measurement duration	Inconsistent measurement duration may introduce variability in data.	Measurement duration is set at 10 s in Logger Pro 3 software (Vernier, <i>Logger pro 3</i>)
Coil specifications	Coil properties (resistance, net area, diameter, etc.) need to be kept unchanged to ensure the changes in data logged is due to change in independent variables	Use the same coil throughout the experiment

	only and not due to any other factor	
Permeability of free space μ_0	Magnetic field is also affected by the constant μ_0 according to the Biot-Savart's Law (Gegersen)	Conducting the entire experiment in air, thus keeping the μ_0 constant, the effect of which can hence be ignored

5. Apparatus-

Table 5: Apparatus and materials

Instrument/Apparatus	Uncertainty
12V 500 rpm DC motor	-
Ball bearing	-
2 teethed gear of diameter 48 mm and 54 mm	-
AC/DC Power Supply	-
Crocodile clips	-
Connecting wires	-
Insulated copper wire of thickness 0.6 mm	-
PVC pipe	-
Stopwatch	± 0.01 s
Wooden platform of diameter 22.86 cm (9 inch), thickness 8 mm	± 0.001 m
24 Ferrite magnets of dimensions 4 cm x 2 cm x 1 cm	-

Logger Pro 3 with Voltage Differential Probe	$\pm 0.000001 \text{ V}$
Gaussmeter	$\pm 0.1 \text{ mT} = 0.0001 \text{ T}$
Hot glue gun	-
String	-
Vernier calliper	$\pm 0.00005 \text{ m}$
Laptop with Logger Pro 3	-

6. Experimental Design:

A custom apparatus was designed using a 3D model in Fusion360 CAD software (Autodesk, n.d.), as in Figs.4-7.

Figure 4: Side view of CAD model (Autodesk, n.d.)

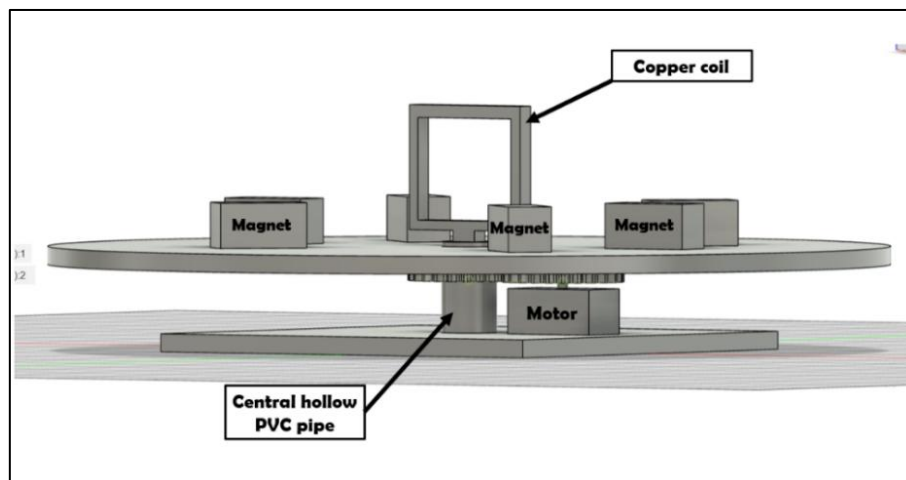


Figure 5: Bottom view (Autodesk, n.d.)

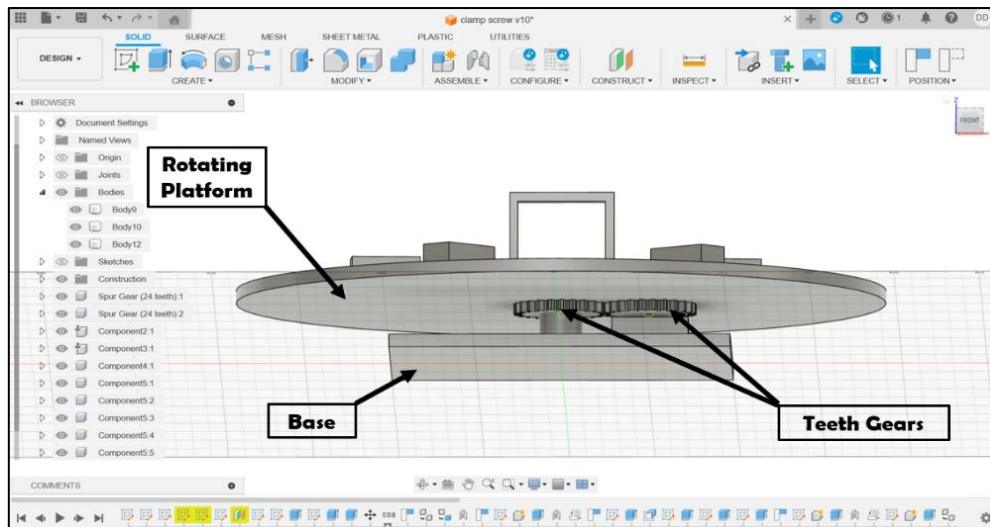
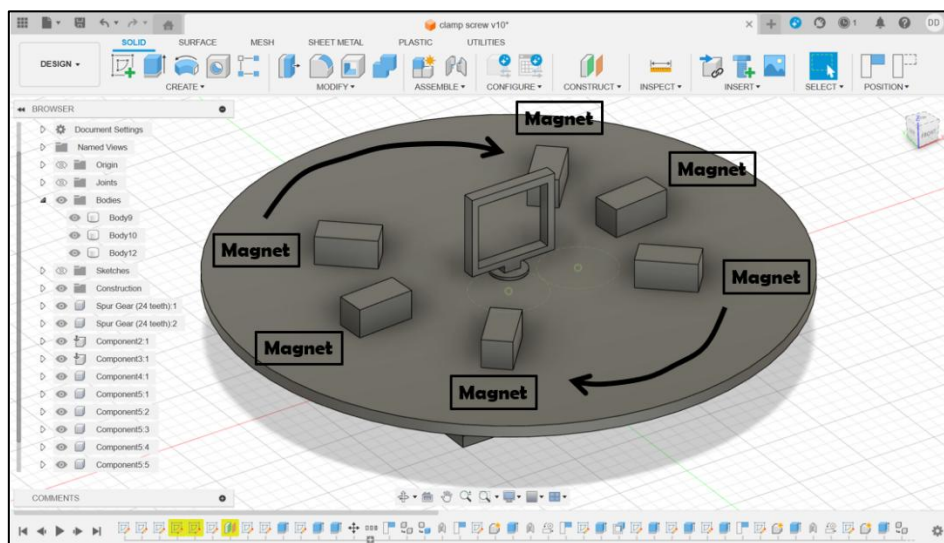


Figure 6: Top view with platform (Autodesk, n.d.)



Different components were sourced from local hardware stores and research was conducted about different methods to optimize this design. The set-up included a mounting bracket welded to a metal base, to which a 500 RPM 12V DC motor was mounted. This motor powered a toothed gear, which spun another gear (ratio=1:1) fixed to a rotating 22.86 cm (9-inch) platform, which was mounted on a hollow PVC pipe with a ball bearing.

Figure 7: Rotating platform set-up



The platform's circumference was calculated (72.0 ± 0.1 cm) and 12 evenly spaced segments were created on the disc. A circle (radius= 12.0 ± 0.1 cm) was made to mark the positions for the magnets, as shown in Fig.8.

Figure 8: Positions for magnets



Each magnet was tied to small wood piece and stuck with hot glue on the platform for easy attachment and removal for readings. This method of attaching the magnets ensures that

the setup remains free of magnetic materials, like screws, which could influence the results. A coil of insulated copper wire was wound with 200 turns, securing it with masking tape (diameter=5.914±0.025 cm) (For measurements, Vernier calliper, as in Fig.9, Table-6, was used). The coil's ends were threaded through a small hole in the hollow pipe, while the coil was positioned at the open end of hollow PVC pipe with modelling clay.

Figure 9: Measuring coil diameter

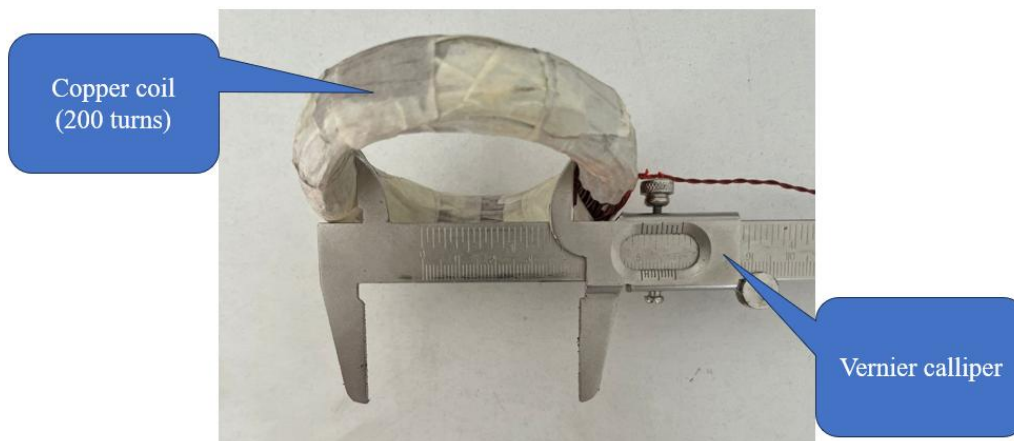


Table 6: Coil diameter raw and processed data

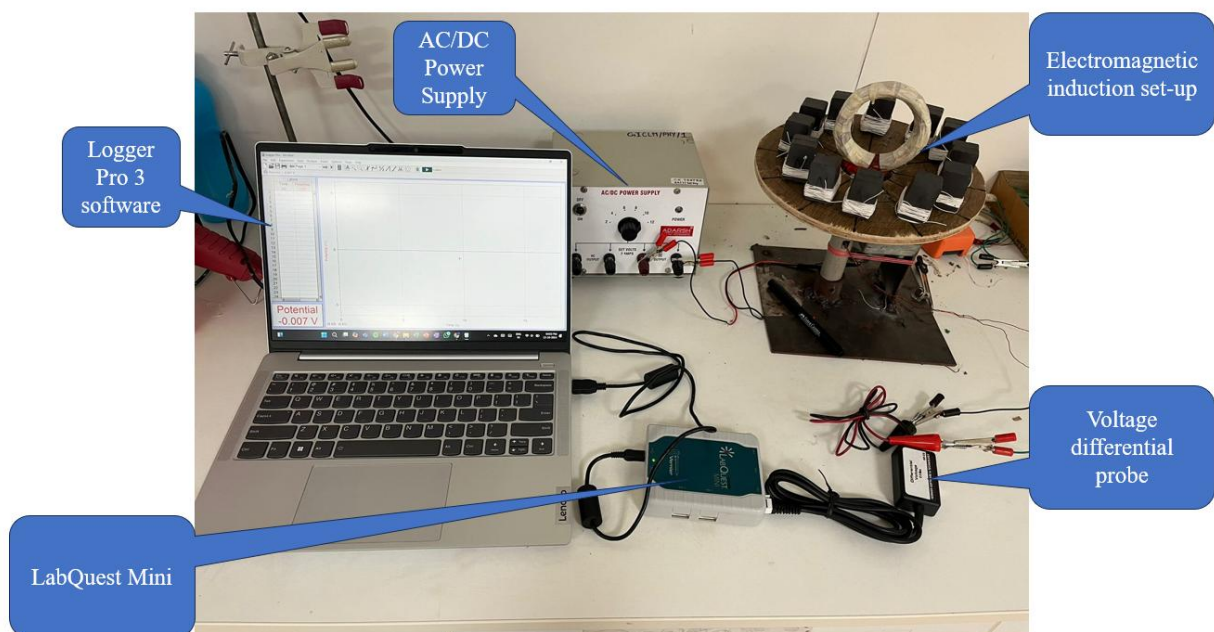
Trial	1	2	3	4	5	Average diameter (cm)
Diameter (cm) (±0.01 cm)	5.88	5.94	5.89	5.92	5.94	5.914

$$\text{Average diameter} = \frac{\sum_{i=1}^5 \text{Trial}_i}{5} = \frac{5.88+5.94+5.89+5.92+5.94}{5} = 5.914 \pm 0.025 \text{ cm}$$

Elastic bands were used to ensure a constant connection between the PVC pipe and mounting bracket, and modelling clay to stabilize the coil. Finally, Logger Pro 3 (Vernier

Software & Technology, n.d.) software was connected to the LabQuest Mini (Vernier Software & Technology, n.d.) and a laptop, setting the sampling rate to 50 readings-per-second, which sufficed for the highest rotations per second (8.33 RPS) without adding excessive data. The setup is shown in Fig.10.

Figure 10:Entire set-up

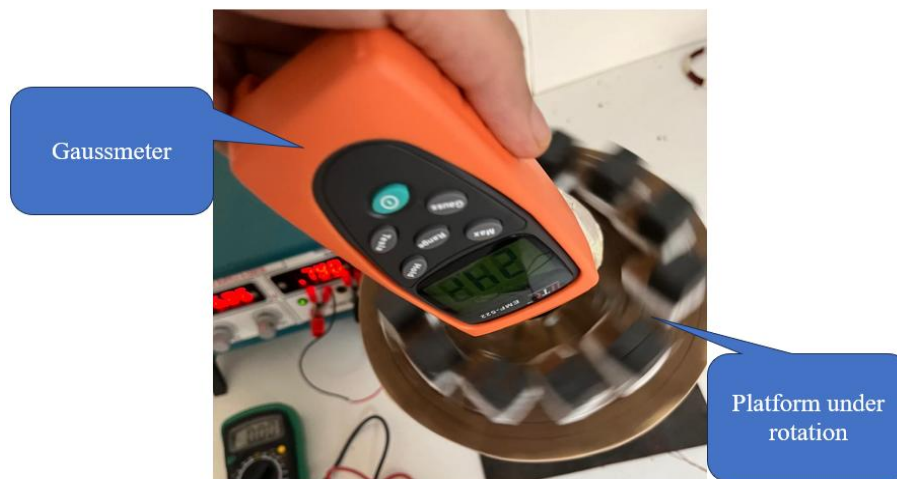


7. Procedure:

1. Connect the wires of the DC motor to the DC terminals of the power source using crocodile clips.
2. Connect the wires of the central coil to the voltage differential probe, which is connected to the laptop via the LabQuest Mini.
3. Set the Logger graph at a time of 10 seconds at a sampling rate of 50 readings per second
4. Switch the power source on and press Start Collect on the graph.

5. Allow the platform to rotate for 10 seconds.
6. Measure the magnetic field strength with a Gaussmeter pointed at the centre of the coil as in Fig.11 below, and record this reading on Microsoft Excel.
7. After completion, switch off the power supply.
8. Save the Logger reading on your computer and start a new one for the next trial.
9. Repeat steps 1 through 8 for 10 trials of each k configuration from $k=0$ to $k=4$

Figure 11: Gaussmeter reading



8. Ethical, environmental and safety considerations

Table 7: Safety, Environmental and Ethical considerations

Consideration type	Significance
Safety	<i>Risk of instability:</i> The platform rotates at high speeds; thus, care was taken to ensure everything was anchored securely
	<i>Injury from welding and cutting:</i> To mitigate these dangers such operations were conducted in the presence of a qualified expert, thereby guaranteeing adherence to safety regulations.

Safety	<i>Hazard due to moving gears:</i> All loose wearables and body appendages were kept clear of rotating parts.
	<i>Potential for splinters:</i> Wooden platform edges were smoothed, and gloves were worn to avoid splinters.
	<i>Electric shock risk:</i> The electrical components posed a risk of getting a shock. Precautions were also taken to ensure that all the electrical parts were handled carefully and only once the system was switched off.
Environmental	<i>Material waste:</i> All materials were sourced from repurposed items to promote sustainable practices and minimize waste
	<i>Improper disposal:</i> Unwanted components may contribute to landfill waste. Responsible disposal practices were followed.
Ethical	All sources of information used were correctly cited in bibliography.

9. Data collection and Analysis

9.1. Data collection and processing

Each trial logged 501 data points (0.02s intervals) in Logger Pro, totalling 5010 data for 10 trials of each array, later processed in Microsoft Excel (Photos of all magnetic arrays have been added in Appendix A)

The rotational speed values were changed from rotations per minute (RPM) to rotations per second (RPS) to conform with proper SI units. This conversion is given in Appendix B in Table 13.

A sample of data points in Logger Pro 3 for 10 trials for a single speed in $k=4$ is shown below in Figure 12.

Figure 12: Raw Data Table in Logger Pro 3 for $k=4$ at 8.33 RPS

	Run 1		Run 2		Run 3		Run 4		Run 5		Run 6		Run 7		Run 8		Run 9		Run 10	
	Time (s)	Potential (V)	Time (s)	Potential (V)	Time (s)	Potential (V)	Time (s)	Potential (V)	Time (s)	Potential (V)	Time (s)	Potential (V)	Time (s)	Potential (V)	Time (s)	Potential (V)	Time (s)	Potential (V)	Time (s)	Potential (V)
1	0.00	-0.004	0.00	-0.101	0.00	-0.006	0.00	-0.006	0.00	-0.025	0.00	-0.004	0.00	-0.159	0.00	-0.004	0.00	-0.004	0.00	-0.013
2	0.02	-0.006	0.02	-0.104	0.02	-0.004	0.02	-0.004	0.02	-0.016	0.02	-0.006	0.02	-0.055	0.02	-0.001	0.02	-0.006	0.02	-0.022
3	0.04	-0.006	0.04	0.066	0.04	-0.004	0.04	-0.006	0.04	-0.016	0.04	-0.006	0.04	0.179	0.04	-0.006	0.04	-0.004	0.04	-0.031
4	0.06	-0.006	0.06	0.121	0.06	-0.004	0.06	-0.006	0.06	-0.004	0.06	-0.013	0.06	-0.116	0.06	-0.001	0.06	-0.006	0.06	-0.046
5	0.08	-0.010	0.08	-0.037	0.08	0.006	0.08	-0.013	0.08	0.002	0.08	-0.006	0.08	-0.089	0.08	-0.001	0.08	-0.004	0.08	-0.049
6	0.10	-0.004	0.10	-0.143	0.10	0.018	0.10	-0.006	0.10	0.009	0.10	-0.006	0.10	0.170	0.10	-0.001	0.10	-0.010	0.10	-0.040
7	0.12	-0.013	0.12	-0.006	0.12	0.021	0.12	-0.006	0.12	0.009	0.12	-0.006	0.12	-0.089	0.12	-0.001	0.12	-0.013	0.12	-0.019
8	0.14	-0.006	0.14	0.133	0.14	0.012	0.14	-0.006	0.14	0.009	0.14	-0.004	0.14	-0.113	0.14	0.006	0.14	-0.016	0.14	0.009
9	0.16	-0.006	0.16	0.002	0.16	0.006	0.16	-0.013	0.16	0.012	0.16	-0.010	0.16	0.188	0.16	0.006	0.16	-0.031	0.16	0.042
10	0.18	-0.004	0.18	-0.149	0.18	-0.013	0.18	-0.006	0.18	0.009	0.18	-0.006	0.18	-0.052	0.18	-0.004	0.18	-0.037	0.18	0.057
11	0.20	-0.010	0.20	-0.031	0.20	-0.040	0.20	-0.006	0.20	0.012	0.20	-0.006	0.20	-0.161	0.20	0.021	0.20	-0.031	0.20	0.045
12	0.22	-0.006	0.22	0.136	0.22	-0.061	0.22	-0.006	0.22	0.012	0.22	-0.006	0.22	0.154	0.22	0.027	0.22	-0.025	0.22	-0.004
13	0.24	-0.006	0.24	0.015	0.24	-0.064	0.24	-0.010	0.24	0.012	0.24	-0.004	0.24	0.006	0.24	0.027	0.24	-0.004	0.24	-0.058
14	0.26	-0.013	0.26	-0.159	0.26	-0.040	0.26	-0.004	0.26	0.009	0.26	-0.006	0.26	-0.174	0.26	0.012	0.26	0.021	0.26	-0.085
15	0.28	-0.006	0.28	-0.004	0.28	0.009	0.28	-0.006	0.28	0.009	0.28	-0.004	0.28	0.121	0.28	-0.013	0.28	0.045	0.28	-0.061
16	0.30	-0.006	0.30	0.148	0.30	0.051	0.30	-0.006	0.30	0.012	0.30	-0.013	0.30	0.039	0.30	-0.046	0.30	0.048	0.30	0.012
17	0.32	-0.004	0.32	-0.016	0.32	0.063	0.32	-0.006	0.32	-0.010	0.32	-0.013	0.32	-0.195	0.32	-0.070	0.32	0.024	0.32	0.075
18	0.34	-0.006	0.34	-0.155	0.34	0.030	0.34	-0.006	0.34	-0.028	0.34	-0.016	0.34	0.133	0.34	-0.067	0.34	-0.025	0.34	0.069
19	0.36	-0.006	0.36	0.012	0.36	-0.037	0.36	-0.006	0.36	-0.052	0.36	-0.028	0.36	0.069	0.36	-0.031	0.36	-0.067	0.36	-0.013
20	0.38	-0.006	0.38	0.145	0.38	-0.089	0.38	-0.013	0.38	-0.064	0.38	-0.037	0.38	-0.195	0.38	0.018	0.38	-0.076	0.38	-0.101
21	0.40	-0.013	0.40	-0.061	0.40	-0.076	0.40	-0.006	0.40	-0.052	0.40	-0.040	0.40	0.057	0.40	0.063	0.40	-0.040	0.40	-0.085
22	0.42	-0.004	0.42	-0.143	0.42	-0.001	0.42	-0.006	0.42	-0.022	0.42	-0.028	0.42	0.118	0.42	0.066	0.42	0.024	0.42	0.018
23	0.44	-0.010	0.44	0.075	0.44	0.075	0.44	-0.006	0.44	0.036	0.44	-0.004	0.44	-0.174	0.44	0.018	0.44	0.075	0.44	0.097
24	0.46	-0.006	0.46	0.106	0.46	0.075	0.46	-0.010	0.46	0.066	0.46	0.024	0.46	0.024	0.46	-0.067	0.46	0.060	0.46	0.069
25	0.48	-0.010	0.48	-0.137	0.48	-0.016	0.48	-0.001	0.48	0.057	0.48	0.045	0.48	0.133	0.48	-0.104	0.48	-0.025	0.48	-0.055
26	0.50	-0.010	0.50	-0.070	0.50	-0.113	0.50	-0.006	0.50	-0.004	0.50	0.051	0.50	-0.186	0.50	-0.049	0.50	-0.101	0.50	-0.119
27	0.52	-0.016	0.52	0.145	0.52	-0.082	0.52	0.002	0.52	-0.073	0.52	0.033	0.52	0.027	0.52	0.045	0.52	-0.076	0.52	-0.040
28	0.54	-0.031	0.54	0.030	0.54	0.042	0.54	0.009	0.54	-0.095	0.54	-0.016	0.54	0.157	0.54	0.094	0.54	0.024	0.54	0.094
29	0.56	-0.034	0.56	-0.167	0.56	0.106	0.56	0.002	0.56	-0.043	0.56	-0.064	0.56	-0.164	0.56	0.045	0.56	0.094	0.56	0.084
30	0.58	-0.046	0.58	0.002	0.58	0.036	0.58	-0.013	0.58	0.048	0.58	-0.080	0.58	-0.055	0.58	-0.064	0.58	0.054	0.58	-0.076

(All data points cannot be displayed in one screenshot. Hence, the first and last page of data collected in Microsoft Excel 2021 has been added in Appendix C)

The Gaussmeter readings were processed as shown below in Table 8:

Table 8: Magnetic field values

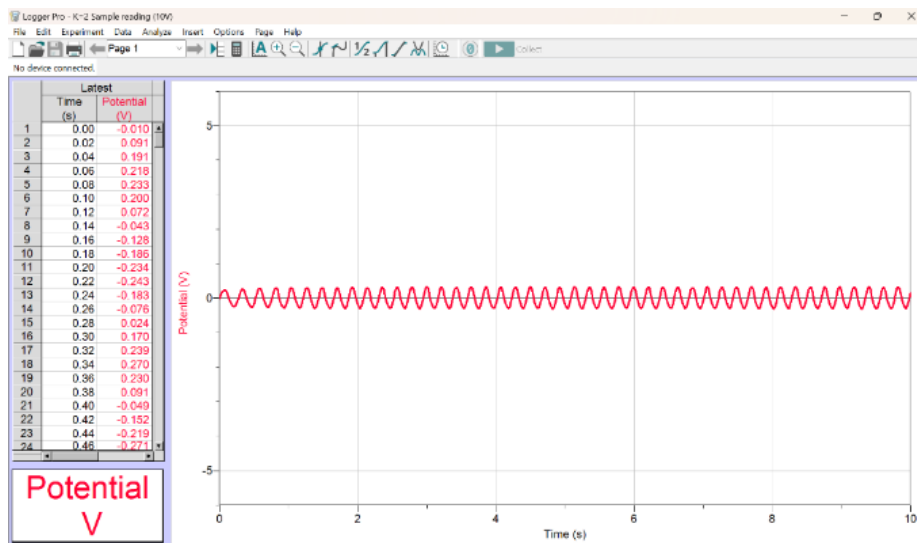
$k =$	B Values in milli-Tesla (± 0.1 mT)					Average B values in milli-Tesla (± 0.5 mT)	Average B values in Tesla (± 0.0005 T)
	T1	T2	T3	T4	T5		
0	1.8	2.3	1.7	2.1	2.4	2.06	0.00206

1	1.5	1.6	1.4	1.7	1.4	1.52	0.00152
2	20.2	19.7	19.6	19.3	20	19.76	0.01976
3	5.9	6.1	6.4	5.7	6	6.02	0.00602
4	9.6	10.1	9.8	9.9	10.5	9.98	0.00998

$$\text{Average B} - \frac{\sum_1^5 \text{Trials}}{5} = \frac{1.8+2.3+1.7+2.1+2.4}{5} = 2.06 \pm 0.5 \text{ mT} = 0.00206 \pm 0.0005 \text{ T}$$

Each trial gave a graph too, as shown in Graph 1 below (Individual analysis of graphs is done later in the section).

Graph 1: Graphical data sample reading for k=4 at 8.33 RPS



Since the induced voltage was AC in nature, the graph is seen to go both above and below the X-axis. After logging all data in Microsoft Excel, the average of the maximum and minimum point was found by the following formula for each speed of rotation and each k value:

$$\frac{(\text{Maximum} - \text{Minimum})}{2}$$

These values are shown in Table 9 below.

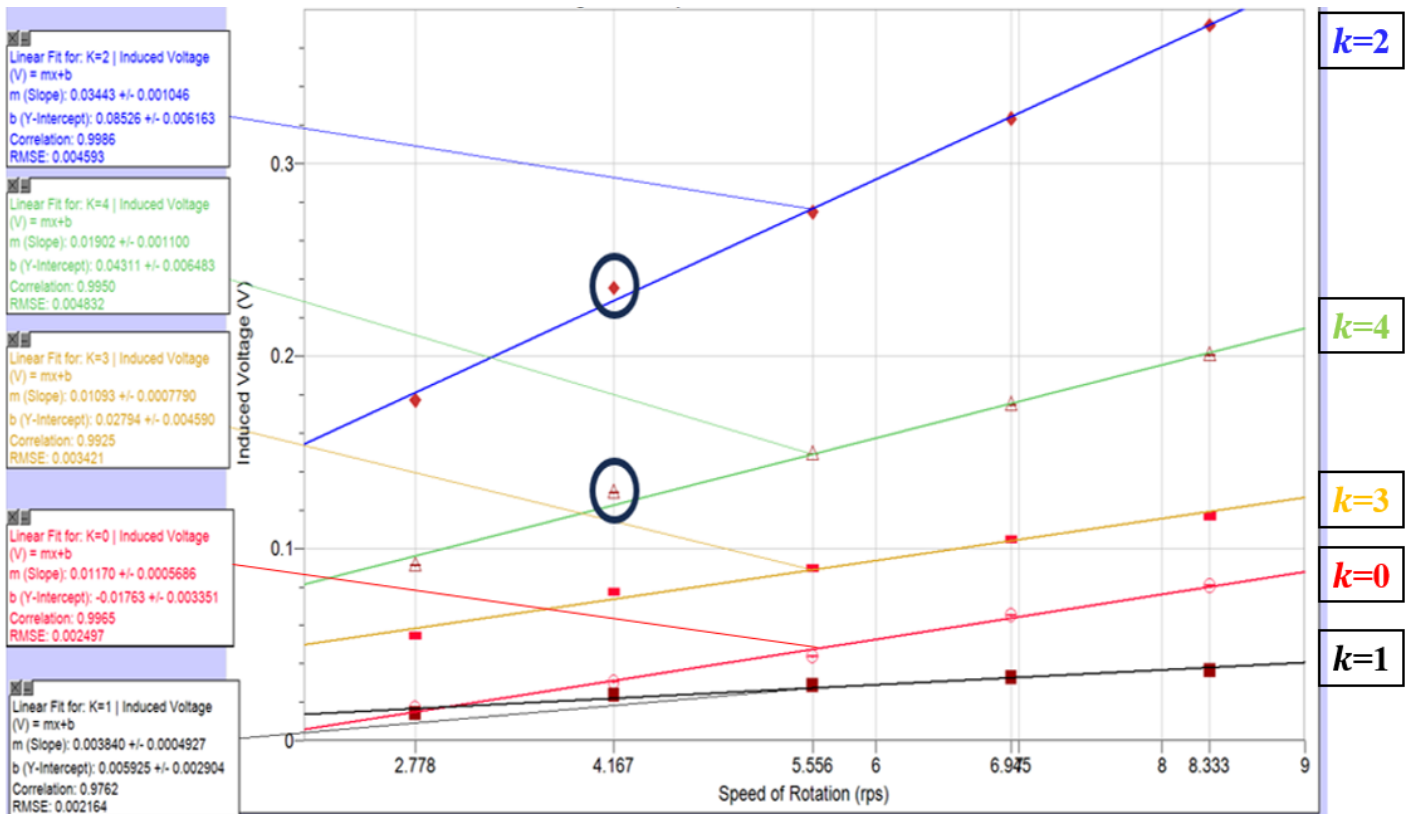
Table 9: Experimental processed values of induced voltage

Speed of rotation (RPS)	2.778	4.167	5.556	6.945	8.333
Halbach Array ($k=$)	Induced voltage (V) (± 0.000001 V)				
0	0.016689	0.030422	0.043964	0.065327	0.080490
1	0.014270	0.023845	0.028896	0.032850	0.036430
2	0.166988	0.235367	0.274849	0.323391	0.372028
3	0.054646	0.077438	0.089645	0.104713	0.116920
4	0.091076	0.129032	0.148869	0.174618	0.200367

Note: (The Logger Pro 3 gives values with more than 10 decimal places. For experimental purposes, 6 decimal places were used, with uncertainty also fixed at the same)

These values were graphed, as shown in Graph 2 below.

Graph 2: Graph of Induced Voltage VS RPS for each k value



(Uncertainty analysis was conducted but not plotted, as error bars were negligible. Hence, maximum and minimum gradient lines have not been plotted as they would overlap the trendlines due to the small difference in slope, however, gradient uncertainty is given in Table 10.)

Table 10: Equations and Statistics from Graph 2

Halbach Array ($k=$)	Slope	Trendline Equation ($y=mx+c$)	RSME (Root mean square error)	Pearson's Correlation coefficient

0	0.01170 (± 0.0005686)	$0.0117x - 0.01763$	0.002497	0.9965
1	0.00384 (± 0.0004927)	$0.00384x + 0.005925$	0.002164	0.9762
2	0.03587 (± 0.001771)	$0.03587x + 0.07526$	0.007778	0.9964
3	0.01093 (± 0.0007790)	$0.01093x + 0.02794$	0.003421	0.9925
4	0.01902 (± 0.001100)	$0.01902x + 0.07526$	0.004832	0.9950

9.2. General results:

Pattern of variation of induced voltage with rotation speed

Induced voltage increases with increase in rotational speed for every k value as justified by the upward sloping trendline and positive value of the slope in Table 10, with a strong positive correlation and low RSME values, thus aligning with hypothesis 2. The slope of consecutive k values demonstrate an increasingly upward trend, indicating that as k values increase, rotation speed has more impact on the induced voltage.

Pattern of induced voltage with k values:

Induced voltage is seen to increase generally with increase in k values as justified by the position of each consecutive k value ($k=0$ to $k=4$) in ascending order in Graph 2 except the $k=1$ and $k=2$ values (reasons for this have been explained later in this section), as

$$k=2 > k=4 > k=3 > k=0 > k=1$$

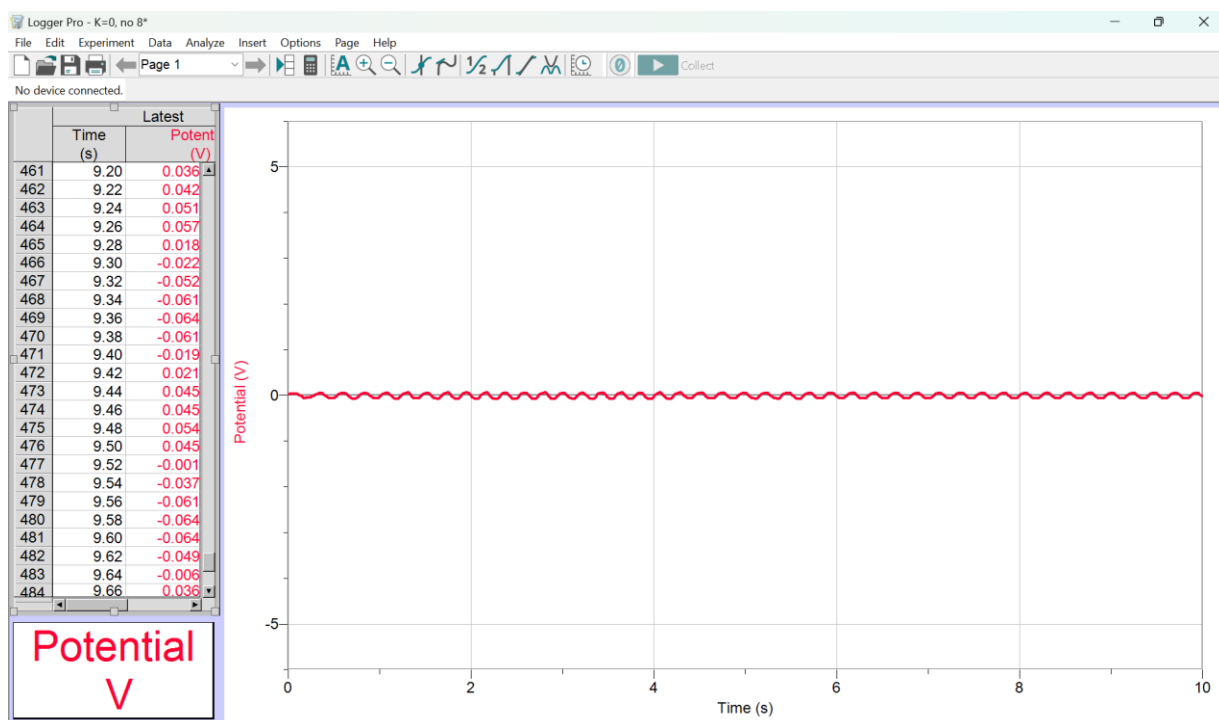
9.3. Specific analysis:

9.3.1. Outliers

The 2 outliers have been circled in Graph 2. These both lie on the same rotational speed 4.167 RPS, for $k=1$ and $k=2$. If magnetic properties were a cause for this deviation, consistent outliers would be expected across all arrays, however, it only occurs for 2. Also, the deviation of these points from the line of best fit is not large enough to consider them as ‘anomalies’. Hence, these can be considered safely to align with their respective trends.

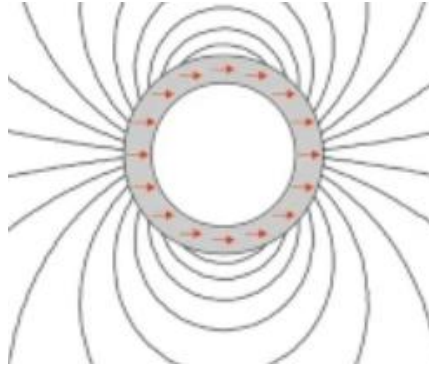
9.3.2. Array $k=0$

Graph 3: $k=0$ graph



This graph shows periodic oscillations which remain consistent without large changes in magnitude. The magnetic field in this array is focused outside the circular arrangement as shown in the Fig.13 (Blümmler and Helmut Soltner):

Figure 13: $k=0$ Magnetic field visual representation



(not to scale) (Provided for visual clarity only)

Thus, the $k=0$ configuration, with identical alignment of all magnets, fails to generate any alternating magnetic field that would induce a significant voltage. The uniform alignment of poles results in a stable but weak magnetic field that minimally affects the coil, leading to low but consistent voltage readings.

9.3.3. Array $k=1$

The 12 magnets arranged 30° apart in a circle with all north polar ends pointing radially outward. The field lines enter from the outside and make their way to the centre where they are parallel, not perpendicular, to the central coil. As shown in Fig.14-15 below.

Figure 14: Side view of field lines

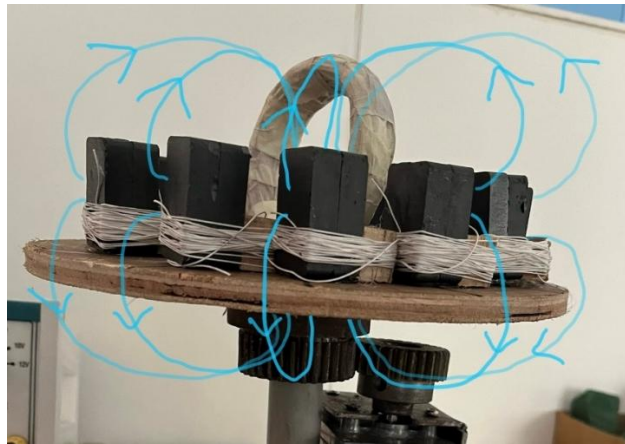
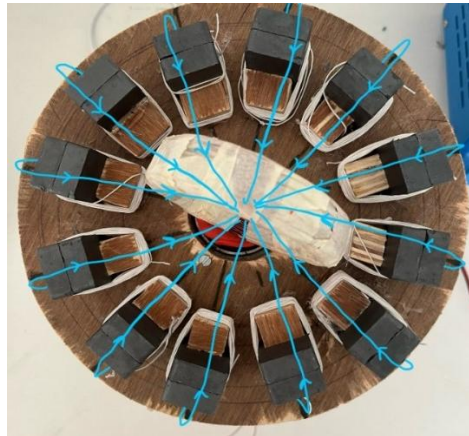


Figure 15: Top view of field lines



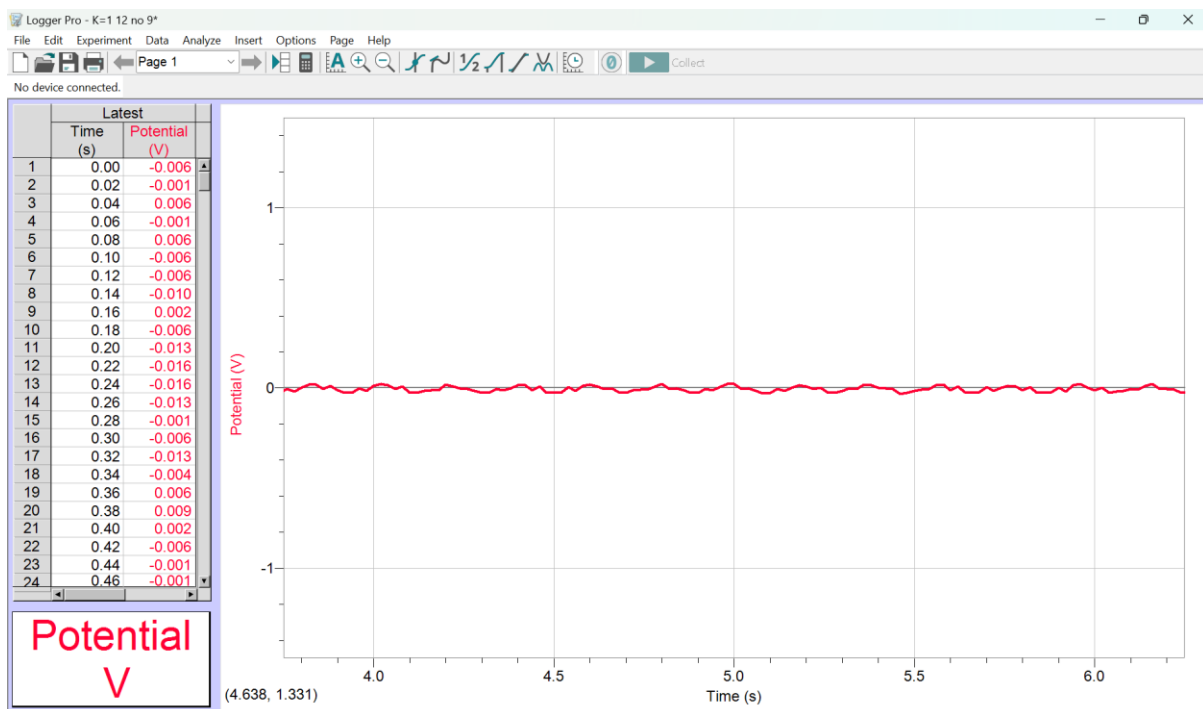
(The field lines appear to overlap, but only due to 3D visualisation, they don't actually intersect.)

Induced EMF is maximized when the field lines run perpendicular to the area of the coil but is minimum when parallel. Though most field lines are parallel to the coil, it is still possible that the coil captures some of the field components at slight angles thus generating a small amount of induced voltage due to the large surface area of the coil. Hence, this configuration limits the flux linkage due to parallel alignment. There is hence a lower induced

voltage. Although all angled field lines in the coil's area contribute marginally, this configuration greatly reduces the array's ability to induce voltage.

Even the graph $k=1$ agrees with this trend, showing very small deviation from the X axis, and voltage doesn't follow a clear sinusoidal pattern like one might expect from a simple rotating magnetic field, as seen in Graph 4.

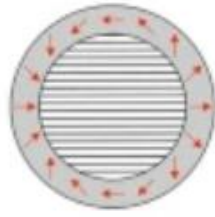
Graph 4: $k=1$ graph



9.3.4. Array $k=2$

The highest induced voltage was produced by the $k=2$ array since it produced a single, uniform, symmetrical dipole that was concentrated inside the circle as shown in the Fig.16 below, and hence had the maximum flux density and flux change.

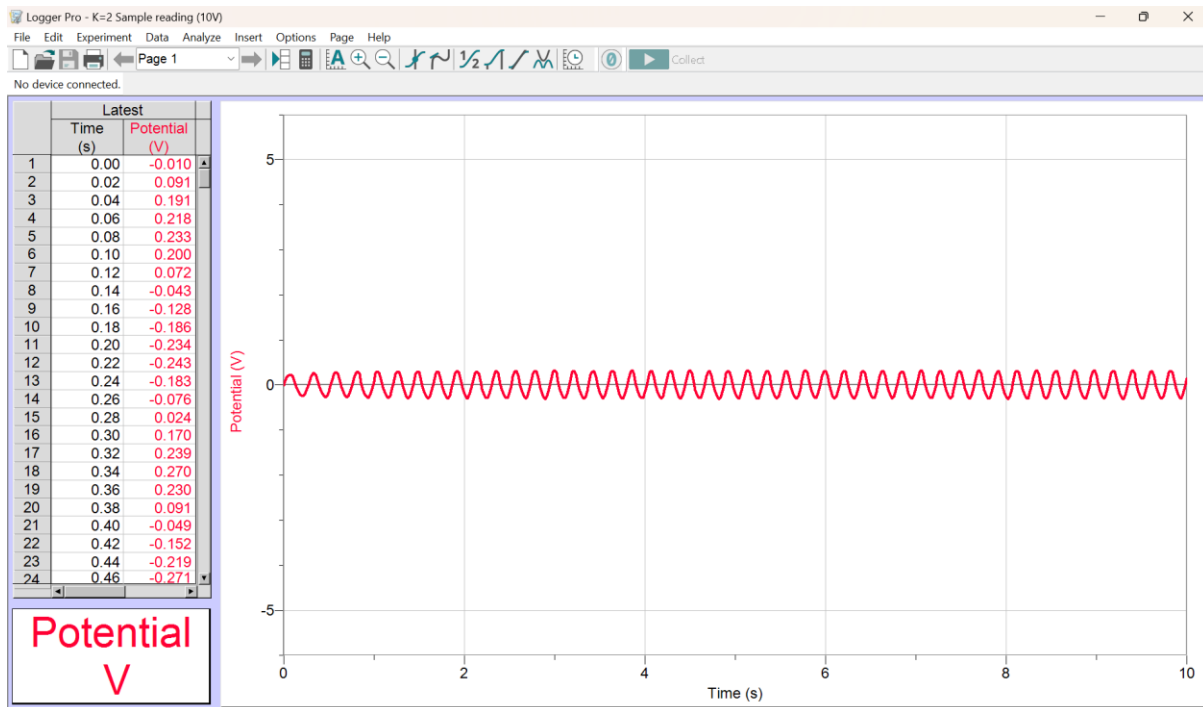
Figure 16: Magnetic field of $k=2$ (Blümler and Helmut Soltner).



Here, the field lines are aligned uniformly perpendicular to the area of the coil producing a single strong dipole, as seen by the Gaussmeter readings of this configuration which are higher than the others, which means that the change in flux is optimized, and a stronger EMF is induced. For other configurations, even though the number of dipoles in the system increases, their interference is destructive as the one dipole cancels the effect of another. As a result, there is a higher frequency of flux change, but the smaller magnitude of the net flux that results will not allow the voltage to rise to its full potential.

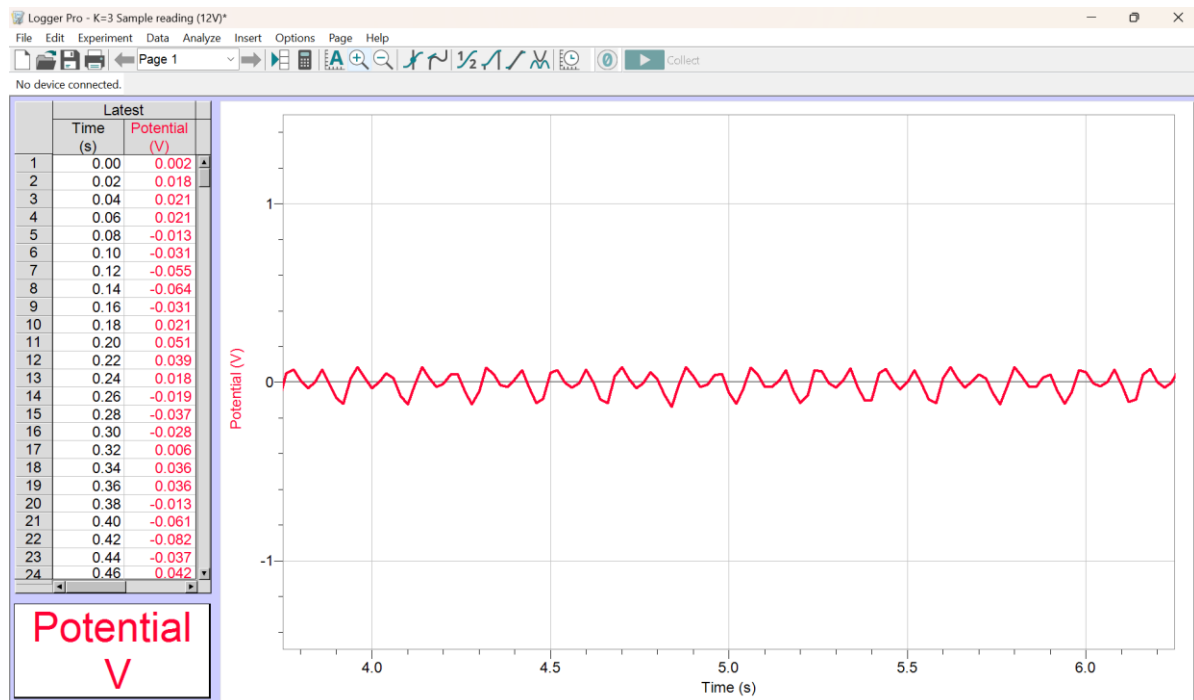
The graph of $k=2$ also agrees with this reasoning. Graph 5 shows a simple, periodic sinusoidal wave with highest peak-to-trough values of induced voltage amongst all the arrays experimented.

Graph 5: $k=2$ graph



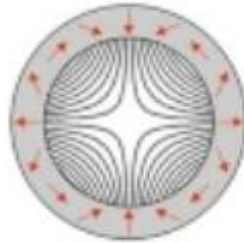
9.3.5. Array $k=3$

Graph 6: $k=3$ Graph



Graph 6 follows a sinusoid with dips in voltage at the peaks. This irregular waveform indicates non-uniform magnetic flux variation due to the array producing two dipoles shown in Fig.17

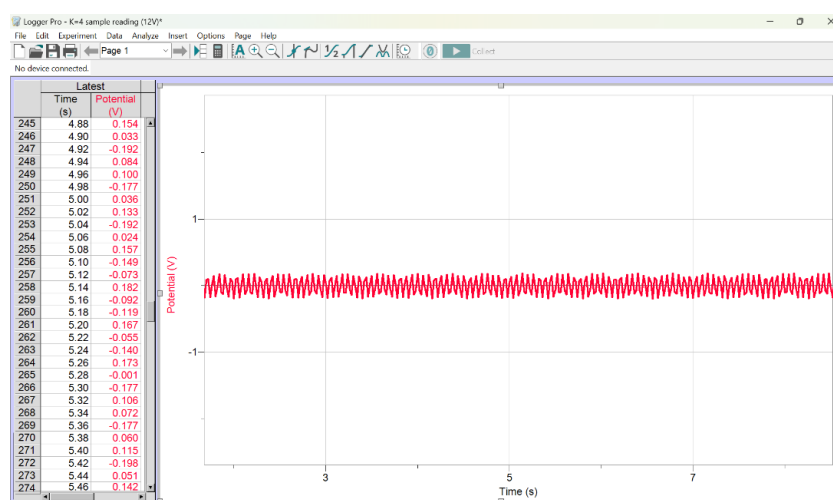
Figure 17: $k=3$ magnetic field (Blümler and Helmut Soltner):



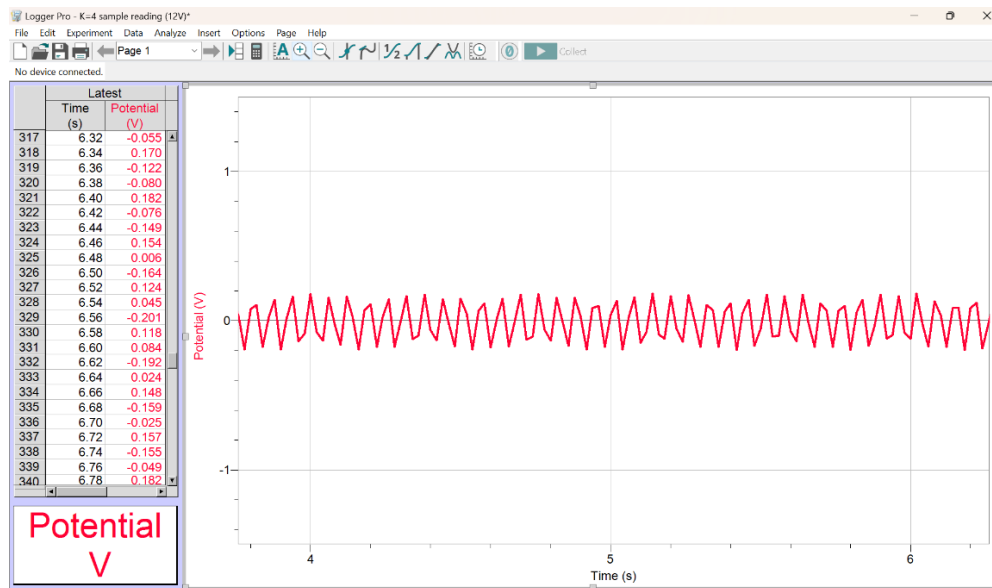
Multiple magnetic dipoles with varied interaction angles are seen in Fig.17. The complex dipole interaction leads to one dipole reducing the strength of the other at the centre as they cancel each other out to some extent, resulting in a lesser magnitude of magnetic strength at the centre, but more frequent change in flux, hence leading to smaller magnitudes of induced voltage, with dips in the peaks.

9.3.6 Array $k=4$

Graph 7: $k=4$ graph



Graph 8: $k=4$ graph zoomed in



Graphs 7-8 exhibit a trend resembling a sine wave, but closer inspection reveals densely packed smaller oscillations within the curve.

Each of the 3 dipoles occupy 120° of the circular array, and at the centre of the coil, their net effect is reduced. This is because the dipoles do not combine their magnetic fields but instead act independently, each directing its field toward its respective dipole region. Voltage changes occur more frequently than in $k=2$, as multiple dipoles interact with the coil simultaneously. However, the overall induced voltage is lower because the magnetic fields from the individual dipoles partly counteract one another, reducing the induced EMF.

9.4. Verifying data using (Eq. 4)

To check accuracy data, it was compared to values obtained by Eq. 4 to get Table 11:

Table 11: Theoretical Values

Theoretical Values					
Speed of rotation (RPS)	2.778	4.167	5.556	6.945	8.333

Halbach Array (k=)	Induced voltage (± 0.000001 V)				
0	0.019442	0.025663	0.039101	0.068005	0.085953
1	0.014345	0.018936	0.028852	0.032695	0.038315
2	0.186491	0.246166	0.346701	0.345478	0.360290
3	0.056816	0.074996	0.105625	0.112419	0.195941
4	0.094190	0.124329	0.175105	0.157670	0.233253

The percentage error was calculated and plotted in Table 12 below:

$$\begin{aligned}
 \% \text{ Error} &= \left| \frac{\text{Theoretical Value} - \text{Experimental Value}}{\text{Theoretical Value}} \right| * 100 \\
 &= \left| \frac{0.019442 - 0.016689}{0.019442} \right| * 100 \\
 &= \mathbf{14.16\%}
 \end{aligned}$$

Table 12: Percentage error

Halbach Array (k=)	Percentage Error (%)				
0	14.16	18.54	12.44	3.94	6.36
1	0.53	25.92	0.16	0.48	4.92
2	10.46	4.39	20.72	6.39	3.26
3	3.82	3.26	15.13	6.85	40.33
4	3.31	3.78	14.98	10.75	14.10

The average error was calculated to be **9.96%**, which could be due to various reasons:

1. A Gaussmeter was hand held during measurement because there was no stable mounting available. This probably led to variations in distance or angle from the intended measurement point, thus giving poor B values, disagreeing with the theoretical. The presence of the motor's magnet nearby and other magnetic parts in the set-up may have caused a discrepancy in measurement.
2. Since measured values already are so small in magnitude, even a slight discrepancy greatly inflates the percentage error, giving high error margins.

10. Conclusion

This investigation clearly established the relationship between Halbach arrays and induced voltage obtained by the process of electromagnetic induction. The experimental data demonstrates a clear positive upward trend with increases in k values and rotational speed. The steeper slope of the k values indicates that rotational speed has a stronger impact on the induced voltage as k increases. Certain potential inadequacies with experimentation, like resource constraints and some inaccuracies stemming from limitations in experimental design, may have introduced variability into the results. Nevertheless, the data fits generally into the hypothesis 1 and completely into hypothesis 2, validating that variations in magnet configuration and rotation speed significantly influence induced voltage, and hence this experiment can be deemed successful, despite moderate reliability concerns.

11. Evaluation

11.1. Limitations and Improvements

1. The configuration must be symmetrical to hold stable rotations, but asymmetrical configurations may lead to wobbly rotations due to uneven weight level; it can be improved using neodymium magnets which are smaller, stronger and lighter.

2. Resonance of the disc may occur if the frequency of rotation falls within the disc's natural frequency, which could have caused over-vibration within the platform, thus exerting greater bending forces onto the central pipe. Such action may have forced the disc to lean off centre, changing the relative distance of magnets and the coil, thus changing induced voltage.
3. Halbach configurations of varying distances would have given a non-uniform magnetic field distribution within the central region. Hence, a well-calibrated, sensitive Gaussmeter would detect slight changes within the magnetic field.
4. Losses due to air resistance and eddy currents might be some affect the precision of the measurements. Using thin laminated cores can minimize the losses this way (Dr. Steve Arar).
5. Using a feedback mechanism to alter the rotational speed based on real-time measurements of voltage output might perfect the design for maximum energy capture.

11.2. Extension for further research

1. *Using magnetic field strength and permeability calculations* - Using the equation $B \approx (\mu_0 \cdot M)/2\pi r$, calculate approximate values for the magnetic field strength at different distances and see how this compares with induced voltage.
2. *Using straight Halbach array units placed in a circular pattern* – Halbach arrays work differently in a straight line and in circular patterns (Marchio), hence this merging of 2 concepts may yield interesting results.
3. *Exploring arrays $k=1$ and $k=2$ in particular*- Since these two values of k are found to deviate from the trend due to interesting reasons, these can be researched more with different variables like radial separation, magnetic strength varied via electromagnets to understand induction more in depth, etc.

4. *Different coil core materials and number of windings* - Test iron, neodymium, ferrite or other conductive metals and see at what point it reaches maximum efficiency by checking the induced voltage.
5. *Use bar, disc, or ring-shaped magnets* to examine the effect of the shape on the distribution of the field, maybe also in Halbach arrays.
6. *Use an electromagnet rather than a permanent magnet*. Control dynamically the magnetic field strength to investigate how it impacts induction.

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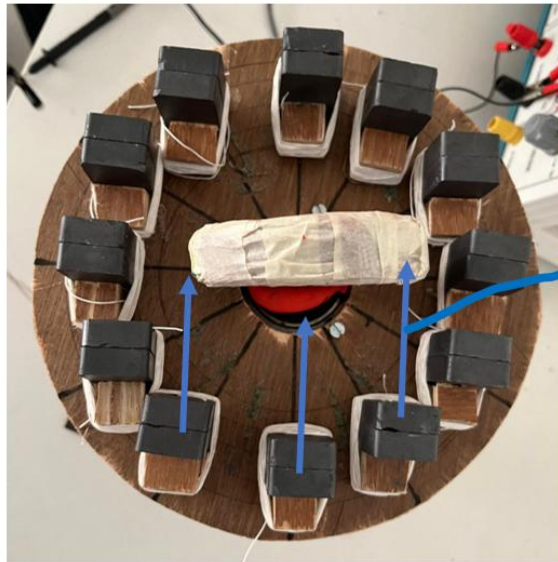
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13. Appendix

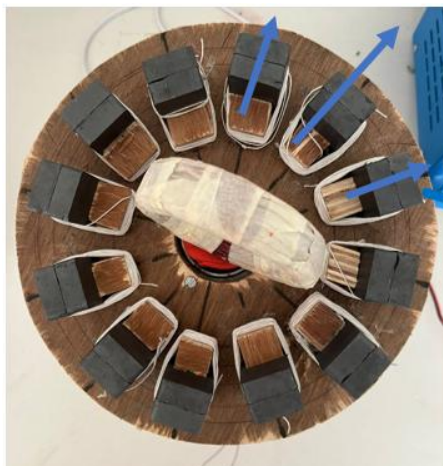
13.1. Appendix A

Image 1: $k=0$ Halbach Array



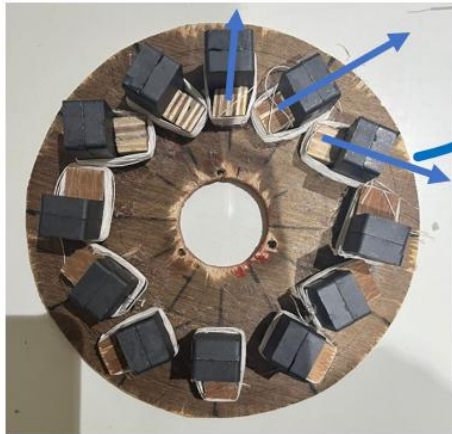
All magnets rotated 0°
clockwise from the last one

Image 2: $k=1$ Halbach Array



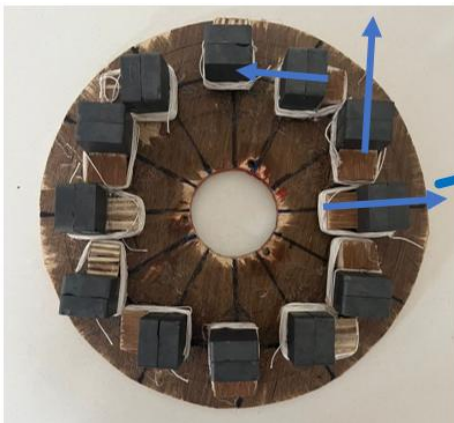
All magnets rotated 30°
clockwise from the last one

Image 3: $k=2$ Halbach Array



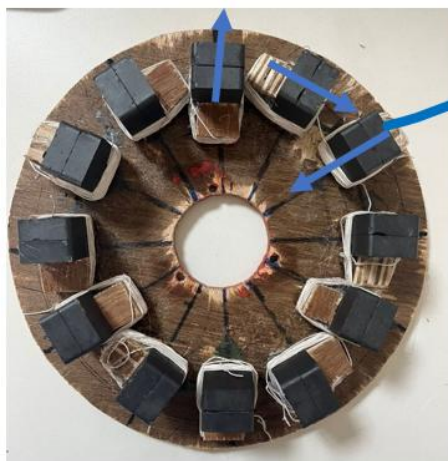
All magnets rotated 60°
clockwise from the last one

Image 4: $k=3$ Halbach Array



All magnets rotated 90°
clockwise from the last one

Image 5: $k=4$ Halbach Array



All magnets rotated 120°
clockwise from the last one

13.2. Appendix B

Formula used to convert RPM to RPS:

$$\text{Value in rps} = \frac{\text{Value in rpm}}{60}$$

The processed values:

RPM	RPS
166.670	2.778
250.000	4.167
333.330	5.556
416.670	6.945
500.000	8.333

Table 13: Processed values of RPS

13.3. Appendix C

Data Points for $k=0$

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	12V= 8.333 RPS												10V= 6.945 RPS									
2	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
3	0	0.02079	-0.04005	-0.0248	0.02384	0.033	0.02689	0.02995	0.02689	0.01469	-0.00648	0	-0.0309	0.01163	-0.04005	-0.01259	-0.01564	-0.00648	-0.05207	0.04501	0.03586	-0.00362
4	0.02	0.02995	-0.04005	-0.02785	0.033	0.04196	0.033	0.02995	0.02079	0.01469	0.00248	0.02	-0.0309	0.01163	-0.01869	-0.02785	0.00248	-0.01564	-0.05207	0.02079	0.03586	-0.02174
5	0.04	0.033	-0.02785	-0.02785	0.02384	0.04501	0.03586	0.02384	0.01163	0.02079	0.01469	0.04	-0.03395	-0.00057	-0.00954	-0.0309	0.02079	-0.03395	-0.05512	-0.00362	0.02384	-0.037
6	0.06	0.033	-0.01869	-0.037	0.01163	0.05112	0.04501	0.01774	-0.00362	0.02689	0.02079	0.06	-0.0309	-0.00648	0.01163	-0.04005	0.02995	-0.04597	-0.05512	-0.0309	0.00248	-0.04902
7	0.08	0.03586	-0.00954	-0.03395	-0.00057	0.04196	0.04196	-0.00057	-0.0248	0.02995	0.02079	0.08	-0.02785	-0.02174	0.02689	-0.04005	0.033	-0.04597	-0.05207	-0.05207	-0.01869	-0.04902
8	0.1	0.033	0.00248	-0.037	-0.00954	-0.00362	0.02079	-0.01869	-0.037	0.02995	0.02384	0.1	-0.01259	-0.0248	0.02079	-0.04292	0.033	-0.04902	-0.0248	-0.05207	-0.0309	-0.05207
9	0.12	0.00858	0.00858	-0.037	-0.0248	-0.0248	-0.00648	-0.037	-0.04902	0.02079	0.02995	0.12	-0.00057	-0.0309	0.033	-0.04902	0.04196	-0.04597	0.00553	-0.05512	-0.04597	-0.06123
10	0.14	-0.01259	0.02079	-0.0309	-0.04005	-0.05207	-0.03395	-0.04902	-0.04902	0.00248	0.033	0.14	0.01163	-0.04005	0.03891	-0.04597	0.03586	-0.05207	0.02995	-0.05512	-0.04902	-0.04292
11	0.16	-0.03395	0.02079	-0.01564	-0.04292	-0.05207	-0.04902	-0.04902	-0.00648	0.033		0.16	0.02079	-0.04005	0.033	-0.0309	0.01163	-0.02785	0.03586	-0.04005	-0.04902	-0.01259
12	0.18	-0.05817	0.02689	-0.00648	-0.04902	-0.06123	-0.05512	-0.04902	-0.05207	-0.0248	0.01774	0.18	0.01469	-0.04005	0.02689	-0.01869	-0.00648	-0.00362	0.04501	-0.00954	-0.05207	0.00858
13	0.2	-0.05512	0.033	0.00858	-0.04597	-0.06428	-0.05817	-0.04902	-0.03395	-0.04597	-0.00057	0.2	0.02384	-0.04292	0.00553	0.00553	-0.0309	0.01163	0.04501	0.02384	-0.04597	0.02995
14	0.22	-0.05207	0.02689	0.01774	-0.05207	-0.037	-0.06123	-0.04597	-0.01259	-0.04902	-0.01259	0.22	0.02689	-0.04902	-0.02174	0.02384	-0.04597	0.033	0.04196	0.04196	-0.02174	0.03586
15	0.24	-0.05207	0.01163	0.02384	-0.0248	0.00248	-0.037	-0.0248	0.00858	-0.04902	-0.0309	0.24	0.033	-0.03395	-0.0309	0.02689	-0.04902	0.02995	0.01774	0.03586	-0.00057	0.04196
16	0.26	-0.02785	-0.00362	0.02689	-0.00648	0.03891	-0.00057	-0.00362	0.033	-0.04902	-0.04005	0.26	0.033	-0.01259	-0.04597	0.033	-0.05817	0.03586	-0.01259	0.05112	0.02079	0.04196
17	0.28	-0.00362	-0.0248	0.033	0.01774	0.04501	0.02689	0.02079	0.03586	-0.04902	-0.04597	0.28	0.01774	-0.00057	-0.05512	0.03891	-0.05207	0.03586	-0.04005	0.04501	0.03586	0.04196
18	0.3	0.02384	-0.04005	0.03586	0.033	0.05417	0.04196	0.03586	0.04501	-0.02785	-0.05207	0.3	0.00248	0.01774	-0.05207	0.04196	-0.04902	0.04196	-0.05817	0.01774	0.03586	0.02079
19	0.32	0.03586	-0.05207	0.02995	0.03586	0.05722	0.04501	0.03891	0.04501	-0.00362	-0.04902	0.32	-0.01259	0.02689	-0.05207	0.033	-0.02174	0.02079	-0.05512	-0.01259	0.04196	-0.00954
20	0.34	0.04501	-0.05512	0.01774	0.03891	0.02384	0.05722	0.04501	0.03586	0.02079	-0.04597	0.34	-0.0248	0.033	-0.04597	0.00858	0.00248	-0.00362	-0.06428	-0.037	0.03891	-0.02785
21	0.36	0.04807	-0.05512	-0.00057	0.04196	-0.02174	0.04501	0.04501	0.00248	0.03586	-0.0248	0.36	-0.04005	0.02689	-0.01259	-0.00954	0.01774	-0.02785	-0.05512	-0.05512	0.033	-0.04597
22	0.38	0.04196	-0.05512	-0.01259	0.03586	-0.04902	0.00858	0.02689	-0.0248	0.033	0.00248	0.38	-0.05207	0.03586	0.00858	-0.02785	0.03586	-0.04597	-0.02174	-0.05817	0.00858	-0.05207
23	0.4	0.00858	-0.02785	-0.03395	-0.00362	-0.06428	-0.0248	-0.00362	-0.04902	0.04196	0.02384	0.4	-0.04597	0.03586	0.02689	-0.05207	0.03891	-0.04902	0.00858	-0.06428	-0.02174	-0.06123
24	0.42	-0.01564	-0.00057	-0.04292	-0.01259	-0.06428	-0.05512	-0.0309	-0.05817	0.04501	0.02995	0.42	-0.04597	0.02079	0.03586	-0.04902	0.04501	-0.06123	0.033	-0.05512	-0.037	-0.06123
25	0.44	-0.05207	0.02384	-0.04597	-0.037	-0.07038	-0.06123	-0.05512	-0.06428	0.02384	0.03891	0.44	-0.04902	-0.00057	0.04501	-0.04902	0.04501	-0.05207	0.03586	-0.02174	-0.05207	-0.04902
26	0.46	-0.06123	0.03586	-0.04597	-0.04902	-0.0248	-0.06733	-0.05512	-0.07038	-0.00362	0.04196	0.46	-0.037	-0.01564	0.04501	-0.05817	0.02079	-0.04902	0.04807	0.00858	-0.04902	-0.00954
27	0.48	-0.05817	0.03891	-0.05512	-0.05207	0.00858	-0.07038	-0.05817	-0.03395	-0.0309	0.03586	0.48	-0.02174	-0.037	0.04196	-0.04902	-0.00362	-0.0248	0.04807	0.03586	-0.05512	0.01774

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
476	9.46	0.04501	-0.07343	-0.06123	-0.02174	0.05722	-0.02785	0.07248	-0.07954	-0.037	-0.06123	9.46	0.02079	-0.05207	-0.01259	0.04807	0.033	-0.05512	-0.01564	0.04501	0.00248	0.00248
477	9.48	0.05417	-0.07038	-0.01564	0.00858	0.06638	-0.07648	0.04807	-0.0824	0.02079	-0.06428	9.48	0.03586	-0.01259	0.02079	0.01163	0.04196	-0.06123	-0.04597	0.04501	-0.01564	0.03586
478	9.5	0.04501	-0.07648	0.02384	0.04196	0.05112	-0.07648	-0.01869	-0.06733	0.06027	-0.06428	9.5	0.03891	0.02079	0.04501	-0.01869	0.04807	-0.06123	-0.06428	0.05722	-0.037	0.04196
479	9.52	-0.00057	-0.0248	0.04501	0.04196	-0.00648	-0.07954	-0.06733	-0.00648	0.06332	-0.04902	9.52	0.04501	0.03891	0.04807	-0.04902	0.04807	-0.06123	-0.06123	0.033	-0.05817	0.05112
480	9.54	-0.037	0.02689	0.04807	0.05112	-0.04902	-0.04597	-0.0824	0.04807	0.06943	-0.00057	9.54	0.04196	0.04807	0.04807	-0.06123	0.033	-0.02785	-0.06123	-0.01259	-0.05207	0.05417
481	9.56	-0.06123	0.05722	0.05417	0.04807	-0.07343	0.01163	-0.08545	0.06027	0.01774	0.03586	9.56	0.03586	0.05112	0.04807	-0.06123	-0.00362	0.00248	-0.05207	-0.037	-0.05512	0.03586
482	9.58	-0.06428	0.06332	0.04196	0.01163	-0.07343	0.05722	-0.06123	0.07248	-0.03395	0.04501	9.58	0.00553	0.04807	0.00248	-0.06123	-0.0309	0.03586	-0.00954	-0.06123	-0.06123	-0.00057
483	9.6	-0.06428	0.06027	-0.00057	-0.01869	-0.07343	0.06638	0.00553	0.04501	-0.07954	0.04501	9.6	-0.02174	0.01163	-0.0248	-0.04005	-0.05207	0.03891	0.02079	-0.07038	-0.0309	-0.04005
484	9.62	-0.04902	0.00858	-0.04005	-0.05207	-0.01564	0.07248	0.05722	-0.01259	-0.07954	0.05722	9.62	-0.037	-0.02174	-0.05512	-0.00057	-0.05817	0.05112	0.04196	-0.06428	-0.00362	-0.06428
485	9.64	-0.00648	-0.037	-0.06123	-0.06123	0.03891	0.02689	0.06638	-0.06123	-0.07954	0.02689	9.64	-0.05207	-0.04597	-0.06123	0.033	-0.05817	0.05112	0.04501	-0.05512	0.02079	-0.06123
486	9.66	0.03586	-0.07038	-0.06428	-0.06123	0.05722	-0.0248	0.07248	-0.07648	-0.02785	-0.01259	9.66	-0.05817	-0.06428	-0.06123	0.04501	-0.06123	0.033	0.04807	-0.01869	0.03586	-0.06123
487	9.68	0.04807	-0.07343	-0.07038	-0.06428	0.06943	-0.07343	0.03586	-0.07648	0.02689	-0.04597	9.68	-0.05207	-0.06123	-0.06428	0.04807	-0.03395	-0.00648	0.04807	0.02079	0.03891	-0.06123
488	9.7	0.04501	-0.07648	-0.04902	-0.037	0.05722	-0.07648	-0.02174	-0.05512	0.05722	-0.06428	9.7	-0.05207	-0.06428	-0.02785	0.05722	0.00248	-0.037	0.02079	0.04501	0.04501	-0.02174
489	9.72	0.05417	-0.03395	-0.00362	-0.00057	0.00553	-0.0885	-0.07038	0.00553	0.06638	-0.06428	9.72	-0.0309	-0.037	0.00858	0.03586	0.02689	-0.06123	-0.01564	0.04196	0.04196	0.01163
490	9.74	0.02384	0.02079	0.02689	0.03586	-0.04292	-0.04292	-0.07648	0.04501	0.06638	-0.06428	9.74	-0.00057	0.00248	0.03891	-0.00057	0.04196	-0.06123	-0.037	0.05417	0.033	0.04196
491	9.76	-0.01259	0.05722	0.04501	0.03891	-0.07343	0.02079	-0.07648	-0.04597	0.07248	-0.037	9.76	0.02384	0.02689	0.04501	-0.03395	0.03586	-0.06123	-0.06123	0.04501	-0.00057	0.04501
492	9.78	-0.04597	0.06027	0.05417	0.04807	-0.07343	0.06027	-0.02174	-0.07648	0.07248	-0.0309	9.78	0.03586	0.04501	0.05417	-0.05817	0.05112	-0.06428	-0.06428	0.00858	-0.0248	0.05112
493	9.8	-0.06123	0.06638	0.05417	0.04196	-0.07648	0.06638	0.01469	0.03586	-0.07038	0.04196	9.8	0.03891	0.04807	0.04501	-0.06123	0.03586	-0.01259	-0.06123	-0.02174	-0.04902	0.05112
494	9.82	-0.06123	0.02384	0.02689	0.053	-0.01869	0.06638	0.06027	-0.02174	-0.07648	0.04196	9.82	0.04501	0.05417	0.02689	-0.06123	0.00858	0.01774	-0.05512	-0.04902	-0.04902	0.02384
495	9.84	-0.06428	-0.02785	-0.01259	-0.00648	0.03586	0.01774	0.06638	-0.06733	-0.07954	0.05417	9.84	0.03891	0.03891	-0.01564	-0.06123	-0.02174	0.04501	-0.01869	-0.06428	-0.05207	-0.01259
496	9.86	-0.0309	-0.07648	-0.04597	-0.04292	-0.03395	0.06943	-0.07954	-0.0248	-0.0248	0.05112	9.86	0.02995	-0.00362	-0.04292	-0.01564	-0.04597	-0.04807	0.01774	-0.06123	-0.05512	-0.04597
497	9.88	0.00858	-0.07648	-0.06123	-0.05817	0.06332	-0.07648	0.02689	-0.0885	0.02384	0.00858	9.88	-0.00057	-0.0309	-0.06123	0.01469	-0.05817	0.05417	0.04196	-0.06428	-0.05512	-0.07038
498	9.9	0.04501	-0.07648	-0.06123	-0.06428	0.06332	-0.07343	-0.02785	-0.04292	0.05722	-0.0248	9.9	-0.02785	-0.06428	-0.06123	0.04501	-0.05817	0.05112	0.04196	-0.037	-0.0248	-0.06428
499	9.92	0.04807	-0.04292	-0.06428	-0.06123	0.00858	-0.07954	-0.07343	0.01469	0.06943	-0.05512	9.92	-0.04292	-0.06123	-0.06123	0.04501	-0.06123	0.01163	0.05112	0.00553	0.00553	-0.06123
500	9.94	0.05112	-0.01623	-0.0309	-0.05207	-0.04005	-0.02785	-0.07648	0.06027	0.06332	-0.0248	9.94	-0.05207	-0.06123	-0.04902	0.05417	-0.04597	-0.01564	0.04807	0.03586	0.02689	-0.04902
501	9.96	0.04501	0.05417	0.00248	-0.01259	-0.0824	0.02384	-0.07954	0.06638	0.00553	-0.07038	9.96	-0.05817	-0.06123	-0.00648	0.04807	-0.00954	-0.05207	0.02689	0.04501	0.03586	-0.00648
502	9.98	0.00248	0.06027	0.04196	0.02079	-0.07648	0.05112	-0.037	0.07248	-0.04292	-0.06123	9.98	-0.05512	-0.01564	0.02689	0.02079	0.02079	-0.05817	-0.00648	0.05417	0.03586	0.02689
503	10	-0.0248	0.06943	0.04807	0.04196	-0.07648	0.06332	0.01774	0.01774	-0.07343	-0.01564	10	-0.04902	0.00858	0.04501	-0.01564	0.04196	-0.06123	-0.04597	0.05417	0.04501	0.04501

	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR
1	8V= 5.556 RPS											6V= 4.167 RPS										
2	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
3	0	-0.02785	0.01163	-0.01564	0.01774	0.01469	0.01163	0.00553	-0.0248	0.01163	0.02689	0	0.01774	-0.02785	-0.02174	-0.00954	-0.00648	0.01163	-0.01259	-0.00648	0.00858	-0.00362
4	0.02	-0.0309	0.01469	-0.0248	0.00858	0.01774	0.01774	0.00858	-0.0248	0.00248	0.033	0.02	0.01163	-0.02785	-0.02174	-0.00648	-0.00954	0.00858	-0.01259	-0.00648	0.01163	0.00858
5	0.04	-0.03395	0.01774	-0.0248	0.02079	0.01774	0.01774	0.00858	-0.0248	-0.00362	0.02995	0.04	0.00858	-0.02785	-0.02785	-0.00362	-0.01564	0.00858	-0.01869	-0.00362	0.00858	0.00553
6	0.06	-0.03395	0.01469	-0.0309	0.02079	0.01774	0.01774	0.00553	-0.0248	-0.00954	0.02079	0.06	0.00248	-0.037	-0.02174	-0.00057	-0.02174	0.00248	-0.02785	0.00248	0.01163	0.00553
7	0.08	-0.037	0.01774	-0.0309	0.02079	0.01469	0.01469	0.00858	-0.0248	-0.00954	0.00553	0.08	-0.00648	-0.03395	-0.02174	0.00553	-0.01564	-0.00057	-0.0248	-0.00057	0.01163	-0.00057
8	0.1	-0.037	0.02079	-0.0309	0.01774	0.00858	0.00553	0.00858	-0.0248	-0.01259	-0.00362	0.1	-0.00648	-0.0248	-0.01564	0.00858	-0.02174	-0.00648	-0.0248	-0.00057	0.00248	-0.00362
9	0.12	-0.037	0.02079	-0.0309	0.01163	-0.00057	-0.00362	0.00858	-0.02174	-0.01564	-0.01564	0.12	-0.01564	-0.02174	-0.01259	0.00858	-0.0248	-0.00362	-0.0248	0.00858	0.00858	-0.00362
10	0.14	-0.037	0.01774	-0.0309	0.00248	-0.00648	-0.00362	0.01163	-0.01259	-0.02174	-0.02785	0.14	-0.0248	-0.01259	-0.01259	0.00248	-0.0248	-0.00954	-0.0248	0.00858	-0.00553	-0.00362
11	0.16	-0.02174	0.01774	-0.0309	-0.00648	-0.00648	-0.00954	0.01469	-0.00954	-0.0248	-0.037	0.16	-0.0248	-0.00648	-0.00954	0.00553	-0.0248	-0.01869	-0.03395	0.00858	0.00248	-0.00362
12	0.18	-0.0648	0.01774	-0.02785	-0.00648	-0.01259	-0.01564	0.01163	-0.00362	-0.02785	-0.04292	0.18	-0.02785	-0.00362	-0.00648	0.01163	-0.02785	-0.01259	-0.0248	0.00858	-0.00362	-0.00954
13	0.2	-0.00057	0.00858	-0.02174	-0.01259	-0.02174	-0.02174	0.00858	-0.00057	-0.0248	-0.04902	0.2	-0.0309	0.00248	-0.00362	0.01163	-0.0248	-0.01869	-0.0248	0.00858	-0.00648	-0.00648
14	0.22	0.00858	0.00553	-0.01259	-0.0248	-0.0248	-0.02785	0.00553	0.00553	-0.02785	-0.04597	0.22	-0.02785	0.00858	-0.00057	0.01163	-0.0248	-0.01869	-0.0248	0.00858	-0.00362	-0.00954
15	0.24	0.01774	-0.00362	-0.00954	-0.0248	-0.02785	-0.02785	0.00248	0.01163	-0.02785	-0.05207	0.24	-0.0309	0.01163	-0.00057	0.01469	-0.0248	-0.02174	-0.0248	0.01163	-0.01259	-0.00954
16	0.26	0.01774	-0.00648	-0.00248	-0.02785	-0.03395	-0.02785	-0.00362	0.01163	-0.02785	-0.04005	0.26	-0.0309	0.00858	0.00248	0.01469	-0.0248	-0.0248	-0.0248	0.01163	-0.01259	-0.01259
17	0.28	0.02079	-0.01259	0.00858	-0.02785	-0.02785	-0.0309	-0.00648	0.00858	-0.02785	-0.01869	0.28	-0.02785	0.01163	0.00553	0.01163	-0.0248	-0.0248	-0.01564	0.00858	-0.01564	-0.02174
18	0.3	0.02384	-0.01564	0.00858	-0.0309	-0.02785	-0.0309	-0.01259	0.00858	-0.0309	-0.00362	0.3	-0.0309	0.01774	0.00858	0.01163	-0.02785	-0.02785	-0.01564	0.01163	-0.02174	-0.01564
19	0.32	0.02079	-0.02174	0.01774	-0.02785	-0.0309	-0.0309	-0.01564	0.01163	-0.0309	0.00858	0.32	-0.02785	0.02079	0.00858	0.00858	-0.01869	-0.02174	-0.00954	0.00858	-0.02174	-0.01564
20	0.34	0.02079	-0.02785	0.01774	-0.0309	-0.03395	-0.02785	-0.0248	0.00858	-0.0248	0.02079	0.34	-0.01564	0.02079	0.00553	0.00858	-0.01259	-0.0248	-0.00362	0.01163	-0.02174	-0.01564
21	0.36	0.02384	-0.0309	0.01163	-0.04005	-0.04005	-0.02785	-0.02785	0.00858	-0.01259	0.02384	0.36	-0.01259	0.02079	0.00248	0.0248	-0.00954	-0.02785	-0.00057	0.00553	-0.02174	-0.02174
22	0.38	0.01774	-0.02785	0.02079	-0.0309	-0.0248	-0.02785	-0.02785	0.00858	-0.00954	0.02689	0.38	-0.01259	0.01774	0.00858	-0.00648	-0.00648	-0.0248	0.00553	0.00858	-0.02174	-0.02174
23	0.4	0.00858	-0.02785	0.02079	-0.0248	-0.01869	-0.01869	-0.02785	0.00858	-0.00362	0.023	0.4	0.00248	0.01163	0.00858	-0.01259	-0.00362	-0.0248	0.00248	0.0248	-0.02174	-0.01869
24	0.42	-0.00362	-0.02785	0.01774	-0.01564	-0.01259	-0.00954	-0.02785	0.00858	0.00248	0.033	0.42	0.00858	0.00858	0.00858	-0.00954	-0.00954	-0.0248	0.00858	-0.00057	-0.0248	-0.01869
25	0.44	-0.00954	-0.02785	0.02079	-0.00954	-0.00648	-0.00057	-0.02785	0.00858	0.00553	0.02689	0.44	0.00858	-0.00057	0.01163	-0.00648	-0.00057	-0.02174	0.01163	-0.00362	-0.0248	-0.01869
26	0.46	-0.02174	-0.0309	0.01163	-0.00362	-0.00648	0.00858	-0.02785	0.00248	0.00858	0.01469	0.46	0.01774	-0.01259	0.00553	0.01259	0.00858	-0.01259	0.00858	-0.00362	-0.0248	-0.02174
27	0.48	-0.02785	-0.0309	0.00057	0.00248	0.00553	0.01469	-0.02785	-0.00057	0.01163	-0.00057	0.48	0.01469	-0.01259	0.00858	-0.01564	0.00858	-0.01259	0.01163	-0.00648	-0.0248	-0.02174

	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC
1	4V= 2.778 RPS														
2	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
3	-0.01259	-0.00648	0.00858	-0.00362	0	-0.00362	-0.01259	-0.01564	-0.01564	-0.00954	-0.01564	-0.00648	-0.01564	0.00248	-0.01259
4	-0.01259	-0.00648	0.01163	0.00858	0.02	-0.00362	-0.01259	-0.01259	-0.01259	-0.00648	-0.01259	-0.00362	-0.01869	0.00248	-0.00648
5	-0.01869	-0.00362	0.00858	0.00553	0.04	-0.00362	-0.01259	-0.01869	-0.01869	-0.00648	-0.01259	-0.00954	-0.01564	-0.00057	-0.01259
6	-0.02785	0.00248	0.01163	0.00553	0.06	-0.00362	-0.01259	-0.01564	-0.01564	-0.00648	-0.01564	-0.00648	-0.01869	0.00248	-0.00954
7	-0.0248	-0.00057	0.01163	-0.00057	0.08	-0.00362	-0.01564	-0.01259	-0.01259	-0.00954	-0.01259	-0.00648	-0.01564	0.00248	-0.01259
8	-0.0248	-0.00057	0.00248	-0.00362	0.1	-0.00362	-0.01259	-0.00954	-0.00954	-0.00648	-0.01259	-0.00648	-0.01869	0.00248	-0.00954
9	-0.0248	0.00858	0.00858	-0.00362	0.12	-0.00057	-0.01259	-0.01259	-0.01259	-0.00648	-0.01259	-0.00648	-0.01869	0.00248	-0.01259
10	-0.0248	-0.00362	0.00553	-0.00362	0.14	-0.00362	-0.01259	-0.00954	-0.00954	-0.00648	-0.01259	-0.01259	-0.02174	0.00248	-0.01259
11	-0.03395	0.00858	0.00248	-0.00362	0.16	-0.00648	-0.01259	-0.00648	-0.00648	-0.00648	-0.00954	-0.00648	-0.01564	0.00553	-0.01259
12	-0.0248	0.00858	-0.00362	-0.00954	0.18	-0.00362	-0.01564	-0.00954	-0.00954	-0.00648	-0.00954	-0.00648	-0.02174	0.00248	-0.01259
13	-0.0248	0.00858	-0.00648	-0.00648	0.2	-0.01259	-0.00954	-0.00954	-0.00954	-0.00648	-0.01259	-0.01564	-0.01259	0.00553	-0.01259
14	-0.0248	0.00858	-0.00362	-0.00954	0.22	-0.00362	-0.01259	-0.00954	-0.00954	-0.00648	-0.00648	-0.00648	-0.01869	0.00248	-0.01259
15	-0.0248	0.01163	-0.01259	-0.00954	0.24	-0.00648	-0.01259	-0.00648	-0.00648	-0.00362	-0.00954	-0.00648	-0.00954	0.00248	-0.01259
16	-0.0248	0.01163	-0.01259	-0.01259	0.26	-0.00362	-0.01259	-0.00648	-0.00648	-0.00362	-0.00648	-0.01259	-0.01564	0.00248	-0.01259
17	-0.01564	0.00858	-0.01564	-0.02174	0.28	-0.00362	-0.01564	-0.00648	-0.00648	-0.00648	-0.01564	-0.00648	-0.00954	0.00248	-0.01259
18	-0.01564	0.01163	-0.02174	-0.01564	0.3	-0.00362	-0.00954	-0.00954	-0.00954	-0.00362	-0.00648	-0.00648	-0.01564	-0.00057	-0.01259
19	-0.00954	0.00858	-0.02174	-0.01564	0.32	-0.00362	-0.00954	-0.01259	-0.01259	-0.00362	-0.00648	-0.00648	-0.00648	0.00248	-0.01259
20	-0.00362	0.01163	-0.02174	-0.01564	0.34	-0.00057	-0.00954	-0.01259	-0.01259	-0.00362	-0.00362	-0.00648	-0.01259	-0.00057	-0.01259
21	-0.00057	0.00553	-0.02174	-0.02174	0.36	-0.00362	-0.01259	-0.00648	-0.00648	-0.00362	-0.01259	-0.01259	-0.00362	0.00248	-0.01259
22	0.00553	0.00858	-0.02174	-0.02174	0.38	-0.00057	-0.01259	-0.00648	-0.00648	-0.00362	-0.00648	-0.00954	-0.00954	-0.00057	-0.01259
23	0.00248	0.00248	-0.02174	-0.01869	0.4	-0.00362	-0.00954	-0.00057	-0.00057	-0.00362	-0.00362	-0.00648	-0.00057	-0.00057	-0.01564
24	0.00858	-0.00057	-0.0248	-0.01869	0.42	-0.00954	-0.00954	-0.00362	-0.00362	-0.00362	-0.00362	-0.01564	-0.00648	-0.00362	-0.01259
25	0.01163	-0.00362	-0.0248	-0.01869	0.44	-0.00362	-0.00954	-0.00362	-0.00362	-0.00362	-0.00362	-0.00954	-0.00057	-0.00648	-0.01564
26	0.00858	-0.00362	-0.0248	-0.02174	0.46	-0.00362	-0.00648	-0.00057	-0.00057	-0.00362	-0.00362	-0.00648	-0.00362	-0.00362	-0.01259
27	0.01163	-0.00648	-0.0248	-0.02174	0.48	-0.00362	-0.00954	-0.00057	-0.00057	-0.00362	-0.00362	-0.00648	0.00248	-0.00648	-0.01259

	AO	AP	AQ	AR	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC
476	-0.00362	0.00858	-0.00362	0.00858	9.46	-0.00362	-0.01869	-0.01259	-0.01259	-0.01259	-0.00362	-0.00954	-0.00648	-0.01564	-0.00057
477	-0.00362	0.00858	-0.00648	0.00553	9.48	-0.01259	-0.00954	-0.01564	-0.01564	-0.01259	-0.00954	-0.00954	-0.00648	-0.01259	-0.00057
478	-0.01259	0.00858	-0.01259	0.00248	9.5	-0.00362	-0.00954	-0.01259	-0.01259	-0.01259	-0.00648	-0.01869	-0.00648	-0.01869	0.00248
479	-0.01259	0.00858	-0.01259	0.00553	9.52	-0.00362	-0.01259	-0.00954	-0.00954	-0.00648	-0.00648	-0.00954	-0.01259	-0.01259	-0.00057
480	-0.01564	0.00858	-0.01259	0.00553	9.54	-0.00362	-0.01259	-0.00954	-0.00954	-0.00362	-0.01259	-0.00954	-0.00954	-0.02174	-0.00057
481	-0.01869	0.00858	-0.0248	-0.00057	9.56	-0.00648	-0.01259	-0.01259	-0.01259	-0.00362	-0.00648	-0.00954	-0.01869	-0.01259	-0.00057
482	-0.02174	0.01163	-0.02174	-0.00362	9.58	-0.00648	-0.01259	-0.00954	-0.00954	-0.00648	-0.00362	-0.00954	-0.01259	-0.01869	-0.00057
483	-0.0309	0.01163	-0.02174	-0.00362	9.6	-0.00362	-0.01259	-0.01259	-0.01259	-0.00362	-0.00954	-0.01869	-0.01564	-0.01259	-0.00362
484	-0.0248	0.00858	-0.02785	-0.00362	9.62	-0.00954	-0.01259	-0.01259	-0.01259	-0.00648	-0.00648	-0.01259	-0.01259	-0.01564	-0.00057
485	-0.0248	0.01163	-0.02174	-0.00648	9.64	-0.00362	-0.01259	-0.01259	-0.01259	-0.00362	-0.00362	-0.00954	-0.01869	-0.00954	-0.00648
486	-0.0248	0.00858	-0.02174	-0.00648	9.66	-0.00362	-0.00954	-0.01259	-0.01259	-0.00362	-0.00362	-0.00954	-0.01564	-0.01259	-0.00057
487	-0.0248	0.01163	-0.0248	-0.00648	9.68	-0.00362	-0.02174	-0.01259	-0.01259	-0.00362	-0.01259	-0.01259	-0.01259	-0.00648	-0.00954
488	-0.03395	0.00248	-0.02174	-0.00648	9.7	-0.00362	-0.01259	-0.01259	-0.01259	-0.00362	-0.00648	-0.01869	-0.01564	-0.01259	-0.00362
489	-0.0248	0.00858	-0.02174	-0.01259	9.72	-0.01259	-0.01259	-0.01259	-0.01259	-0.00057	-0.00362	-0.00954	-0.01259	-0.00648	-0.01259
490	-0.0248	0.00553	-0.01869	-0.00954	9.74	-0.00362	-0.00954	-0.01259	-0.01259	-0.00057	-0.00954	-0.00954	-0.01564	-0.01259	-0.00362
491	-0.0248	-0.00057	-0.0248	-0.01259	9.76	-0.00362	-0.01259	-0.01259	-0.01259	-0.00648	-0.00648	-0.00954	-0.01564	-0.00648	-0.01259
492	-0.02174	-0.00362	-0.02174	-0.01259	9.78	-0.00362	-0.00648	-0.01259	-0.01259	-0.00954	-0.00648	-0.01259	-0.01564	-0.00954	-0.00362
493	-0.02174	-0.00362	-0.01869	-0.01869	9.8	-0.00648	-0.01259	-0.00954	-0.00954	-0.00648	-0.00648	-0.01259	-0.01259	-0.00648	-0.01259
494	-0.01259	-0.00648	-0.0248	-0.01564	9.82	-0.00057	-0.00648	-0.00954	-0.00954	-0.00362	-0.00648	-0.00954	-0.02174	-0.00954	-0.00362
495	-0.00648	-0.01564	-0.01564	-0.01564	9.84	-0.00362	-0.00648	-0.01259	-0.01259	-0.00362	-0.00648	-0.01259	-0.01564	-0.00362	-0.01564
496	-0.00362	-0.00954	-0.00954	-0.01564	9.86	-0.00954	-0.00954	-0.01564	-0.01564	-0.00057	-0.00362	-0.01259	-0.02174	-0.00954	-0.00648
497	0.00248	-0.01564	-0.01259	-0.0248	9.88	-0.00362	-0.00648	-0.01564	-0.01564	-0.00362	-0.01259	-0.00954	-0.01564	-0.00362	-0.01564
498	-0.00057	-0.01564	-0.00648	-0.01564	9.9	-0.00362	-0.01259	-0.01259	-0.01259	-0.00362	-0.00648	-0.00954	-0.01564	-0.00057	-0.00648
499	0.00858	-0.01869	-0.00362	-0.01564	9.92	-0.00362	-0.00954	-0.01259	-0.01259	-0.00057	-0.00648	-0.00648	-0.01564	-0.00362	-0.01259
500	0.01163	-0.01869	-0.00057	-0.01869	9.94	-0.00648	-0.00648	-0.01259	-0.01259	-0.00362	-0.00954	-0.01259	-0.01564	-0.00057	-0.00954
501	0.00858	-0.02174	0.00248	-0.02174	9.96	-0.01259	-0.00648	-0.00648	-0.00648	-0.00057	-0.00362	-0.00648	-0.01564	-0.00362	-0.01564
502	0.01163	-0.0248	0.00858	-0.01564	9.98	-0.00362	-0.00648	-0.00648	-0.00648	-0.00057	-0.00362	-0.00954	-0.01564	-0.00057	-0.01259
503	0.00858	-0.02174	0.00553	-0.01869	10	-0.00362	-0.00648	-0.00648	-0.00648	-0.00057	-0.00362	-0.00648	-0.01259	-0.00362	-0.01259
504															

Data points for $k=1$

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	
1	12V= 8.333 RPS											10V= 6.945 RPS											
2	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11
3	0	-0.01564	0.00248	-0.00648	-0.01564	-0.00057	-0.01564	-0.01564	-0.00648	-0.00648	-0.01564	0	-0.01564	-0.00954	0.00248	-0.01564	0.00248	0.00248	-0.00648	-0.01259	-0.00362	-0.01259	
4	0.02	-0.00954	-0.00648	-0.00954	-0.0248	-0.01259	-0.0248	-0.01564	-0.00057	-0.00057	-0.00362	0.02	-0.01259	-0.01869	0.00248	-0.00648	-0.00648	0.00248	0.008583	-0.02174	-0.00954	-0.01564	
5	0.04	-0.00954	0.00248	-0.01869	-0.01564	-0.01564	-0.01869	-0.01259	-0.01259	0.005531	-0.00954	0.04	-0.01259	-0.01564	-0.00057	-0.00648	-0.00362	-0.00362	0.008583	-0.01869	-0.00648	-0.01259	
6	0.06	-0.00362	0.00248	-0.01869	-0.01564	-0.01564	-0.01564	-0.01259	-0.01259	-0.00057	0.014687	0.06	-0.00648	-0.01259	-0.00362	-0.00362	-0.00362	0.00248	0.00248	-0.02174	-0.00648	-0.01564	
7	0.08	-0.00057	-0.00954	-0.01259	-0.00954	-0.01869	-0.01259	-0.01259	-0.01564	0.005531	0.02079	0.08	-0.01259	-0.01564	-0.00954	-0.00954	-0.00362	-0.00648	-0.00648	-0.00648	-0.00648	-0.00648	
8	0.1	0.00248	-0.00057	-0.00648	-0.01259	-0.01259	-0.01259	-0.01259	-0.01564	-0.00648	-0.00362	0.1	-0.00648	-0.00648	-0.00648	-0.01259	-0.00648	-0.02174	0.00248	-0.01259	-0.00362	-0.00648	
9	0.12	-0.00057	-0.00648	-0.00648	-0.01259	-0.00954	-0.00362	-0.00954	-0.01564	-0.00648	-0.00362	0.12	0.00248	-0.00648	-0.00057	-0.01259	-0.00057	-0.0248	-0.00362	0.00248	-0.00362	-0.00057	
10	0.14	-0.00362	-0.00954	-0.01259	-0.00648	-0.00648	-0.00057	-0.00648	-0.00954	-0.00954	-0.00362	0.14	0.00248	-0.00954	-0.00648	-0.00057	-0.00362	-0.01259	-0.01564	0.00248	-0.00057	0.005531	
11	0.16	-0.01259	-0.01564	-0.00057	-0.00362	-0.01259	0.00248	-0.00057	-0.00648	0.00248	-0.0248	0.16	-0.00057	0.00248	-0.01259	-0.00362	-0.00362	-0.01564	-0.0309	-0.00057	-0.01259	-0.00057	
12	0.18	-0.01259	-0.01564	0.00248	-0.00057	-0.01259	0.00248	-0.00648	-0.01259	-0.00648	-0.02785	0.18	-0.00057	0.00248	-0.01564	-0.00648	-0.01259	-0.00954	-0.01869	0.005531	-0.00362	0.005531	
13	0.2	-0.00362	-0.01564	0.00248	-0.00362	-0.00362	-0.00362	-0.00057	-0.00362	-0.01259	-0.00362	0.2	-0.00954	-0.00057	-0.02174	-0.00648	-0.01259	0.011635	-0.00648	-0.01564	-0.00362	-0.00954	
14	0.22	-0.00648	-0.00954	0.00248	-0.00057	0.00248	-0.00648	-0.00362	0.005531	-0.01564	-0.01564	0.22	-0.00057	0.008583	-0.01259	-0.01259	-0.01259	0.005531	-0.00954	0.00248	-0.00362	-0.00648	
15	0.24	-0.01564	-0.00648	-0.00648	-0.00954	-0.00057	-0.01259	-0.00954	-0.00648	-0.01564	0.00248	0.24	0.00248	-0.01259	-0.01259	-0.00362	-0.01259	0.00248	0.00248	-0.00057	-0.00057	-0.00362	
16	0.26	-0.02174	-0.00954	-0.00362	-0.00954	-0.00057	0.00248	-0.01259	-0.00057	-0.01259	0.017738	0.26	-0.00648	-0.00057	-0.01259	-0.00362	-0.01564	-0.00362	0.00248	-0.00954	-0.00648	-0.00648	
17	0.28	-0.0248	0.005531	-0.00362	-0.00057	-0.00648	-0.00954	-0.00362	0.00248	-0.00057	0.017738	0.28	-0.01259	0.00248	-0.00057	-0.00954	-0.00057	0.005531	-0.0248	-0.00954	-0.00362	-0.01259	
18	0.3	-0.01869	-0.00362	-0.01259	-0.00648	-0.00648	-0.01869	-0.01564	-0.00954	-0.00648	-0.00648	0.3	-0.01564	-0.00648	0.005531	-0.00362	-0.00362	-0.00362	0.008583	-0.01564	-0.00057	-0.02174	
19	0.32	-0.01259	0.005531	-0.02174	-0.01869	0.00248	-0.01259	-0.01564	-0.00057	-0.01259	0.00248	0.32	-0.01259	-0.01259	-0.00057	-0.00648	-0.00648	0.00248	-0.01259	-0.00648	-0.01564	-0.00362	
20	0.34	-0.00362	0.005531	-0.02174	-0.02174	-0.00648	-0.01259	-0.01564	-0.00648	-0.00362	-0.01869	0.34	-0.01564	-0.0248	0.00248	-0.00648	-0.00362	-0.0248	-0.00057	-0.00648	0.00248	-0.00648	
21	0.36	-0.01259	-0.00954	-0.01259	-0.01869	-0.01259	-0.00954	-0.01259	-0.01869	0.005531	-0.0248	0.36	-0.00954	-0.01564	-0.00954	-0.00954	0.00248	-0.02174	-0.01259	-0.00954	-0.00057	-0.00362	
22	0.38	-0.00362	-0.00362	-0.00648	-0.01259	-0.01869	-0.00648	-0.01259	-0.01564	0.008583	-0.0309	0.38	-0.00648	-0.00954	-0.00057	-0.01259	-0.00057	-0.00648	-0.02174	-0.00057	-0.00954	-0.01259	
23	0.4	-0.00057	-0.00648	-0.00648	-0.01259	-0.01869	-0.01564	-0.00648	-0.01564	0.00248	-0.00057	0.4	-0.00057	-0.01259	-0.00362	-0.01259	-0.00057	-0.00057	-0.0248	-0.00362	-0.01259	-0.00362	
24	0.42	-0.00648	-0.01259	-0.00648	-0.01869	-0.01869	-0.00362	-0.01259	-0.01869	-0.00648	-0.01564	0.42	0.008583	-0.00648	-0.00648	-0.01259	0.005531	-0.01564	-0.01564	-0.00057	-0.00362	0.005531	
25	0.44	-0.00057	-0.01259	0.00248	-0.00362	-0.00954	-0.00057	-0.00648	-0.00954	-0.00057	0.011635	0.44	-0.00648	0.005531	-0.01259	-0.01564	-0.00648	0.017738	-0.01259	0.00248	-0.00362	-0.00057	
26	0.46	-0.00954	-0.01564	-0.00057	0.00248	-0.00648	0.005531	0.005531	-0.00648	-0.00057	0.02079	0.46	0.00248	-0.00057	-0.0248	-0.01259	-0.00362	0.008583	-0.00648	-0.00954	-0.00057	-0.00362	
27	0.48	-0.01259	-0.01564	-0.00648	0.00248	-0.01259	0.00248	-0.00362	-0.00954	-0.01564	-0.00057	0.48	-0.00248	0.00248	-0.01259	-0.00954	0.00248	0.008583	0.011635	0.005531	-0.00362	-0.00362	
28	0.5	-0.00057	-0.00362	0.008583	-0.00648	-0.00362	-0.00362	-0.00362	-0.00648	-0.01564	-0.00954	0.5	0.00248	0.008583	-0.00954	-0.00362	-0.00362	-0.01259	0.008583	-0.00954	-0.00954	0.00248	
29	0.52	-0.00954	-0.01259	-0.01259	0.00248	0.00248	-0.00362	-0.00057	0.005531	-0.0248	0.005531	0.52	0.00248	-0.01259	-0.00648	-0.01259	-0.01259	0.005531	0.005531	-0.01564	-0.01564	-0.00648	
30	0.54	-0.01869	-0.00057	-0.00362	-0.00954	0.005531	-0.00362	-0.00648	0.005531	-0.00648	-0.02785	0.54	-0.01564	0.005531	-0.00648	-0.00362	-0.01564	-0.00648	0.00248	-0.0248	-0.01259	-0.01564	
31	0.56	-0.02174	0.005531	-0.00057	-0.00648	-0.00057	-0.00057	-0.00057	-0.00057	-0.00057	-0.0248	0.56	-0.01564	-0.00362	-0.005531	-0.00362	-0.01738	-0.01564	-0.00362	-0.01738	-0.01738	-0.01738	
Physics EE readings K=0 K=1 K=2 K=3 K=4 +																							

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
474	9.42	-0.00954	-0.01869	-0.00057	-0.00648	-0.01564	0.008583	-0.01259	-0.01259	-0.0309	0.008583	9.42	-0.00057	-0.00057	-0.02174	-0.00362	-0.01259	-0.01259	-0.0248	-0.00057	-0.00648	-0.00648
475	9.44	-0.01869	-0.01259	-0.00648	-0.01259	-0.00954	0.00248	-0.00057	-0.01869	-0.01564	0.011635	9.44	-0.00648	-0.00362	-0.0309	0.00248	-0.02174	0.011635	-0.01259	-0.00057	-0.01259	-0.01259
476	9.46	-0.01564	-0.01564	-0.02174	0.00248	-0.00954	-0.00362	0.008583	-0.01869	-0.00954	0.005531	9.46	-0.02785	-0.00648	-0.01259	0.008583	-0.0248	0.008583	-0.00057	-0.01259	-0.02174	-0.0248
477	9.48	-0.01564	-0.00648	-0.0248	0.005531	-0.00954	-0.00954	0.00248	-0.01564	-0.00954	-0.00954	9.48	-0.0248	-0.01869	-0.00362	0.008583	-0.01259	0.008583	-0.00362	-0.02174	-0.02174	-0.02174
478	9.5	-0.01259	-0.01259	-0.01259	0.005531	-0.00648	-0.00648	-0.00362	-0.00362	0.005531	-0.00362	9.5	-0.01564	-0.0248	-0.00648	0.008583	-0.00362	-0.00057	0.014687	-0.02785	-0.01259	-0.00648
479	9.52	-0.01564	-0.00648	-0.00954	0.005531	0.008583	-0.00057	-0.00954	-0.01259	0.014687	-0.02785	9.52	-0.00648	-0.0248	0.014687	-0.01259	-0.00954	-0.00362	-0.00362	-0.00648	-0.00362	-0.00648
480	9.54	-0.01259	0.005531	-0.00954	-0.00954	0.008583	-0.01259	-0.00362	-0.00057	0.008583	-0.0248	9.54	-0.00648	-0.00954	0.008583	-0.00057	0.02079	-0.00057	0.011635	-0.00362	-0.00954	-0.00057
481	9.56	0.00248	0.011635	-0.01259	-0.00648	0.00248	-0.02174	-0.00362	0.008583	-0.01259	-0.01259	9.56	0.00248	-0.00648	0.008583	-0.01259	0.00248	-0.01564	-0.00648	-0.00362	0.011635	0.011635
482	9.58	0.008583	0.00248	-0.00057	-0.00362	-0.00362	-0.02174	-0.01259	0.005531	-0.00362	-0.00057	9.58	0.008583	-0.00057	-0.01564	-0.02174	0.008583	-0.02174	0.008583	0.014687	0.005531	0.005531
483	9.6	0.00248	-0.00057	0.008583	-0.01259	-0.00648	-0.01259	-0.02785	0.00248	-0.02174	-0.00648	9.6	0.008583	0.011635	0.00248	-0.0248	-0.00954	-0.0248	-0.01259	0.008583	0.011635	0.008583
484	9.62	-0.00057	-0.01259	-0.00362	-0.01564	-0.00362	-0.00648	-0.01564	-0.00954	-0.02785	0.014687	9.62	0.00248	0.008583	-0.00954	-0.01564	0.00248	-0.01259	-0.02174	0.00248	-0.00362	-0.00648
485	9.64	-0.01564	-0.00362	0.011635	-0.01564	-0.00362	-0.00954	-0.00954	-0.00648	-0.00954	0.02079	9.64	-0.00362	0.008583	-0.02174	-0.00362	-0.01259	-0.00362	-0.0248	-0.00648	0.00248	0.00248
486	9.66	-0.00057	-0.00648	-0.00362	-0.01564	-0.01564	-0.01259	-0.00954	-0.00648	-0.00362	-0.00954	9.66	-0.00057	-0.00954	-0.02785	-0.01259	-0.01564	-0.00362	-0.01564	0.00248	-0.01259	-0.01259
487	9.68	-0.01259	-0.01259	-0.00648	-0.00648	-0.0248	-0.00362	-0.01259	-0.01564	-0.00362	-0.00057	9.68	-0.01259	-0.00057	-0.01564	-0.02079	-0.0309	0.014687	-0.00648	-0.01259	-0.0248	-0.02174
488	9.7	-0.01564	-0.01869	0.00248	-0.01259	-0.02174	0.00248	-0.00057	-0.01869	-0.02079	-0.01259	9.7	-0.02174	-0.01259	-0.00648	0.005531	-0.01564	0.008583	-0.01259	-0.02174	-0.0248	-0.0309
489	9.72	-0.01869	-0.01274	-0.00648	-0.01869	-0.00648	-0.01259	-0.008583	-0.0248	-0.02079	-0.01357	9.72	-0.03395	-0.0248	-0.00648	0.008583	-0.00954	0.008583	0.008583	-0.02785	-0.01869	-0.02174
490	9.74	-0.01869	-0.01259	-0.00648	-0.00362	-0.00648	0.00248	-0.00248	-0.00648	-0.00057	-0.0248	9.74	-0.00954	-0.02785	0.014687	-0.01259	-0.00648	-0.00954	0.008583	-0.01564	-0.00648	-0.00954
491	9.76	-0.01564	-0.01259	-0.01564	-0.00248	-0.00954	-0.01259	-0.00362	-0.00362	-0.00648	-0.00057	9.76	-0.00954	-0.00954	0.008583	0.005531	0.011635	-0.00248	0.008583	-0.00648	-0.00648	-0.01259
492	9.78	-0.00954	-0.01564	-0.01869	-0.00057	-0.00362	-0.00362	-0.01564	-0.01259	-0.00648	-0.02174	9.78	-0.00057	-0.01869	-0.00248	-0.00648	0.008583	-0.01259	-0.00362	-0.00954	0.011635	0.011635
493	9.8	-0.01259	-0.00362	-0.00648	-0.00057	0.005531	-0.00648	-0.00057	-0.00362	-0.02174	0.011635	9.8	0.014687	-0.00362	-0.00362	-0.01564	0.00248	-0.01564	-0.00057	0.017738	0.00248	0.008583
494	9.82	-0.00248	0.00248	-0.00648	-0.00954	-0.00362	-0.01564	-0.00954	0.008583	-0.02785	0.02079	9.82	-0.00057	0.017738	-0.00057	-0.02785	0.00248	-0.02785	-0.00648	0.008583	0.008583	0.011635
495	9.84	0.005531	0.00248	-0.01259	-0.00057	0.00248	-0.01259	-0.01869	0.005531	-0.00954	0.017738	9.84	0.008583	0.00248	-0.00057	-0.01564	-0.00362	-0.01259	-0.01869	0.005531	-0.00362	-0.00648
496	9.86	0.00248	0.00248	0.00248	-0.00057	-0.00057	-0.01564	-0.01564	0.00248	-0.00954	-0.00648	9.86	-0.00954	0.011635	-0.01564	-0.00954	-0.00057	-0.01564	-0.02174	-0.00362	-0.00057	0.008583
497	9.88	-0.00057	-0.01259	0.017738	-0.01259	-0.00648	-0.01564	-0.01869	-0.00648	-0.00362	-0.00057	9.88	0.00248	-0.01259	-0.0248	-0.00648	-0.01564	-0.00362	-0.02174	-0.00057	-0.00648	-0.00954
498	9.9	-0.01564	-0.00057	0.00248	-0.02174	-0.00057	-0.01259	-0.00648	-0.00057	0.014687	-0.0248	9.9	-0.01259	0.00248	-0.01564	-0.011635	-0.02174	0.008583	-0.01564	-0.00648	-0.01564	-0.01564
499	9.92	-0.00057	-0.00362	-0.00057	-0.02174	-0.00648	-0.00648	-0.00954	-0.00057	0.017738	-0.0248	9.92	-0.0248	-0.01259	-0.00954	0.008583	-0.0248	0.008583	-0.00362	-0.01564	-0.02785	-0.0248
500	9.94	-0.00954	-0.01564	-0.01259	-0.01259	-0.02174	-0.01869	-0.00362	-0.01259	-0.00648	-0.01869	9.94	-0.02785	-0.01259	-0.00362	0.008583	-0.01259	0.008583	-0.00362	-0.0309	-0.02785	-0.01564
501	9.96	-0.01869	-0.01428	0.00248	-0.00362	-0.0248	-0.00057	-0.00648	-0.02174	-0.00057	-0.00057	9.96	-0.00954	-0.0309	0.00248	-0.00057	-0.00648	-0.00362	0.014687	-0.01564	-0.00648	-0.00362
502	9.98	-0.01869	-0.01564	-0.00648	-0.00954	-0.02174	0.008583	0.00248	-0.0248	-0.01564	-0.00954	9.98	-0.01259	-0.01259	0.005531	-0.00057	-0.00648	-0.00648	0.008583	-0.00954	-0.00362	-0.01259
503	10	-0.01564	-0.00648	-0.01259	-0.00954	-0.00648	0.00248	0.005531	-0.00057	-0.02785	0.011635	10	-0.00362	-0.00648	0.008583	0.00248	0.017738	-0.00057	-0.0248	-0.00648	0.00248	0.011635
504																						
505																						

1	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR
2	8V= 5.56 RPS												6V= 4.167 RPS										
3	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
3	-0.01259	0	-0.00057	-0.00057	-0.01259	-0.00648	-0.00362	-0.00362	-0.00057	-0.01564	-0.00648	-0.00648	0	-0.00362	-0.00954	-0.00362	-0.00648	-0.00648	-0.00057	-0.00648	-0.00362	-0.00648	-0.00362
4	-0.01564	0.02	-0.00248	-0.00362	-0.00362	-0.00362	-0.005531	0.00248	-0.01259	-0.00954	-0.00648	-0.00648	0.02	-0.00648	-0.00362	-0.00057	-0.00648	-0.00954	-0.00057	-0.00954	-0.00362	-0.00648	-0.00648
5	-0.01259	0.04	-0.00057	-0.01259	-0.00648	-0.00362	-0.00648	-0.00057	-0.00648	-0.01259	-0.00057	-0.01259	0.04	-0.00648	-0.00648	-0.00362	-0.00057	-0.00648	0.00248	-0.00057	-0.01259	-0.00648	-0.00648
6	-0.01564	0.06	-0.00248	-0.01259	-0.00057	-0.00362	-0.00248	-0.01259	-0.00954	-0.01259	-0.00648	-0.01564	0.06	-0.00954	-0.01259	-0.00057	-0.00057	-0.00648	0.00248	-0.00057	-0.01259	-0.01259	-0.00362
7	-0.00648	0.08	-0.00362	-0.01259	0.005531	-0.00057	-0.01259	-0.01564	-0.00057	-0.01259	-0.00057	-0.00954	0.08	-0.02174	-0.00362	-0.00362	-0.00362	-0.00057	-0.00057	-0.00362	-0.01259	-0.02174	-0.00362
8	-0.00648	0.1	-0.00362	-0.01259	0.00248	-0.00362	-0.00648	-0.01564	-0.00057	-0.00648	-0.00057	-0.01564	0.1	-0.01259	-0.00648	-0.00362	-0.00954	0.00248	0.00248	-0.00362	-0.01259	-0.01259	-0.00362
9	-0.00057	0.12	-0.00248	-0.00648	0.00248	-0.00648	0.00248	-0.01564	-0.01259	-0.00057	-0.00362	-0.01259	0.12	-0.01259	-0.01259	-0.00362	-0.01564	-0.00057	-0.00648	-0.00362	-0.00954	-0.01259	-0.00648
10	0.005531	0.14	-0.00057	-0.00648	-0.00648	-0.00362	-0.00362	-0.00648	-0.01564	-0.00248	-0.01564	-0.01259	0.14	-0.01259	-0.00362	-0.01259	-0.01564	-0.00057	-0.01259	-0.00954	-0.01564	-0.01259	-0.01259
11	-0.00057	0.16	-0.00362	-0.00648	-0.00362	-0.00362	-0.00954	-0.00362	-0.01564	-0.00248	-0.00057	-0.01564	0.16	-0.00954	-0.00648	-0.00362	-0.01259	-0.00057	0.00248	-0.00057	-0.01259	-0.00954	-0.00362
12	0.005531	0.18	-0.01259	-0.00057	0.00248	-0.00057	-0.01564	-0.00648	-0.01259	-0.00057	-0.00057	-0.00362	0.18	-0.00362	-0.00954	0.00248	-0.01259	-0.00648	-0.00057	-0.00648	-0.00954	-0.00362	-0.00362
13	-0.00954	0.2	-0.02174	0.00248	-0.00362	-0.00648	-0.01259	-0.00362	-0.01259	-0.00057	-0.00954	-0.00648	0.2	-0.00648	-0.00362	-0.00362	-0.00954	-0.00954	-0.00648	-0.00648	-0.00362	-0.00362	-0.00648
14	-0.00648	0.22	-0.02174	0.00248	-0.00954	-0.00648	-0.01259	-0.00057	-0.00648	-0.00362	-0.00648	-0.00648	0.22	-0.00057	-0.00362	-0.00648	-0.00648	-0.00057	-0.01259	-0.00648	-0.00362	-0.00362	-0.00362
15	-0.00362	0.24	-0.01564	0.008583	-0.01259	-0.00648	-0.00648	-0.00362	-0.00648	-0.00362	-0.00648	-0.00362	0.24	-0.00362	-0.00057	-0.01259	-0.01259	-0.00362	-0.01259	-0.01259	-0.00954	0.005531	-0.00648
16	-0.00648	0.26	-0.00648	-0.00362	-0.01259	-0.00648	-0.00362	-0.00362	-0.00362	0.00248	-0.02174	-0.00362	0.26	-0.00057	-0.00362	-0.01259	-0.00362	-0.00648	-0.01259	-0.01869	0.00248	0.00248	-0.00362
17	-0.01259	0.28	-0.00362	-0.00362	-0.01259	-0.00648	-0.00362	-0.01564	-0.00362	-0.00362	-0.01564	-0.00057	0.28	-0.00057	-0.00362	-0.01564	-0.00362	-0.00648	-0.01564	-0.01259	0.00248	-0.00057	-0.00362
18	-0.02174	0.3	-0.00954	0.00248	-0.00362	-0.00362	0.005531	-0.00057	-0.00057	-0.00954	-0.01259	-0.00362	0.3	0.005531	-0.00648	-0.01564	-0.00057	-0.01259	-0.01564	-0.00057	-0.00057	0.00248	-0.01259
19	-0.01259	0.32	-0.00057	-0.00362	-0.00362	-0.00057	0.00248	-0.00057	-0.00057	-0.01564	-0.01259	0.008583	0.32	-0.00648	-0.01259	-0.01259	0.00248	-0.01564	-0.00954	-0.01564	-0.00057	-0.00648	-0.00362
20	-0.00648	0.34	-0.00057	-0.00648	-0.00954	-0.00362	-0.00057	-0.00362	-0.00057	-0.01564	-0.00362	-0.00954	0.34	-0.00954	-0.00648	-0.00954	-0.00057	-0.01259	-0.00648	-0.00362	-0.00362	-0.00648	-0.00648
21	-0.00362	0.36	-0.00057	-0.01259	0.00248	-0.00362	0.00248	-0.01259	-0.00648	-0.01259	-0.00648	-0.00954	0.36	0.00248	-0.00954	-0.00954	0.00248	-0.02174	-0.01259	-0.00648	-0.00362	0.00248	-0.01259
22	-0.01259	0.38	0.005531	-0.01564	0.00248	-0.00648	-0.00648	-0.01564	-0.00057	-0.01259	-0.01259	0.00248	0.38	-0.00362	-0.01259	-0.00648	-0.00648	-0.00648	-0.00362	-0.00362	-0.00057	-0.00362	-0.00362
23	-0.00362	0.4	-0.00648	-0.01259	-0.00057	-0.00954	0.005531	-0.01564	-0.00057	-0.00648	-0.00057	-0.00648	0.4	-0.00648	-0.00362	-0.00362	-0.00648	-0.00362	-0.00057	0.00248	-0.00954	-0.00648	-0.00362
24	0.005531	0.42	-0.00057	-0.00648	0.00248	-0.01259	-0.00362	-0.01259	-0.00648	-0.00362	0.00248	-0.00954	0.42	-0.01564	-0.00648	-0.00362	-0.00057	-0.01259	-0.00362	-0.00057	-0.00648	-0.00954	-0.00362
25	-0.00057	0.44	-0.00248	-0.00648	-0.00648	-0.01259	-0.01259	-0.00648	-0.01259	-0.00648	0.00248	-0.01259	0.44	-0.01259	-0.00648	0.00248	-0.00362	-0.00362	-0.00057	0.005531	-0.01259	-0.01259	-0.00362
26	-0.00362	0.46	-0.00648	-0.01259	0.00248	-0.01259	-0.01564	-0.00648	-0.01564	0.008583	-0.00362	-0.01869	0.46	-0.01564	-0.00648	-0.00057	-0.01259	0.00248	-0.00057	-0.00362	-0.01259	-0.01564	-0.01259
27	-0.00362	0.48	-0.01564	-0.00057	-0.00057	-0.01259	-0.01564	-0.00362	-0.01564	0.00248	-0.00057	-0.01259	0.48	-0.01259	-0.00362	-0.00362	-0.01564	-0.00248	0.00248	-0.01259	-0.01564	-0.01564	-0.00648
28	-0.00248	0.5	-0.0248	-0.00057	-0.00954	-0.01259	-0.01564	0.00248	-0.01564	0.00248	-0.00954	-0.01564	0.5	-0.01259	-0.00362	-0.00648	-0.01564	-0.00362	-0.00648	-0.00057	-0.02174	-0.00954	-0.00648
29	-0.00648	0.52	-0.01869	-0.00057	-0.01259	-0.00648	-0.00648	-0.00057	-0.00648	0.008583	-0.00362	-0.01259	0.52	-0.00362	-0.00057	-0.00648	-0.01564	-0.00362	-0.00648	-0.00057	-0.01259	-0.00362	-0.00954
30	-0.01564	0.54	-0.01259	0.00248	-0.01564	-0.00362	-0.00362	0.00248	-0.00954	-0.00954	-0.00057	-0.00362	0.54	-0.00648	-0.00057	0.00248	-0.02174	-0.00648	0.00248	-0.00057	-0.01259	-0.00362	-0.00362
31	-0.01259	0.55	-0.00648	-0.01564	-0.01259	-0.00648	-0.00362	-0.00362	-0.00954	-0.00648	-0.00362	-0.00362	0.55	-0.00648	-0.00648	-0.00648	-0.00954	-0.00362	-0.00648	-0.00648	-0.00057	-0.00362	-0.00648

	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR
474	-0.00648	9.42	-0.01259	0.00248	-0.00362	-0.0248	0.005531	-0.01259	-0.01564	-0.01259	-0.02174	0.008583	9.42	0.00248	0.008583	-0.01259	-0.01259	-0.0248	-0.01869	-0.01869	0.008583	-0.00648	-0.00362
475	-0.01259	9.44	-0.01869	0.011635	0.00248	-0.01259	0.008583	-0.01259	-0.01259	-0.01564	-0.01259	0.008583	9.44	0.005531	-0.00362	-0.00954	-0.00648	-0.02174	-0.01869	-0.01869	0.00248	-0.00057	-0.00648
476	-0.0248	9.46	-0.02785	-0.00057	-0.00954	-0.00362	-0.00954	-0.00057	-0.00362	-0.01869	-0.00362	0.00248	9.46	0.005531	-0.01259	-0.01564	0.008583	-0.01869	-0.01564	-0.01564	-0.00057	-0.00362	-0.00648
477	-0.02174	9.48	-0.0248	0.011635	-0.01259	-0.00648	-0.00057	0.011635	-0.00648	-0.02174	-0.00954	-0.01259	9.48	0.008583	-0.00057	-0.02174	0.005531	-0.01564	-0.01259	-0.01259	-0.00362	-0.01564	-0.00648
478	-0.00648	9.5	-0.01259	-0.00954	-0.02174	-0.00362	-0.00057	0.005531	0.00248	-0.01564	0.005531	-0.00648	9.5	-0.01564	-0.00057	-0.01259	0.00248	-0.00648	-0.00362	-0.00362	-0.00362	-0.01564	-0.01259
479	-0.00648	9.52	-0.00362	-0.00057	-0.01564	0.011635	-0.00954	0.005531	0.017738	-0.01259	0.008583	-0.01259	9.52	0.00248	-0.01259	-0.00648	-0.00648	-0.00057	-0.00648	-0.00362	-0.00057	-0.02174	-0.00648
480	-0.00057	9.54	-0.00954	-0.00954	-0.00954	-0.00248	-0.01564	-0.00057	0.014687	-0.00057	-0.00362	-0.01564	9.54	-0.00362	-0.01564	-0.00648	-0.00362	-0.00057	0.008583	0.00248	-0.01259	-0.01259	-0.00648
481	0.011635	9.56	-0.00057	-0.01564	-0.00362	0.008583	-0.0248	-0.00648	-0.00362	0.005531	0.008583	-0.02174	9.56	-0.01259	-0.01564	-0.00057	-0.00362	0.005531	-0.00248	0.005531	-0.01564	-0.01259	-0.00648
482	0.005531	9.58	0.008583	-0.02174	-0.00954	-0.00648	-0.01564	-0.00057	-0.01564	0.008583	-0.01259	-0.02174	9.58	-0.01259	-0.02174	0.005531	-0.00362	-0.00057	0.005531	0.008583	-0.01564	-0.01259	-0.00648
483	0.008583	9.6	-0.00057	-0.02785	-0.008583	-0.00362	-0.00362	-0.00648	-0.00057	-0.00248	-0.00362	-0.02174	9.6	-0.01564	-0.01259	0.008583	-0.01564	-0.008583	0.00248	0.00248	-0.02174	-0.00362	-0.01259
484	-0.00648	9.62	0.008583	-0.02174	-0.008583	-0.00057	-0.00362	-0.01259	-0.00362	0.00248	-0.00648	-0.00954	9.62	-0.01564	-0.01259	-0.00057	-0.01259	0.008583	-0.00954	-0.01259	-0.01259	0.008583	-0.00362
485	0.00248	9.64	-0.00954	-0.00954	0.005531	-0.01564	-0.00648	-0.0248	-0.01259	-0.01259	-0.00362	-0.00362	9.64	-0.01869	-0.01259	0.00248	-0.01564	-0.01259	0.00248	-0.00648	-0.00954	0.008583	-0.00648
486	-0.01259	9.66	-0.00057	-0.01564	0.008583	-0.01564	0.00248	-0.01869	-0.02174	-0.00057	-0.0248	0.008583	9.66	-0.00362	-0.00362	0.005531	-0.0248	-0.00362	-0.00057	-0.00057	-0.01259	0.005531	-0.00648
487	-0.02174	9.68	-0.00057	-0.01259	-0.00954	-0.0248	-0.008583	-0.01259	-0.0248	-0.0248	-0.02174	0.008583	9.68	-0.00648	0.005531	-0.01564	-0.01259	-0.00648	-0.01259	-0.01259	-0.00362	-0.00057	-0.00648
488	-0.0309	9.7	-0.01259	0.00248	0.00248	-0.01564	0.00248	-0.00648	-0.01259	-0.01259	-0.01564	0.008583	9.7	-0.00057	0.008583	-0.00057	-0.01259	-0.01259	-0.02174	-0.02174	0.008583	0.00248	-0.00648
489	-0.02174	9.72	-0.02174	-0.00362	-0.00648	-0.00648	0.00248	-0.00648	-0.00648	-0.01564	-0.01564	0.00248	9.72	-0.00362	-0.00248	-0.00362	-0.01259	-0.02785	-0.01674	-0.01869	-0.00531	-0.00648	-0.00648
490	-0.00954	9.74	-0.02785	0.00248	-0.01259	-0.00648	-0.00362	0.005531	-0.00648	-0.01869	-0.00362	-0.00954	9.74	0.008583	0.00248	-0.01869	-0.00362	-0.02174	-0.02174	-0.01564	-0.00248	-0.00362	-0.00648
491	-0.01259	9.76	-0.0248	-0.00362	-0.01869	-0.00362	0.00248	0.008583	-0.00648	-0.01259	-0.00362	0.00248	9.76	0.008583	0.00248	-0.01869	0.008583	-0.02174	-0.01259	-0.01564	-0.00362	-0.01259	-0.00954
492	0.011635	9.78	-0.01259	-0.00362	-0.01564	0.008583	-0.00954	-0.00362	0.00248	-0.01564	-0.00248	-0.01259	9.78	0.0248	-0.00648	-0.01869	-0.00362	-0.01869	-0.00648	-0.00648	-0.00362	-0.01259	-0.00362
493	0.008583	9.8	-0.00648	-0.00057	-0.01564	0.005531	-0.01869	-0.00362	0.014687	-0.00362	0.008583	-0.01259	9.8	-0.01564	-0.00057	-0.01564	0.00248	-0.00648	-0.00362	-0.00648	-0.00057	-0.01869	-0.00648
494	0.011635	9.82	-0.00648	-0.00954	-0.00362	0.008583	-0.02785	-0.01259	0.014687	-0.00057	0.008583	-0.02785	9.82	-0.00057	-0.00648	-0.01259	-0.00057	-0.00362	-0.00362	-0.00057	-0.00057	-0.02785	-0.00648
495	-0.00362	9.84	0.005531	-0.01564	-0.00954	-0.00362	-0.02785	-0.00057	-0.00362	-0.008583	-0.00362	-0.02174	9.84	-0.00362	-0.01564	-0.00648	-0.01259	-0.00248	0.008583	0.008583	-0.01259	-0.01259	-0.00648
496	0.008583	9.86	0.00248	-0.0248	0.005531	0.00248	-0.01869	-0.00362	-0.00362	-0.008583	-0.00362	-0.01564	9.86	-0.01259	-0.01564	-0.00362	-0.00362	0.008583	0.00248	-0.00057	-0.01564	-0.01259	-0.00954
497	-0.00954	9.88	0.005531	-0.01564	-0.008583	-0.00057	-0.00954	-0.01259	0.00248	0.008583	-0.00362	-0.00362	9.88	0.02174	-0.0248	-0.00057	-0.00362	-0.00248	-0.00362	0.00248	-0.01869	-0.00648	-0.00362
498	-0.01564	9.9	0.008583	-0.01259	0.00248	-0.01259	-0.01259	-0.01564	-0.00648	-0.00362	-0.02174	0.00648	9.9	-0.01564	-0.02174	0.008583	-0.02174	0.008583	-0.00362	0.005531	-0.02174	-0.00362	-0.00362
499	-0.0248	9.92	-0.00648	-0.00362	-0.008583	-0.01564	0.00248	-0.01564	-0.01564	-0.01259	-0.01564	-0.00648	9.92	-0.01564	-0.01259	-0.00248	-0.01259	-0.00057	-0.00648	-0.01259	-0.01564	-0.00362	-0.00362
500	-0.00248	9.94	-0.00248	0.00248	0.00248	-0.008583	-0.01564	-0.02174	-0.008583	-0.01564	-0.02174	0.008583	9.94	-0.01564	-0.00248	-0.00248	-0.01259	-0.00248	-0.00248	-0.00248	-0.00248	-0.00362	-0.00362
501	-0.00362	9.96	-0.00954	-0.00248	-0.00248	-0.01564	0.00248	-0.00362	-0.00362	-0.00785	-0.00057	-0.02785	0.008583	9.96	-0.00648	-0.00362	-0.008583	-0.02785	-0.00057	-0.00954	-0.00954	-0.00954	-0.00954
502	-0.01259	9.98	-0.01259	0.005531	-0.00648	-0.01259	-0.011635	-0.00648	-0.01564	-0.01259	-0.01564	0.00248	9.98	-0.00362	-0.00248	-0.01564	-0.01259	-0.00057	-0.00954	-0.01259	-0.00362	0.00248	-0.00648
503	0.011635	10	-0.01869	0.00248	-0.01259	-0.00954	-0.00954	0.00248	-0.00362	-0.01869	-0.00954	-0.01259	10	-0.00648	0.008583	-0.00362	-0.00954	-0.01259	-0.00954	-0.01564	0.008583	-0.00362	-0.00362
504																							

AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC
9.42	-0.00362	-0.01259	0.00248	-0.00057	-0.00648	-0.00362	-0.00057	-0.01564	-0.00057	-0.01259
9.44	-0.00954	-0.01564	-0.00648	-0.00362	-0.01564	-0.01564	-0.00648	-0.01564	-0.00362	-0.01869
9.46	-0.00954	-0.00648	-0.00362	-0.01259	-0.00954	-0.00057	-0.00057	-0.01259	0.00248	-0.01259
9.48	-0.00954	-0.00954	0.00248	-0.00362	-0.01259	-0.00057	-0.00954	-0.01564	-0.00954	-0.00362
9.5	-0.00057	-0.00648	-0.00362	-0.00057	-0.01259	-0.00648	-0.00362	-0.00057	-0.00648	-0.00362
9.52	-0.00648	-0.00362	-0.01259	-0.00057	-0.01259	-0.00954	-0.00362	-0.00648	-0.00648	-0.00648
9.54	-0.00648	-0.00362	-0.00057	-0.00648	-0.02174	-0.01259	-0.00954	-0.00648	-0.00057	-0.00648
9.56	-0.00648	-0.00057	-0.00057	-0.01259	-0.00954	-0.01259	-0.00362	-0.00362	-0.00362	-0.00057
9.58	-0.00954	-0.00057	-0.00648	-0.01259	-0.00648	-0.01259	-0.00057	-0.00057	-0.00362	-0.00362
9.6	-0.01259	0.00248	-0.00648	-0.01869	-0.00362	-0.01259	-0.00648	0.005531	-0.00648	-0.00057
9.62	-0.01259	-0.00954	-0.00954	-0.01259	-0.00954	-0.01259	-0.01259	-0.00362	-0.01259	-0.00362
9.64	-0.01259	-0.00057	-0.01259	-0.01259	-0.00648	-0.01259	-0.00954	0.00248	-0.01564	-0.00057
9.66	-0.01564	-0.00362	-0.01259	-0.01259	0.00248	-0.00648	-0.01259	-0.00057	-0.02174	-0.00057
9.68	-0.00954	-0.00954	-0.02174	-0.01259	0.00248	-0.01259	-0.01564	-0.00362	-0.00954	-0.00362
9.7	-0.01259	-0.00057	-0.00954	-0.00648	-0.00362	-0.00362	-0.01259	-0.00648	-0.01259	-0.01259
9.72	-0.00362	-0.00057	-0.00954	-0.00362	-0.00057	-0.00362	-0.02174	-0.00954	-0.00954	-0.00057
9.74	-0.00648	-0.00648	-0.00648	-0.00648	-0.00057	-0.00057	-0.01259	-0.00057	-0.00648	0.00248
9.76	-0.00362	-0.01564	-0.00648	-0.00057	-0.00362	-0.00057	-0.00362	-0.00362	-0.00362	-0.00954
9.78	-0.00362	-0.00954	-0.00954	0.00248	-0.01259	-0.00648	-0.00362	-0.00362	-0.00362	-0.00954
9.8	0.00248	-0.01259	-0.00362	-0.00057	-0.00362	0.00248	-0.00648	-0.01259	-0.00648	-0.01564
9.82	-0.00057	-0.01259	-0.00362	-0.00057	-0.00362	-0.00057	-0.00648	-0.01259	-0.00057	-0.01259
9.84	-0.00648	-0.01259	0.00248	-0.00648	-0.00057	-0.00057	-0.00057	-0.01564	0.00248	-0.01259
9.86	0.00248	-0.00954	-0.00057	0.00248	-0.00362	-0.00648	-0.00362	-0.01869	0.00248	-0.01564
9.88	-0.00954	-0.00954	-0.00057	-0.00362	-0.01564	-0.01259	-0.00057	-0.00648	-0.00362	-0.01259
9.9	-0.00954	-0.01259	-0.00362	-0.00362	-0.01259	-0.00362	-0.00362	-0.01259	-0.00057	-0.00954
9.92	-0.00648	-0.00648	-0.00362	-0.00362	-0.01259	0.00248	0.00248	-0.00954	-0.00362	-0.00954
9.94	-0.00057	-0.00954	-0.00362	-0.00954	-0.01259	-0.00648	-0.00648	-0.00362	-0.01564	-0.00954
9.96	-0.00362	-0.00362	-0.01259	-0.00362	-0.01259	-0.01259	-0.01259	-0.00648	-0.00057	-0.00362
9.98	-0.00648	-0.00362	0.00248	-0.00648	-0.01564	-0.01259	-0.00362	-0.00648	-0.00057	-0.01259
10	-0.00954	0.00248	-0.00057	-0.00648	-0.01259	-0.01259	-0.00362	-0.00954	-0.00648	0.00248

Data points for $k=2$

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V		
1	12V= 8.333 RPS												10V= 6.945 RPS											
2	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T	
3		0	-0.00362	-0.19798	-0.0309	0.01774	0.19073	0.20294	0.14839	-0.00648	-0.00954	-0.00362	0	0.13313	0.0906	0.25463	-0.00648	0.26379	0.29106	0.16956	-0.00954	-0.0885	-0.00362	
4		0.02	-0.01259	-0.16441	0.02689	0.03586	0.17567	0.06332	0.18482	-0.00362	-0.00954	-0.00648	0.02	0.15144	0.00248	0.30327	-0.00362	0.20294	0.15736	0.00858	-0.00362	-0.1524	-0.00648	
5		0.04	-0.00648	-0.09766	0.0967	0.05112	0.14534	-0.06123	0.20294	-0.00362	-0.00362	-0.00362	0.04	0.15736	-0.06123	0.2758	-0.00362	0.05722	-0.04005	-0.09766	-0.00362	-0.19798	-0.00362	
6		0.06	-0.00362	-0.0248	0.16346	0.06943	0.05722	-0.1524	0.19684	-0.00648	-0.00362	-0.00954	0.06	0.14534	-0.11292	0.09975	-0.00648	-0.06428	-0.1585	-0.17357	-0.00362	-0.23136	-0.00648	
7		0.08	-0.00362	0.06332	0.19379	0.06027	-0.0309	-0.23136	0.15144	-0.00362	-0.00362	-0.00362	0.08	0.12093	-0.1524	-0.06733	-0.00954	-0.1524	-0.23441	-0.23136	-0.00362	-0.22221	-0.00362	
8		0.1	-0.01259	0.17262	0.20905	0.09365	-0.0885	-0.26474	0.04196	-0.00648	-0.00362	-0.00954	0.1	0.05722	-0.18578	-0.17662	-0.00362	-0.2161	-0.28591	-0.25864	-0.00362	-0.1524	-0.00362	
9		0.12	-0.00362	0.22125	0.18787	0.11482	-0.1524	-0.24033	-0.037	-0.00362	-0.00648	-0.00648	0.12	-0.00648	-0.21	-0.25558	-0.00362	-0.26779	-0.24643	-0.21	-0.00362	-0.06123	-0.00362	
10		0.14	-0.00362	0.23937	0.13008	0.13313	-0.20695	-0.14935	-0.10986	-0.00362	-0.00648	-0.00362	0.14	-0.06123	-0.19188	-0.28896	-0.00362	-0.23727	-0.11902	-0.09766	-0.00362	-0.02995	-0.00362	
11		0.16	-0.00362	0.21515	0.02384	0.13008	-0.22221	-0.0309	-0.16136	-0.00362	-0.01259	-0.00362	0.16	-0.10376	-0.13103	-0.22831	-0.00362	-0.13409	0.02689	0.02079	-0.00648	0.16651	-0.00362	
12		0.18	-0.01564	0.12093	-0.05512	0.11482	-0.21	0.10891	-0.19798	-0.00362	-0.00648	-0.00362	0.18	-0.14324	-0.05512	-0.0885	-0.00648	-0.02174	0.21515	0.18482	-0.00362	0.23022	-0.00648	
13		0.2	-0.01259	-0.01564	-0.12798	0.0967	-0.14935	0.25158	-0.23136	-0.00648	-0.00648	-0.01259	0.2	-0.17357	0.02079	0.06027	-0.00648	0.12093	0.29411	0.26073	-0.00362	0.25463	-0.00362	
14		0.22	-0.0248	-0.11292	-0.19188	0.03891	-0.06733	0.29106	-0.20084	-0.00362	-0.00362	-0.00362	0.22	-0.19798	0.13313	0.24853	-0.01259	0.23937	0.30022	0.28191	-0.00954	0.22736	-0.00362	
15		0.24	-0.03395	-0.19188	-0.22221	-0.01259	0.03586	0.26684	-0.12188	-0.00362	-0.00362	-0.00057	0.24	-0.19798	0.20294	0.30327	-0.00648	0.28191	0.18177	0.2243	0.00858	0.11787	-0.00362	
16		0.26	-0.04902	-0.24033	-0.22831	-0.05512	0.16346	0.12398	-0.0309	-0.00954	-0.00362	0.00858	0.26	-0.1585	0.23022	0.28801	-0.00362	0.25158	-0.26785	0.05722	-0.02799	-0.01564	-0.00362	
17		0.28	-0.04292	-0.24643	-0.18578	-0.10681	0.23937	-0.04292	0.07534	-0.00362	-0.00362	0.02689	0.28	-0.0946	0.22736	0.13313	-0.00362	0.11787	-0.14935	-0.07648	0.02689	-0.10986	-0.00362	
18		0.3	-0.06123	-0.20084	-0.09766	-0.13714	0.26684	-0.16136	0.19989	-0.00648	-0.01259	0.04196	0.3	-0.0248	0.1543	-0.05207	-0.00362	-0.04005	-0.23441	-0.17357	0.04501	-0.18272	-0.00648	
19		0.32	-0.07954	-0.10681	-0.00057	-0.17357	0.23022	-0.25253	0.24548	-0.00057	-0.01869	0.05722	0.32	0.06332	0.01163	-0.17052	-0.00362	-0.1524	-0.28591	-0.24033	0.04807	-0.23441	-0.00362	
20		0.34	-0.0946	0.00858	0.11787	-0.19188	0.11196	-0.28286	0.26379	-0.00362	-0.037	0.06332	0.34	0.16956	-0.07954	-0.24643	-0.00362	-0.21915	-0.26169	-0.27676	0.04501	-0.24948	-0.00362	
21		0.36	-0.11902	0.15736	0.2182	-0.18883	-0.037	-0.24338	0.19989	-0.00648	-0.04902	0.08144	0.36	0.20905	-0.1524	-0.28896	-0.00648	-0.27084	-0.11902	-0.22526	0.03891	-0.19188	-0.00362	
22		0.38	-0.12493	-0.12493	0.26073	0.26073	-0.1585	-0.13714	-0.11902	0.033	-0.00954	-0.06428	0.38	0.23327	-0.20084	-0.24033	-0.00954	-0.25253	0.033	-0.09766	0.01774	-0.08545	-0.00362	
23		0.4	-0.12493	0.28496	0.25158	-0.0946	-0.22526	0.02384	-0.08545	-0.00362	-0.07648	0.0845	0.4	0.20599	-0.23727	-0.10071	-0.00648	-0.13714	0.2182	0.033	-0.00648	0.02995	-0.00362	
24		0.42	-0.12188	0.24548	0.16651	-0.02785	-0.25864	0.21515	-0.17967	-0.00362	-0.07343	0.07534	0.42	-0.12398	-0.22526	0.05112	-0.00362	-0.01564	0.30022	0.20294	-0.02174	0.18177	-0.01259	
25		0.44	-0.10376	0.0845	-0.00648	0.06638	-0.24948	0.29411	-0.23136	-0.01259	-0.08545	0.05112	0.44	-0.00057	-0.14324	0.25158	-0.00362	0.1543	0.30022	0.2758	-0.04597	0.25158	-0.01259	
26		0.46	-0.07038	-0.07038	-0.11292	0.16956	-0.16441	0.30327	-0.26474	-0.00362	-0.0946	0.01774	0.46	-0.08545	-0.04597	0.30918	-0.00648	0.26379	0.16956	0.28496	-0.06733	0.22725	-0.0248	
27		0.48	-0.03395	-0.17357	-0.20084	0.21515	-0.04005	0.19073	-0.22221	-0.00648	-0.09155	-0.02174	0.48	-0.14935	0.06943	0.29106	-0.00648	0.29106	-0.037	0.19989	-0.08545	0.22736	-0.04597	
28		0.5	0.00858	-0.25253	-0.25864	0.23632	0.10281	-0.0248	-0.10681	-0.01259	-0.07648	-0.04005	0.5	-0.19798	0.20599	0.13008	-0.00362	0.24548	-0.1524	0.01163	-0.10681	0.07148	-0.05512	
29		0.52	0.06332	-0.2739	-0.26474	0.20599	0.25463	-0.1585	0.01469	-0.00954	-0.05207	-0.07343	0.52	-0.23136	0.26684	-0.05207	-0.01259	0.06027	-0.23727	-0.11292	-0.12493	-0.06428	-0.05817	
30		0.54	0.13008	-0.23136	-0.19188	0.13313	0.28801	-0.26169	0.18177	-0.01564	-0.02785	-0.09766	0.54	-0.22221	0.25463	-0.17357	-0.00648	-0.07648	-0.29507	-0.21	-0.14629	-0.16136	-0.05817	

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	
474	9.42	-0.29202	0.31223	-0.31338	0.02689	-0.01259	0.35477	-0.18578	-0.19493	-0.17052	-0.35286		9.42	-0.00057	0.29411	0.28191	-0.30727	-0.07038	-0.00362	0.16041	0.27275	-0.29507	0.16651
475	9.44	-0.0885	0.03586	-0.34676	0.29411	-0.19493	0.18177	0.00248	-0.31624	-0.29202	-0.23441		9.44	0.20294	0.14534	0.32749	-0.24948	0.11196	-0.14629	0.29106	0.32444	-0.28896	-0.04292
476	9.46	0.1543	-0.16136	-0.22526	0.36392	-0.3315	-0.0946	0.26379	-0.34676	-0.3437	-0.02785		9.46	0.29716	-0.05817	0.2697	-0.10376	0.28496	-0.24033	0.32444	0.28801	-0.15545	-0.17357
477	9.48	0.3458	-0.30422	-0.02174	0.31223	-0.3437	-0.25864	0.34866	-0.19798	-0.23136	0.24548		9.48	0.30632	-0.18578	0.06332	0.05417	0.32749	-0.30117	0.23632	0.08755	0.00248	-0.25864
478	9.5	0.36392	-0.34676	0.24242	0.02689	-0.19493	-0.35286	0.32444	0.00553	-0.0248	0.36392		9.5	0.19073	-0.27084	-0.10376	0.26073	0.26379	-0.26779	0.00858	-0.08545	0.2121	-0.30422
479	9.52	0.17872	-0.24338	0.35782	-0.17052	0.01163	-0.31624	0.08144	0.27885	0.24548	0.36665		9.52	-0.02785	-0.30422	-0.21	0.31223	0.04807	-0.11597	-0.13409	-0.20695	0.30022	-0.23441
480	9.54	-0.08545	-0.04005	0.3397	-0.31338	0.27885	-0.11292	-0.12798	0.36392	0.35477	0.06943		9.54	-0.16747	-0.22221	-0.29202	0.28801	-0.11292	0.04196	-0.23441	-0.28286	0.30327	-0.07954
481	9.56	-0.24643	0.2182	0.0967	-0.3437	0.36392	0.11787	-0.25864	0.31834	0.3397	-0.14019		9.56	-0.25864	-0.07038	-0.29202	0.12093	-0.21915	0.24853	-0.29812	-0.30117	0.17872	0.09365
482	9.58	-0.3437	0.34866	-0.13714	-0.23136	0.31834	0.3336	-0.3376	0.04196	0.0906	-0.27981		9.58	0.30727	0.10281	-0.17357	-0.06428	-0.28896	0.31528	-0.2739	-0.18578	-0.037	0.28191
483	9.6	-0.31033	0.3458	-0.27981	-0.0309	0.05417	0.37308	-0.27981	-0.17357	-0.13409	-0.36182		9.6	-0.24643	0.27885	-0.01564	-0.18578	-0.28896	0.29411	-0.13714	-0.0248	-0.16747	0.32749
484	9.62	-0.11292	0.14534	0.34676	0.23022	-0.15545	0.24242	-0.09155	-0.28896	-0.27084	-0.25558		9.62	-0.10071	0.32444	0.18177	-0.27084	-0.16136	0.13313	0.02995	0.17872	-0.25253	0.27275
485	9.64	0.12398	-0.10376	-0.27981	0.36087	-0.31033	-0.04902	0.12703	-0.34981	-0.3437	-0.04902		9.64	0.06943	0.2697	0.30022	-0.30727	-0.00057	-0.06428	0.23937	0.30022	-0.30422	0.06638
486	9.66	0.14839	-0.26779	-0.07343	0.3397	-0.34676	-0.22831	0.31528	-0.23441	-0.27084	0.21515		9.66	0.2697	0.06027	0.31834	-0.20695	0.20599	-0.18578	0.30327	0.31834	-0.24948	-0.10071
487	9.68	0.34275	-0.35286	0.16956	0.10891	-0.24338	-0.3376	0.36087	-0.0309	-0.06428	0.35477		9.68	0.31834	-0.10071	0.2121	-0.05512	0.30327	-0.27084	0.30022	0.21515	-0.09766	-0.21
488	9.7	0.36697	-0.30422	0.34275	-0.13103	-0.05207	-0.32845	0.23632	0.23022	0.18787	0.34275		9.7	0.28801	-0.2161	-0.01564	0.11787	0.31223	-0.31033	0.15736	-0.00057	0.06332	-0.28591
489	9.72	0.20905	-0.10681	0.35782	-0.27981	0.2121	-0.1585	-0.0309	0.35172	0.34275	0.11482		9.72	0.0967	-0.28896	-0.14629	0.27885	0.18787	-0.21	-0.04902	0.14935	0.26684	-0.29507
490	9.74	-0.07648	0.11787	0.19684	-0.3437	0.35477	0.05417	-0.20084	0.3397	0.35477	-0.12188		9.74	-0.0885	-0.28896	-0.24033	0.31528	-0.0428	-0.05512	0.17357	-0.24643	0.31834	-0.17357
491	9.76	-0.24033	0.32749	-0.07343	0.27981	0.3397	0.31528	-0.30422	0.0967	0.06436	-0.27084		9.76	-0.20695	-0.17052	-0.30117	0.25768	0.16136	0.11787	-0.26474	0.30727	0.28801	-0.15564
492	9.78	-0.34065	0.36087	-0.24033	0.0768	0.13313	0.38223	-0.3437	-0.13103	-0.19346	0.35286		9.78	-0.28286	-0.01564	-0.26169	0.05417	-0.25358	0.29106	-0.2422	-0.26474	0.10891	0.18484
493	9.8	-0.31624	0.23937	-0.3437	0.16346	-0.11292	0.28496	-0.18578	-0.2739	-0.23338	-0.27676		9.8	-0.29812	0.17872	-0.11597	-0.10376	-0.30727	0.32749	-0.22526	-0.10681	-0.07343	0.30022
494	9.82	-0.12493	-0.04597	-0.31929	-0.3458	-0.27981	-0.06048	0.00248	-0.3437	-0.3376	0.27676		9.82	-0.18578	0.29411	0.04196	-0.2161	-0.23338	0.25463	-0.07648	0.05722	-0.19493	0.31834
495	9.84	0.09975	-0.22221	-0.13714	0.36392	-0.34676	-0.19349	0.25768	-0.2739	-0.30422	0.17872		9.84	-0.03395	0.31223	0.24548	-0.28896	-0.0946	0.03586	0.0967	0.26379	-0.27084	0.21515
496	9.86	0.33665	-0.33455	0.08545	0.19379	-0.29202	-0.23234	0.34275	-0.06733	-0.11292	0.34866		9.86	0.16651	0.2515	0.31223	-0.29507	0.07534	-0.11597	0.27885	0.31528	-0.30422	-0.06648
497	9.88	0.37918	-0.3315	0.31834	-0.07954	-0.0946	-0.34065	0.32444	0.18177	0.10891	0.35782		9.88	0.30022	-0.00057	0.29716	-0.16747	0.26684	-0.22831	0.31528	0.28801	-0.2039	-0.14935
498	9.9	0.25158	-0.17052	0.36392	-0.24033	0.14534	-0.19798	0.09565	0.33054	0.32749	0.15736		9.9	0.32444	-0.14629	0.31008	-0.1259	0.3239	-0.30117	0.2697	0.11787	-0.04902	-0.24643
499	9.92	-0.04597	0.03891	0.2758	-0.34065	0.34275	0.20279	-0.12493	0.34866	0.36697	0.09766		9.92	0.23022	-0.24033	-0.06428	0.28177	0.28191	-0.28286	0.07248	-0.27343	0.13313	-0.30422
500	9.94	-0.22526	0.30022	-0.01259	-0.31624	0.36087	0.29106	-0.25864	0.18177	0.23937	0.25558		9.94	-0.00057	-0.29507	-0.18578	0.29106	0.09365	-0.14935	0.10376	-0.2039	0.28801	-0.25864
501	9.96	-0.3376	0.36697	-0.19798	-0.12798	0.2121	0.37613	-0.34065	-0.07648	-0.05207	0.34676		9.96	-0.14324	-0.25864	-0.27084	0.30918	-0.09155	0.00248	-0.21305	-0.28286	0.32444	-0.10986
502	9.98	-0.3437	0.31223	-0.32539	0.0967	-0.06733	0.32139	-0.2705	-0.22221	-0.22526	-0.29507		9.98	-0.24033	-0.1586	-0.31033	0.22125	-0.20695	0.2182	-0.28896	-0.30727	0.25158	-0.05722
503	10	-0.17622	0.02995	-0.34065	0.33054	-0.23727	0.03586	-0.0946	-0.29812	-0.32234	-0.0946		10	-0.30422	0.03388	-0.29155	0.00248	-0.28698	0.30022	-0.29202	-0.19188	0.03891	0.25463

U		V	W	X		Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR																					
				8V= 5.556 RPS															6V= 4.167 RPS																											
1																																														
2			Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10																						
3	0885	-0.00362		0	0.16651	0.00248	0.04196	0.0906	-0.07343	-0.00954	-0.05207	0.19073	0.04807	0.0967	0	-0.13103	-0.12188	-0.21915	-0.07038	0.22736	0.15736	-0.11292	-0.04902	0.07248	-0.01564																					
4	1524	-0.00648		0.02	0.18177	-0.01564	0.06943	0.05722	-0.05512	-0.01869	-0.05207	0.20294	0.10281	0.12703	0.02	-0.05512	-0.14019	-0.18272	-0.07648	0.19684	0.2121	-0.16747	0.04196	0.18177	-0.0946																					
5	9798	-0.00362		0.04	0.17872	-0.04005	0.0845	0.00858	-0.0309	-0.02785	-0.04597	0.17872	0.13313	0.14534	0.04	0.02079	-0.1524	-0.10681	-0.0885	0.12398	0.23022	-0.20695	0.1543	0.21515	-0.15545																					
6	3136	-0.00648		0.06	0.15144	-0.06123	0.0967	-0.0248	-0.00362	-0.02785	-0.04597	0.10586	0.14839	0.16041	0.06	0.13924	-0.14935	-0.02785	-0.10376	0.00858	0.21515	-0.22221	0.20599	0.23022	-0.19798																					
7	2221	-0.00362		0.08	0.07534	-0.08545	0.10281	-0.05512	0.02689	-0.04005	-0.04597	0.00553	0.15736	0.15736	0.08	0.19989	-0.13409	0.06332	-0.11902	-0.07038	0.14534	-0.19493	0.22736	-0.08545	-0.22221																					
8	1524	-0.00362		0.1	-0.00362	-0.10986	0.10891	-0.0885	0.07534	-0.06428	-0.04292	-0.06123	0.1543	0.14229	0.1	0.22125	-0.10681	0.16956	-0.12493	-0.13409	0.02079	-0.12188	0.21515	0.11196	-0.21																					
9	6123	-0.00362		0.12	-0.06123	-0.12493	0.10891	-0.10986	0.12093	-0.07648	-0.04292	-0.11902	0.13313	0.09365	0.12	0.21515	-0.06733	0.20905	-0.12798	-0.17662	-0.06733	-0.04292	0.15736	-0.00057	-0.1524																					
10	9295	-0.00362		0.14	-0.11292	-0.14935	0.09975	-0.13409	0.14229	-0.08545	-0.04005	-0.1585	0.07534	0.02079	0.14	0.16651	-0.0248	0.22736	-0.11902	-0.21305	-0.12798	0.04196	0.03586	-0.07954	-0.07648																					
11	6651	-0.00362		0.16	-0.14935	-0.1585	0.0845	-0.1524	0.1543	-0.10071	-0.04005	-0.19188	0.00553	-0.0309	0.16	0.04501	0.02384	0.20905	-0.09766	-0.21915	-0.17052	0.1543	-0.05512	-0.13714	0.00248																					
12	3022	-0.00648		0.18	-0.17967	-0.1524	0.04807	-0.16747	0.1543	-0.12188	-0.037	-0.20695	-0.04597	-0.07648	0.18	-0.04597	0.0845	0.13924	-0.06123	-0.17357	-0.20695	0.2121	-0.12798	-0.17967	0.10586																					
13	5463	-0.00362		0.2	-0.20084	-0.12493	0.00858	-0.1585	0.14839	-0.13103	-0.04005	-0.18578	-0.0885	-0.11292	0.2	-0.11292	0.13618	0.01163	-0.0309	-0.0946	-0.21915	0.23022	-0.18272	-0.2161	0.19073																					
14	2736	-0.00362		0.22	-0.19188	-0.0885	-0.02785	-0.12493	0.11787	-0.13714	-0.04292	-0.12188	-0.12798	-0.14629	0.22	-0.16747	0.15736	-0.07038	0.00248	-0.01564	-0.18578	0.21515	-0.2161	-0.21915	0.21515																					
15	1787	-0.00362		0.24	-0.14324	-0.04005	-0.06123	-0.07648	0.05112	-0.12798	-0.05512	-0.04902	-0.1585	-0.17357	0.24	-0.20084	0.17262	-0.13409	0.04501	0.08144	-0.11292	0.1543	-0.22221	-0.17357	0.22125																					
16	1564	-0.00362		0.26	-0.07038	0.00248	-0.0946	-0.0248	-0.01259	-0.10071	-0.06733	0.033	-0.18578	-0.18578	0.26	-0.21915	0.17262	-0.17967	0.0967	0.18482	-0.0309	0.02384	-0.18578	-0.10071	0.18787																					
17	0986	-0.00362		0.28	0.00858	0.06638	-0.11597	0.033	-0.06428	-0.06428	-0.06733	0.14229	-0.19188	-0.17357	0.28	-0.19188	0.15144	-0.21305	0.13313	0.21515	0.06638	-0.06123	-0.10986	-0.01869	0.0906																					
18	8272	-0.00648		0.3	0.09365	0.13313	-0.14019	0.11482	-0.10376	-0.0309	-0.06733	0.20599	-0.16136	-0.12493	0.3	-0.12493	0.09975	-0.2161	0.1543	0.22736	0.17872	-0.12493	-0.02785	0.07839	-0.02174																					
19	3441	-0.00362		0.32	0.17872	0.16651	-0.1585	0.16956	-0.13714	0.00858	-0.05512	0.23022	-0.10681	-0.06123	0.32	-0.04902	0.02079	-0.17052	0.16651	0.19684	0.21515	-0.17357	0.06638	0.18177	-0.0071																					
20	4948	-0.00362		0.34	0.20905	0.18177	-0.16441	0.18482	-0.16747	0.06027	-0.037	0.2182	-0.04005	-0.0057	0.34	0.033	-0.037	-0.09766	0.16041	0.11482	0.23327	-0.20695	0.17567	0.21515	-0.15545																					
21	9188	-0.00362		0.36	0.2182	0.18787	-0.14935	0.19684	-0.18883	0.12093	-0.01564	0.15144	0.02689	0.08144	0.36	0.14534	-0.0824	-0.02174	0.14229	-0.00362	0.20599	-0.21915	0.21515	0.23022	-0.19188																					
22	8545	-0.00362		0.38	0.19073	0.16651	-0.11902	0.17262	-0.18578	0.15144	0.00858	0.02384	0.12093	0.16651	0.38	0.20599	-0.12188	0.07248	0.09365	-0.12398	0.17967	0.23022	0.19684	-0.2161																						
23	1295	-0.00362		0.4	0.09975	0.0967	-0.06428	0.10891	-0.14324	0.16956	0.03891	-0.06428	0.18177	0.19073	0.4	0.23022	-0.14935	0.17872	0.01774	-0.14324	0.00553	-0.10681	0.20905	0.10586	-0.20695																					
24	8177	-0.01259		0.42	-0.00648	0.00553	-0.01564	0.01163	-0.0824	0.17872	0.08144	-0.12798	0.20599	0.20599	0.42	0.21515	-0.17967	0.21515	-0.0309	-0.08248	-0.07038	-0.20695	0.13714	-0.00362	-0.1524																					
25	5158	-0.01259		0.44	-0.08545	-0.05512	0.04501	-0.05207	-0.01869	0.16651	0.11196	-0.17967	0.21515	0.19073	0.44	0.1543	-0.18578	0.23327	-0.07648	-0.21915	-0.13409	0.06027	0.00248	-0.07954	-0.07648																					
26	7275	-0.0248		0.46	-0.14324	-0.10986	0.13008	-0.10681	0.05722	0.11787	0.13008	-0.12915	0.18482	0.12703	0.46	0.033	-0.16747	0.20599	-0.10986	-0.21	-0.18272	0.17262	-0.08545	-0.13714	-0.00057																					
27	2736	-0.04597		0.48	-0.18578	-0.14935	0.17567	-0.14935	0.14534	0.033	0.14534	-0.22831	0.0906	0.02079	0.48	-0.05207	-0.11902	0.12093	-0.14019	-0.1585	-0.21915	0.21515	-0.14935	-0.17967	0.09975																					
28	17248	-0.05512		0.5	-0.21915	-0.18578	0.19379	-0.18578	0.18787	-0.0309	0.1543	-0.17662	-0.01869	-0.05512	0.5	-0.11597	-0.05802	-0.00057	-0.16747	-0.07954	-0.22526	0.23327	-0.18883	-0.2161	0.9073																					
29	6428	-0.05817		0.52	-0.21915	-0.20084	0.19379	-0.21	0.20599	-0.0885	0.14839	-0.0885	-0.09155	-0.12188	0.52	-0.16441	0.00248	-0.0824	-0.18578	-0.00362	-0.17967	0.20599	-0.21915	-0.2161	0.21515																					
30	6136	-0.05817		0.54	-0.16136	-0.18883	0.16956	-0.19493	0.19684	-0.13714	0.12398	0.00248	-0.14935	-0.16747	0.54	-0.20084	0.07534	-0.14324	-0.17357	0.09975	-0.0946	0.11787	-0.2161	-0.16747	0.22736																					

	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	
474	-0.29507	0.16651	9.42	0.25463	0.033	-0.1524	-0.09155	0.04501	0.25768	-0.21915	0.04807	-0.17357	0.03586	9.42	-0.20084	-0.16747	-0.04902	0.18177	-0.00362	0.00248	0.10586	-0.18883	-0.12188	0.23632	
475	-0.28896	-0.04292	9.44	0.17872	-0.0824	-0.04597	-0.16747	0.20599	0.26379	-0.25864	0.20294	-0.22831	0.19073	9.44	-0.22526	-0.2039	-0.11292	0.06943	0.0967	0.10891	-0.00954	-0.11902	-0.17357	0.2121	
476	-0.15545	-0.17357	9.46	0.02079	-0.16747	0.07248	-0.22221	0.25463	0.19379	-0.24033	0.25158	-0.26169	0.24548	9.46	-0.21	-0.22221	-0.16747	-0.0309	0.19073	0.19379	-0.0824	-0.04005	-0.21	0.12398	
477	0.00248	-0.25864	9.48	-0.0824	-0.22526	0.2182	-0.25558	0.26684	0.033	-0.13714	0.26684	0.20221	0.26073	9.48	-0.14324	-0.19798	-0.20695	-0.10071	0.21215	0.2182	-0.14019	0.05417	-0.22526	0.00553	
478	0.2121	-0.30422	9.5	-0.16441	-0.25864	0.25158	-0.22221	0.20905	-0.08545	-0.0248	-0.09055	-0.21902	0.2121	9.5	-0.05817	-0.13409	-0.23136	-0.1585	0.23022	0.2243	-0.18578	0.17262	-0.19493	-0.07343	
479	0.30022	-0.23441	9.52	-0.22221	-0.22221	0.25768	-0.12493	0.04501	-0.17357	0.10586	0.04196	-0.06048	0.08755	9.52	0.02995	-0.05207	-0.20695	-0.19798	0.19379	0.19073	-0.21915	0.20905	-0.12188	-0.13409	
480	0.30327	-0.07954	9.54	-0.25864	-0.12188	0.18787	-0.01564	-0.07648	-0.22831	0.23022	-0.07648	0.14229	-0.04005	9.54	0.14534	0.03586	-0.13409	-0.22221	0.0845	0.07839	-0.21915	0.23022	-0.04292	-0.17967	
481	0.17872	0.09365	9.56	-0.22221	-0.00954	0.02384	0.12398	-0.16136	-0.25864	0.26379	-0.1585	0.23937	-0.10986	9.56	0.20905	0.14534	-0.05512	-0.20695	-0.0248	-0.0248	-0.16441	0.20294	0.04501	-0.21305	
482	-0.037	0.28191	9.58	-0.12188	0.13313	-0.0885	0.24242	-0.22221	-0.22831	0.25158	-0.21915	0.2697	-0.1585	9.58	0.23022	0.20294	0.03586	-0.14019	-0.09766	-0.10376	-0.07954	0.12398	0.1543	-0.21915	
483	-0.16747	0.32749	9.6	-0.01259	0.23937	-0.16747	0.27275	-0.25864	-0.12798	0.16651	-0.25558	0.24242	-0.20084	9.6	0.2182	0.22125	0.16041	-0.05512	-0.1524	-0.1585	-0.00057	0.00248	0.20294	-0.17357	
484	-0.25253	0.27275	9.62	0.12703	0.26684	-0.22526	0.25158	-0.23727	-0.01564	0.00553	-0.23136	0.13313	-0.22221	9.62	0.16346	0.21515	0.2121	0.02689	-0.19188	-0.19798	0.10891	-0.07343	0.22125	-0.09766	
485	-0.30422	0.06638	9.64	0.23937	0.24242	-0.25558	0.13924	-0.13714	0.12093	-0.10376	0.13714	0.02174	-0.19798	9.64	0.04501	0.16346	0.23022	0.14229	-0.2161	-0.21915	0.19684	-0.13043	0.25115	-0.18689	
486	-0.24948	-0.10071	9.66	0.26684	0.14229	-0.21915	-0.18689	-0.0248	0.23632	-0.18883	-0.21742	0.12188	-0.13103	9.66	-0.04902	0.03891	0.2121	0.20294	-0.20695	-0.20084	0.22125	-0.17357	0.1543	0.0845	
487	-0.09766	-0.21	9.68	0.24242	-0.01564	-0.12188	-0.12188	0.10891	0.26684	0.24338	0.11787	-0.19493	-0.2057	9.68	-0.11902	-0.04902	0.14839	0.22125	-0.14935	-0.13714	0.2243	-0.20695	0.03586	0.19379	
488	0.36332	-0.28591	9.7	0.13924	-0.12493	-0.01259	-0.18188	0.23632	0.24548	-0.26747	0.23937	-0.24948	0.26869	9.7	-0.17662	-0.12188	0.02689	0.2121	-0.07648	-0.06123	0.18787	-0.2161	-0.05207	0.21515	
489	0.26684	-0.29507	9.72	-0.01564	-0.19493	0.12703	-0.24643	0.26684	0.15144	-0.2303	0.27275	-0.25253	0.13313	9.72	-0.2161	-0.17052	-0.05817	0.1585	0.00858	0.20799	0.08144	-0.17357	-0.12188	0.22736	
490	0.31834	-0.17357	9.74	-0.01986	-0.24643	0.23937	-0.25864	0.25158	0.06684	-0.07966	0.24548	-0.18272	0.19073	9.74	-0.23136	-0.20695	-0.12188	-0.03203	0.12703	0.13313	-0.02174	-0.10071	-0.17357	0.19379	
491	0.28801	-0.01564	9.76	-0.18578	-0.25558	0.27275	-0.19493	0.1543	-0.12192	-0.04149	0.13924	-0.04248	0.20599	9.76	-0.19493	-0.21915	-0.17052	-0.05207	0.20599	0.19889	-0.0946	-0.21742	-0.20695	0.09365	
492	0.10891	0.18482	9.78	-0.24033	-0.19188	0.24853	-0.0824	-0.03062	-0.18578	0.16956	-0.02174	0.04501	0.19889	9.78	-0.11597	-0.18883	-0.21	-0.11902	0.23022	0.22736	-0.1524	0.07839	-0.21915	-0.18689	
493	-0.07343	0.30022	9.8	-0.25558	-0.08545	0.13924	0.033	-0.10376	-0.24338	0.23937	-0.11597	0.09804	0.16041	9.8	-0.03395	-0.11902	-0.22221	-0.17052	0.22125	0.2243	-0.19493	0.19073	-0.18578	-0.0946	
494	-0.19493	0.31834	9.82	-0.19798	0.26689	-0.01869	0.19073	0.18188	-0.25864	0.26684	-0.18578	0.24548	0.0845	9.82	0.06332	-0.04005	-0.18883	-0.20695	0.17262	0.17262	-0.21915	0.2182	-0.11292	-0.19385	
495	-0.27084	0.25158	9.84	-0.0885	-0.17357	-0.11902	-0.15243	-0.24033	-0.19798	0.23632	-0.24338	0.2679	-0.00362	9.84	0.17262	-0.04501	-0.10681	-0.22221	0.05112	0.05112	-0.20695	0.23022	-0.04005	-0.19188	
496	-0.20695	0.26684	9.86	0.26684	0.24643	-0.04501	-0.0946	-0.0946	-0.05056	-0.25558	0.23136	-0.04005	0.20599	9.86	0.2121	0.15706	-0.13714	-0.04005	-0.04005	-0.04501	-0.16409	0.19684	0.04501	-0.21305	
497	-0.2039	0.14935	9.88	0.16255	0.2697	-0.04338	0.22125	-0.06995	0.02074	-0.04092	-0.19493	0.05112	-0.10891	9.88	0.23022	0.20599	0.06943	-0.11902	-0.10681	-0.11597	-0.06428	0.10586	0.16041	-0.22221	
498	-0.04902	0.24643	9.9	0.24242	0.23022	-0.25558	0.06684	-0.0946	-0.18177	-0.13714	-0.07954	-0.07343	-0.14935	9.9	0.20599	0.23022	-0.18717	-0.04005	-0.1585	-0.14747	0.02079	-0.06048	0.20599	-0.16747	
499	0.13313	-0.30422	9.92	0.26684	0.07955	-0.19188	-0.08545	-0.16129	0.17163	0.24548	-0.20695	-0.2995	-0.16136	-0.16747	9.92	0.13313	0.21515	0.25115	0.05417	-0.19798	-0.2039	0.13008	-0.07954	-0.22736	
500	0.28801	-0.25864	9.94	0.23022	-0.04902	-0.0885	-0.14629	0.10262	0.2697	-0.24948	0.19073	0.22221	-0.18272	9.94	0.11663	0.14839	0.23022	0.16651	-0.22221	-0.22221	0.19989	-0.13714	0.25115	-0.03632	
501	0.32444	-0.10986	9.96	0.09975	-0.14324	0.07879	-0.20695	0.24548	0.23022	-0.24338	0.25158	-0.25864	-0.17357	9.96	-0.06733	-0.02029	0.20294	0.23005	-0.20084	-0.18188	0.22125	-0.18272	0.14839	0.10586	
502	0.25158	0.05112	9.98	-0.04292	-0.20695	0.18272	-0.25253	0.2697	0.0845	-0.16441	0.2697	-0.23136	0.13714	9.98	-0.13714	-0.06478	0.22398	0.23002	-0.23022	-0.12798	-0.12188	0.2182	-0.2161	0.02689	0.19379
503	0.03891	-0.25463	10	-0.13409	-0.24948	0.18188	-0.24338	0.22736	-0.05207	0.20212	-0.08105	0.2182	-0.04501	10	-0.18578	-0.17357	0.00858	0.20905	-0.04597	-0.04597	0.17872	-0.21915	-0.05817	-0.21515	
504																									

AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC
9.44	-0.01259	0.14839	0.1543	-0.04597	-0.1524	-0.00362	0.14839	0.14229	0.0845	-0.07954
9.46	0.02995	0.15736	0.15144	-0.07648	-0.1585	-0.04005	0.14839	0.12093	0.02079	-0.10681
9.48	0.0845	0.14839	0.13618	-0.10071	-0.1524	-0.07343	0.14229	0.06638	-0.02174	-0.12188
9.5	0.13008	0.13313	0.09975	-0.12188	-0.12493	-0.10071	0.10586	0.00858	-0.06123	-0.14324
9.52	0.14534	0.0845	0.04196	-0.13714	-0.0885	-0.12188	0.04807	-0.03395	-0.0885	-0.1585
9.54	0.1543	0.02384	-0.00648	-0.1524	-0.04902	-0.14629	-0.00362	-0.06733	-0.10986	-0.1524
9.56	0.1543	-0.02174	-0.04005	-0.15545	-0.01259	-0.1585	-0.04005	-0.09766	-0.13103	-0.14324
9.58	0.13924	-0.05817	-0.07343	-0.13714	0.02689	-0.14935	-0.07648	-0.12188	-0.14935	-0.09766
9.6	0.10586	-0.0885	-0.09766	-0.10681	0.08755	-0.13714	-0.10071	-0.14324	-0.1585	-0.05817
9.62	0.05112	-0.10986	-0.11902	-0.06733	0.12703	-0.10376	-0.12493	-0.1524	-0.14935	-0.02174
9.64	-0.00362	-0.12798	-0.13714	-0.0248	0.14229	-0.06733	-0.14324	-0.1585	-0.12493	0.02079
9.66	-0.04005	-0.14629	-0.14935	0.01469	0.1543	-0.02785	-0.1524	-0.14324	-0.08545	0.08144
9.68	-0.07038	-0.15545	-0.1524	0.06332	0.1543	0.01163	-0.1524	-0.10986	-0.04005	0.11787
9.7	-0.10071	-0.1524	-0.13714	0.11482	0.14229	0.05722	-0.13714	-0.06733	-0.00362	0.14534
9.72	-0.12493	-0.12798	-0.10681	0.13924	0.11196	0.11482	-0.10071	-0.02785	0.04196	0.15736
9.74	-0.13714	-0.0946	-0.07038	0.14839	0.04501	0.13924	-0.06428	0.01163	0.09975	0.15736
9.76	-0.1524	-0.05207	-0.0309	0.15736	-0.00648	0.14534	-0.02785	0.06332	0.13313	0.14534
9.78	-0.1524	-0.01564	0.00858	0.14839	-0.04292	0.14839	0.01163	0.12093	0.14534	0.11482
9.8	-0.13714	0.02689	0.05417	0.12703	-0.07038	0.14229	0.05722	0.14534	0.1543	0.05112
9.82	-0.10681	0.08144	0.10891	0.08144	-0.09766	0.12398	0.10891	0.15736	0.15736	0.00248
9.84	-0.07038	0.11787	0.13618	0.02079	-0.11902	0.08144	0.13618	0.16346	0.13924	-0.04005
9.86	-0.02785	0.13924	0.14839	-0.02174	-0.13714	0.02079	0.15144	0.14534	0.0967	-0.07648
9.88	0.01163	0.15144	0.15736	-0.05817	-0.1524	-0.0248	0.15736	0.11787	0.02995	-0.10071
9.9	0.05722	0.15144	0.14839	-0.08545	-0.1524	-0.06123	0.14534	0.06027	-0.01564	-0.12798
9.92	0.11482	0.13618	0.13313	-0.10681	-0.13714	-0.09155	0.12703	-0.00057	-0.05512	-0.14629
9.94	0.13924	0.10281	0.0845	-0.12798	-0.10681	-0.11292	0.07534	-0.037	-0.0885	-0.15545
9.96	0.14229	0.04807	0.02689	-0.14324	-0.06733	-0.13409	0.01469	-0.07038	-0.10986	-0.1524
9.98	0.15736	-0.00362	-0.01564	-0.1524	-0.02785	-0.14935	-0.0248	-0.09766	-0.13103	-0.13714
10	0.14839	-0.04292	-0.05512	-0.14935	0.00858	-0.15545	-0.06428	-0.12188	-0.14935	-0.10376

Data points for k=3

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1		12V= 8.333 RPS											10V= 6.945 RPS									
2	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
3	0	-0.03395	-0.00648	0.02079	-0.04005	-0.00648	-0.00362	-0.00648	-0.01869	-0.00648	0.02384	0	-0.00648	-0.00362	-0.00648	-0.04292	-0.02174	-0.01259	-0.00362	0.00858	-0.01259	-0.00362
4	0.02	-0.0248	-0.01259	0.00248	-0.05512	-0.00954	-0.00362	-0.00362	-0.00057	0.00858	0.01774	0.02	0.00858	-0.00362	-0.00362	-0.01869	-0.01259	-0.00057	-0.00362	0.06332	-0.00362	-0.00648
5	0.04	0.00553	-0.00954	-0.0309	-0.05512	-0.00648	-0.00362	-0.00362	0.02079	0.033	-0.00362	0.04	-0.00954	-0.00362	-0.00362	0.01163	-0.00954	-0.00648	-0.00362	0.04501	-0.00362	-0.00954
6	0.06	0.02384	-0.00648	-0.06123	-0.01869	-0.00648	-0.00954	-0.00648	0.02995	0.01774	-0.0248	0.06	-0.0248	-0.00954	-0.00954	0.04196	-0.00362	-0.01259	-0.00362	0.00858	-0.00362	-0.00362
7	0.08	0.01469	-0.00648	-0.07038	0.02689	-0.00362	-0.00648	-0.00648	-0.00648	-0.00954	-0.04292	0.08	-0.0248	-0.00362	-0.00648	0.00553	0.00248	-0.01564	-0.00362	-0.037	-0.00648	-0.00648
8	0.1	-0.01564	-0.00648	-0.04292	0.04501	-0.00648	-0.00648	-0.00954	-0.04005	-0.04005	-0.06428	0.1	-0.02174	-0.00362	-0.00362	-0.04005	0.00858	-0.01259	-0.00362	-0.03395	-0.00362	-0.00648
9	0.12	-0.05207	-0.00648	0.01774	0.03891	-0.00954	-0.00362	-0.01259	-0.06428	-0.06733	-0.04902	0.12	-0.00954	-0.00362	-0.00362	-0.07954	0.00858	-0.01259	-0.00362	0.01163	-0.00954	-0.00648
10	0.14	-0.07648	-0.01259	0.05722	0.01774	-0.00648	-0.00362	-0.00954	-0.07038	-0.06733	-0.00362	0.14	-0.00362	-0.00954	-0.00362	-0.07343	0.00553	-0.00954	-0.00648	0.04501	-0.00954	-0.00954
11	0.16	-0.04292	-0.00362	0.04807	-0.00954	-0.00362	-0.00362	-0.00648	-0.01259	-0.0248	0.03586	0.16	0.01163	-0.00648	-0.00954	0.01774	0.00553	-0.00954	-0.00362	0.02384	-0.00362	-0.00362
12	0.18	0.02384	-0.00648	0.02079	-0.04005	-0.00362	-0.01259	-0.00648	0.04807	0.02995	0.04807	0.18	0.02689	-0.00648	-0.00648	0.06638	-0.00362	-0.01259	-0.00362	-0.04005	-0.00648	-0.00362
13	0.2	0.05722	-0.00362	-0.00954	-0.02785	-0.01259	-0.00648	-0.00648	0.06027	0.04807	0.033	0.2	0.01469	-0.01259	-0.00648	0.04501	0.00858	-0.01259	-0.00362	-0.08545	-0.00362	-0.00648
14	0.22	0.03586	-0.00648	-0.04005	-0.00057	-0.00362	-0.00648	-0.00648	0.02689	0.033	-0.01869	0.22	-0.00954	-0.00648	-0.00362	0.00248	0.00858	-0.01259	-0.00362	-0.07038	-0.00648	-0.00648
15	0.24	0.00553	-0.00648	-0.0248	0.02384	-0.00648	-0.00362	-0.00648	-0.00057	0.00858	0.07839	0.24	-0.0309	-0.00362	0.01163	-0.04005	0.00248	-0.00954	-0.00362	0.02384	-0.00362	-0.01259
16	0.26	-0.02785	-0.00954	-0.00057	0.03891	-0.00648	-0.00362	-0.00648	-0.03395	-0.0248	0.04501	0.26	-0.05207	-0.00648	-0.00954	-0.037	0.00248	-0.01259	-0.00362	0.06332	-0.00648	-0.00362
17	0.28	-0.03395	-0.00648	0.02384	-0.00362	-0.01259	-0.00648	-0.00954	-0.02785	-0.037	-0.01259	0.28	-0.07038	-0.01259	-0.01259	0.00553	0.00248	-0.00954	-0.00362	0.033	-0.00954	-0.00362
18	0.3	-0.00648	-0.00954	0.02689	-0.04005	-0.01564	-0.01259	-0.01259	-0.00362	-0.01564	-0.037	0.3	-0.04005	-0.00648	-0.01259	0.04501	0.00248	-0.00648	-0.00362	-0.01259	-0.00362	-0.00648
19	0.32	0.02689	-0.00362	-0.01259	-0.07648	-0.01259	-0.00362	-0.00648	0.02995	-0.00858	-0.00057	0.32	-0.0163	-0.00648	-0.01564	0.02384	-0.00057	-0.00648	-0.00362	-0.04292	-0.00648	-0.00648
20	0.34	0.03891	-0.00057	-0.06123	-0.06428	-0.01564	-0.00362	-0.00648	0.02384	0.03891	0.05722	0.34	0.04501	-0.00362	-0.01869	-0.04005	-0.00057	-0.00362	-0.00648	-0.01564	-0.00648	-0.01259
21	0.36	-0.02174	0.00248	-0.0824	0.00858	0.01163	-0.00362	-0.00648	-0.0309	0.00248	0.00248	0.36	0.04196	-0.00954	-0.01259	-0.0885	-0.00057	-0.00362	-0.00362	0.02689	-0.00648	-0.00362
22	0.38	-0.06428	0.00858	-0.0309	0.06027	-0.01564	-0.00362	-0.00362	-0.07648	-0.04292	-0.01697	0.38	0.02079	-0.0248	-0.00362	-0.08545	-0.00648	-0.00648	-0.00362	0.04196	-0.00362	-0.00362
23	0.4	-0.08545	0.02384	0.05112	0.04501	0.01469	-0.00954	0.01774	-0.0885	-0.0824	-0.01564	0.4	-0.00362	-0.00648	0.00248	0.02384	-0.00648	-0.00858	-0.00648	-0.02174	0.01469	-0.00648
24	0.42	-0.01869	-0.00362	0.06332	0.01163	-0.00648	-0.00954	-0.01259	-0.00648	-0.07343	-0.00362	0.42	-0.0309	-0.00057	-0.00362	0.06943	0.00248	-0.01259	-0.00362	-0.07343	-0.01259	-0.01259
25	0.44	0.05722	0.02689	0.01774	-0.02785	-0.00057	-0.00648	-0.01259	0.06332	0.01469	0.03845	0.44	-0.0309	-0.00362	-0.00858	0.03891	-0.01259	-0.00648	-0.00362	-0.09766	-0.00648	-0.03586
26	0.46	0.05722	0.01163	-0.01259	-0.04005	-0.00648	-0.00362	-0.01564	0.05417	0.07248	0.03586	0.46	-0.01259	0.00553	0.01163	-0.00362	-0.01259	0.00858	-0.00648	-0.00362	-0.01274	-0.00057
27	0.48	0.01774	0.01774	-0.04005	-0.00362	0.01774	-0.00362	-0.01564	0.01163	0.04501	-0.01869	0.48	0.01469	0.00248	0.00858	-0.04005	-0.01259	-0.00057	-0.00362	0.07534	-0.00362	-0.00553
28	0.5	-0.0248	0.00248	-0.02174	0.033	-0.00954	-0.00362	-0.0309	-0.0309	0.00248	-0.0309	0.5	0.033	-0.00954	0.01163	-0.01564	-0.0309	0.00248	-0.00362	0.04501	0.02079	0.01774
29	0.52	-0.037	-0.00057	0.01469	0.04196	-0.0248	-0.01259	-0.00648	-0.037	-0.04005	-0.00858	0.52	0.01774	-0.00954	-0.00362	0.01774	-0.0248	0.01774	-0.00362	-0.00362	-0.01259	0.01774
30	0.54	-0.00057	-0.00954	0.04196	-0.02785	-0.0309	-0.00362	-0.00362	-0.00362	-0.0248	-0.0309	0.54	-0.02174	0.03586	-0.0248	0.04196	0.01469	0.01774	-0.00648	-0.04597	-0.01259	-0.00057

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
473	9.4	0.04501	-0.01259	-0.02785	-0.00057	-0.02174	0.07839	-0.06733	-0.00362	0.00553	0.04807	9.4	-0.0309	0.04501	-0.01259	-0.07954	0.033	-0.01564	-0.0248	0.01163	-0.10986	-0.00362
474	9.42	0.07534	0.02995	-0.10376	0.0845	-0.00818	0.07733	-0.12493	-0.0885	-0.09155	-0.04005	9.42	-0.0946	-0.0309	0.03586	-0.11292	0.07334	-0.0248	0.01163	-0.05817	-0.06123	-0.03862
475	9.44	0.00588	0.04807	-0.0885	0.02689	0.06943	-0.037	-0.00309	-0.01192	-0.11902	-0.12493	9.44	-0.10681	-0.09155	0.05417	-0.02129	0.02384	0.00588	0.06027	-0.11292	0.05112	0.05417
476	9.46	-0.03395	-0.04005	0.05417	-0.0248	-0.03945	-0.00057	0.08144	0.02384	0.02079	-0.06123	9.46	-0.1163	-0.10376	-0.01564	-0.02748	-0.0248	0.06027	-0.05817	-0.07027	0.06943	-0.02174
477	9.48	-0.05722	-0.12493	-0.00277	-0.02174	-0.10986	0.06638	0.033	0.07534	-0.0845	0.06638	9.48	0.07248	0.02079	-0.0885	-0.04517	-0.0248	0.01163	-0.05817	-0.06123	-0.03862	-0.09155
478	9.5	0.05722	-0.06123	0.01469	0.033	0.08144	0.01774	-0.07174	0.02489	0.05112		9.5	0.02095	0.07534	-0.10986	-0.01259	0.02079	0.06027	-0.11292	0.05817	-0.03862	-0.03862
479	9.52	0.00588	0.06638	0.037	0.04807	0.06027	-0.07648	-0.1564	-0.04005	-0.0309	-0.09354	9.52	-0.01564	-0.02384	-0.03642	-0.00405	0.05417	-0.01986	-0.0309	-0.1163	-0.0362	-0.05817
480	9.54	-0.07648	0.05112	-0.03642	-0.04005	0.06638	-0.13409	0.02689	-0.00057	-0.05057	-0.0309	9.54	-0.0309	-0.1869	0.07248	0.00248	-0.01259	-0.04597	0.06943	-0.04005	0.02995	0.07334
481	9.56	-0.12493	-0.09354	-0.07272	-0.11292	-0.06648	-0.00057	-0.07272	0.05417	0.04807	-0.06648	9.56	0.02048	-0.07648	0.03586	0.0112	-0.07954	-0.06392	-0.04807	-0.00057	-0.05722	-0.03862
482	9.58	-0.09354	-0.0309	0.00588	-0.07648	-0.02785	0.08144	-0.04597	0.02384	0.02384	0.06943	9.58	0.05722	-0.00057	-0.06648	-0.02689	-0.11902	-0.05722	0.00248	0.03891	-0.02174	-0.02174
483	9.6	0.0845	-0.06648	-0.07038	0.06027	-0.00548	0.02995	-0.11902	-0.07338	-0.07648	-0.01259	9.6	0.01774	-0.02772	-0.0309	-0.04902	-0.01259	0.05858	-0.037	0.05112	-0.0885	-0.0248
484	9.62	0.03586	0.06943	-0.12188	0.06638	0.07839	-0.02785	-0.06123	-0.12493	-0.12798	-0.0946	9.62	-0.05512	0.01469	0.05553	-0.10681	0.07248	-0.03395	-0.06648	-0.0248	-0.11292	-0.00057
485	9.64	-0.01869	-0.01259	0.0248	0.05553	-0.00548	-0.09354	0.06943	-0.02174	-0.01564	-0.10986	9.64	-0.10986	-0.04062	-0.05112	-0.07143	0.04501	-0.03642	0.04501	-0.0946	-0.00057	0.06638
486	9.66	-0.01564	-0.0946	0.0845	-0.02785	-0.0885	0.04196	-0.07272	0.0845	0.0845	0.033	9.66	-0.06123	-0.10681	0.01469	-0.05512	-0.04501	-0.06648	0.04196	0.02689	-0.10376	0.06943
487	9.68	0.02689	-0.10986	0.033	0.05553	-0.12188	0.04807	-0.01259	0.04196	0.03586	0.08144	9.68	0.04807	-0.05112	0.05417	0.06943	-0.0309	0.04196	-0.04597	-0.0469	0.03891	-0.05112
488	9.7	0.05722	0.033	-0.037	0.05112	0.02995	-0.06123	-0.0309	-0.02174	-0.01248	0.01774	9.7	0.06943	0.04501	-0.10986	-0.07174	-0.00057	-0.03395	-0.10681	0.06943	-0.01259	-0.11292
489	9.72	-0.04005	0.08144	-0.09354	0.00248	0.04144	-0.12188	-0.0309	-0.01564	-0.01869	-0.0309	9.72	0.02079	0.06638	-0.06123	-0.03395	-0.05417	-0.10376	-0.07534	0.02995	-0.0785	-0.05112
490	9.74	-0.10986	0.01774	0.02689	-0.07954	0.02079	-0.04902	0.07534	0.03586	0.033	-0.06648	9.74	-0.0248	0.01774	0.05417	-0.01259	0.02689	-0.11292	0.04196	-0.02174	-0.00057	-0.05722
491	9.76	-0.07038	-0.0309	0.0401	-0.11902	-0.0309	0.07248	-0.0309	0.05722	0.04501	0.05112	9.76	-0.01564	-0.0248	0.06943	-0.02689	-0.04292	-0.0469	0.07248	-0.01259	0.05722	0.06638
492	9.78	0.06027	-0.06648	-0.04005	0.01163	-0.00057	-0.05112	-0.0986	-0.04902	-0.05112	0.02079	9.78	0.02384	-0.01259	0.00858	-0.05722	-0.10376	0.06943	0.02079	0.00858	0.02079	0.01163
493	9.8	0.06332	0.05112	-0.11292	0.0845	-0.07272	-0.06648	-0.0946	-0.11902	-0.13013	-0.06123	9.8	0.05722	0.02384	-0.07285	-0.08169	-0.07248	0.02384	-0.07285	0.06027	-0.04902	-0.04005
494	9.82	-0.03642	0.02079	-0.07954	0.02995	0.01774	-0.0248	0.04501	-0.05512	-0.04292	-0.12188	9.82	-0.01259	0.06027	-0.06648	-0.0885	0.04501	-0.0248	-0.09354	-0.0248	-0.10986	-0.03862
495	9.84	-0.03395	-0.06123	0.02689	-0.0248	-0.06733	0.00858	0.07248	0.0248	0.08144	-0.02785	9.84	-0.07648	-0.01259	0.03586	-0.0885	0.04501	-0.02174	0.02689	-0.07038	-0.06428	-0.03862
496	9.86	0.0248	-0.12188	-0.06638	-0.01259	-0.12493	0.07534	0.00858	0.05722	0.05417	0.08144	9.86	-0.11902	-0.07648	0.05722	0.00858	0.02079	0.0248	0.05722	-0.12188	0.04501	-0.05112
497	9.88	0.06638	-0.0785	-0.09354	-0.04196	-0.05417	-0.02785	-0.0785	-0.01564	-0.0248	0.03586	9.88	-0.01259	-0.12493	-0.07248	0.07339	-0.048	0.06027	-0.01869	-0.0309	0.06638	-0.0785
498	9.9	-0.01259	0.08144	-0.09354	0.00891	0.07534	-0.06638	-0.05722	-0.0785	-0.05722	-0.06638	9.9	0.02079	0.06638	-0.06123	-0.03395	-0.05417	-0.10376	-0.07534	0.02995	-0.0785	-0.05112
499	9.92	-0.0885	0.03586	-0.06648	-0.04902	0.033	-0.00057	0.05112	0.02079	-0.0112	-0.02785	9.92	-0.04807	0.06638	-0.11292	-0.02174	0.02384	-0.07038	-0.11292	0.05112	-0.0248	-0.10071
500	9.94	-0.11292	-0.01564	0.05722	-0.11902	-0.0248	0.03891	0.033	0.07248	-0.06027	0.01774	9.94	0.02048	-0.04807	0.00858	-0.037	-0.07272	-0.11292	-0.03642	0.05553	-0.01259	-0.0469
501	9.96	0.02995	-0.0785	0.0248	-0.05512	-0.01259	0.07248	-0.04005	-0.0309	-0.04597	0.05722	9.96	0.0309	-0.00057	0.07839	-0.00057	-0.01259	-0.04597	0.06943	0.03586	0.02689	0.07248
502	9.98	0.04814	0.01774	-0.06648	-0.07248	0.03891	0.02079	-0.10986	-0.11292	-0.11597	-0.0785	9.98	-0.06648	-0.0309	0.033	0.05722	-0.0885	0.05112	0.033	-0.03642	-0.05417	0.02995
503	10	0.02079	0.05722	-0.12493	0.04807	0.04807	-0.037	-0.07954	-0.09766	-0.07038	0.02079	10	0.04196	-0.00057	-0.01564	0.0163	-0.12188	0.04501	-0.02174	0.04501	-0.01564	-0.01869

		H																													
		U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR						
1		8V= 5.556 RPS										6V= 4.167 RPS																			
2		T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10						
3	-0.01259	-0.00362	0	0.05417	-0.037	-0.00362	-0.03395	-0.01564	0.02689	0.02689	0.01469	0.01774	0.04501	0	0	-0.00648	-0.03395	-0.02785	-0.0248	-0.04902	-0.0309	-0.00648	0.02079	-0.03395	0.04196						
4	-0.00362	-0.00648	0.02	0.00689	-0.03395	0.03586	-0.07954	0.00858	0.04196	0.033	-0.01564	0.04501	0.00858	0.02	0.00362	-0.0248	0.03586	-0.02785	-0.07648	-0.00954	-0.00648	-0.00362	0.05722								
5	-0.00362	-0.00954	0.04	-0.00057	-0.04005	0.03586	-0.0946	0.04196	0.00248	-0.01564	-0.0248	0.02079	-0.0248	0.04	-0.01259	-0.00362	0.05112	-0.01259	-0.07648	0.02079	-0.01259	-0.0309	0.02079	0.02079							
6	-0.00362	-0.00362	0.06	-0.0248	-0.0309	-0.02174	-0.01564	0.02995	-0.06123	-0.06733	-0.00057	-0.0309	-0.01869	0.06	-0.00648	0.02689	0.03586	0.00858	-0.00648	0.03586	-0.00362	-0.01869	0.033	0.00248							
7	-0.00648	-0.00648	0.08	-0.04597	-0.0309	-0.07038	0.04807	-0.0309	-0.10376	-0.09766	0.033	-0.0824	0.01163	0.08	-0.00648	0.033	0.00553	0.03586	0.04501	0.02995	-0.00362	0.00553	-0.00362	-0.0248							
8	-0.00362	-0.00648	0.1	-0.01564	-0.00648	-0.09766	0.04807	-0.0824	-0.06428	-0.05207	0.03891	-0.0946	0.04196	0.1	-0.00648	-0.00057	-0.02174	0.033	0.05417	-0.02174	-0.00648	0.02689	-0.037	-0.0248							
9	-0.00954	-0.00648	0.12	0.01163	0.01163	-0.04292	0.02384	-0.09766	0.02384	0.03891	-0.01564	-0.01564	0.02079	0.12	-0.00648	-0.04005	-0.0309	-0.02785	0.02384	-0.05817	-0.00362	0.02689	-0.06733	-0.00057							
10	-0.00954	-0.00954	0.14	0.02995	0.02384	0.04196	-0.01564	-0.0248	0.06638	0.06027	-0.06733	0.05112	-0.04005	0.14	-0.00648	-0.07038	-0.00362	-0.05512	-0.00057	-0.07648	-0.00648	-0.00362	-0.07954	-0.02079							
11	-0.00362	-0.00362	0.16	0.02079	0.033	0.06332	-0.0248	0.05112	0.02995	0.02995	-0.10071	0.05722	-0.0885	0.16	-0.00648	-0.0824	0.01774	0.07954	-0.0248	-0.06733	-0.00362	-0.0309	-0.04005	0.033							
12	-0.00648	-0.00362	0.18	-0.01564	0.02995	0.02689	0.00248	0.06027	-0.00648	-0.00648	-0.04902	0.02079	-0.09766	0.18	-0.01259	-0.02785	0.02689	-0.06123	-0.0248	-0.00057	-0.00362	-0.07038	0.02689	0.01774							
13	-0.00362	-0.00648	0.2	-0.05512	0.01774	-0.00648	0.02689	0.02079	-0.03395	-0.02785	0.04196	-0.01259	-0.01564	0.2	-0.00648	0.03586	0.02384	0.05553	-0.00057	0.04807	-0.01259	-0.07648	0.05722	-0.0785							
14	-0.00648	-0.00648	0.22	-0.07954	0.00248	-0.03395	0.03586	-0.01564	-0.00648	-0.00648	-0.06638	-0.02785	0.05417	0.22	-0.00362	0.05722	-0.0309	0.05112	0.02384	0.05722	-0.00648	-0.04005	0.03891	-0.04174							
15	-0.00362	-0.01259	0.24	-0.04902	-0.01259	-0.00954	-0.01259	-0.0309	0.02079	0.033	0.02384	0.00248	0.05112	0.24	-0.00648	0.03586	-0.06428	0.04807	0.03586	0.02384	-0.00362	0.02689	0.00858	-0.08545							
16	-0.00648	-0.00362	0.26	0.01774	-0.02174	0.02384	-0.06123	-0.00648	0.05112	0.04501	-0.01564	0.02689	0.01774	0.26	-0.00648	0.00553	-0.0885	0.01469	0.00553	-0.00362	-0.00648	0.05417	-0.01564	-0.04597							
17	-0.00954	-0.00362	0.28	0.06332	0.04005	0.04501	-0.09766	0.03586	0.01469	-0.00057	-0.03395	0.04196	-0.02174	0.28	-0.00648	-0.02174	-0.05207	-0.00648	-0.037	-0.03395	0.0362	0.04196	-0.0309	0.02384							
18	-0.00362	-0.00648	0.3	0.03586	-0.01869	-0.00057	-0.06123	0.03586	-0.04902	-0.05817	-0.00954	0.00248	-0.0248	0.3	-0.00648	-0.02785	0.01774	0.02785	-0.07038	-0.01564	-0.00648	-0.01774	-0.01259	-0.0517							
19	-0.00648	-0.00648	0.32	0.01163	0.00248	-0.05512	0.02689	-0.01259	-0.0885	-0.10376	0.02384	-0.05207	0.00858	0.32	-0.01259	-0.01259	0.05417	-0.0248	-0.0824	0.00858	-0.00648	-0.01564	0.01469	0.04501							
20	-0.00648	-0.01259	0.34	-0.0248	0.00858	-0.0885	0.06638	-0.06733	-0.0885	-0.07648	0.04501	-0.09155	0.03891	0.34	-0.00362	0.01774	0.04196	0.00858	-0.0309	0.02995	0.00648	-0.0309	0.033	0.0116							
21	-0.00648	-0.00362	0.36	-0.04005	-0.02689	-0.06428	0.02995	-0.09766	0.00858	0.02384	-0.00362	-0.07343	0.03586	0.36	-0.00362	0.033	0.02079	0.02995	0.033	0.03586	-0.01259	-0.01564	0.033	-0.01564							
22	-0.00362	-0.00362	0.38	-0.01564	0.02384	0.02079	-0.00648	-0.05512	0.06332	0.06638	-0.05512	0.01774	-0.02174	0.38	-0.00362	0.02384	-0.01564	0.03586	0.05722	-0.00954	-0.01259	0.00553	-0.02785	-0.0785							
23	0.01469	-0.00648	0.4	0.01774	-0.00057	0.06638	-0.02785	0.03586	0.04501	0.03586	-0.0946	0.06027	-0.07648	0.4	-0.00648	-0.02785	-0.037	-0.01259	0.033	0.04597	0.00248	0.02995	-0.05817	-0.01564							
24	-0.01259	-0.01259	0.42	0.033	-0.037	0.03586	-0.01564	0.06332	0.00858	-0.00362	-0.06733	0.03586	-0.09766	0.42	-0.01564	-0.06428	-0.00648	0.04597	0.00858	-0.07343	-0.00648	-0.03362	-0.07954	-0.04169							
25	-0.00648	-0.03586	0.44	0.00858	-0.04902	0.00248	0.02079	0.02995	-0.02785	-0.02785	0.02689	-0.00362	-0.0309	0.44	-0.00954	-0.07648	0.01469	-0.07648	-0.02174	-0.07954	-0.00362	-0.03362	-0.05817	-0.033							
26	-0.02174	-0.00057	0.46	-0.02785	-0.06428	-0.04005	0.04501	-0.00954	-0.02174	-0.00954	0.06638	-0.03395	0.05112	0.46	-0.00648	-0.05207	0.03586	-0.07648	-0.0309	-0.02174	-0.00057	-0.04597	0.00248	0.03586							
27	-0.00362	-0.00648	0.48	-0.07648	-0.05207	-0.01564	0.02384	-0.02785	0.00858	0.01774	0.03586	-0.02785	0.06027	0.48	-0.00648	0.01774	0.02079	-0.01259	0.00648	0.03891	-0.00057	-0.07038	0.05112	-0.04174							
28	0.02079	0.01774	0.5	0.0824	-0.00362	0.02079	-0.04005	-0.00362	0.04807	0.04196	-0.00057	0.00248	0.05417	0.5	0.00648	0.04501	-0.01869	0.06123	0.01774	0.04501	-0.00362	-0.07648	0.04807	-0.05207							
29	-0.01259	0.01774	0.52	-0.01259	0.03891	0.04501	-0.0885	0.02689	0.02384	0.01163	-0.04005	0.04196	-0.01259	0.52	-0.00648	0.04807	-0.05512	0.04501	0.03586	0.02689	-0.00648	-0.03395	0.02079	0.7954							
30	-0.01259	-0.00057	0.54	0.05112	0.05112	0.00858	-0.09766	0.04196	-0.04005	-0.04005	-0.01564	0.02995	-0.02785	0.54	0.00248	0.02079	-0.07954	0.02384	0.02689	0.00248	-0.01259	0.033	-0.02079	-0.07038							

	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR		
473	-0.10986	-0.00362		9.4	-0.00362	0.01469	0.02384	-0.10071	-0.02785	-0.10071	-0.03395	0.05417	0.00858	-0.03395	9.4	-0.01259	-0.04902	-0.04292	0.08545	0.01469	0.05722	-0.04005	-0.00648	0.05417	-0.06428	
474	-0.06123	0.03586		9.42	0.06027	-0.01869	-0.03395	-0.05817	-0.0824	-0.04902	0.0824	0.01774	-0.04597	-0.00057	9.42	0.01774	-0.07954	-0.07038	-0.04292	0.02689	0.03586	-0.07648	-0.0309	0.04501	-0.00954	
475	0.05112	0.05417		9.44	0.04807	-0.0309	-0.07648	0.033	-0.0946	0.03891	-0.0946	-0.01869	-0.0885	0.02995	9.44	0.03586	-0.07648	-0.07954	0.02689	0.02995	0.05553	-0.07648	-0.01869	0.02079	0.04807	
476	0.06943	-0.02174		9.46	0.01163	-0.00057	-0.09766	0.06332	-0.02174	0.06638	-0.01259	-0.0248	-0.07648	0.04196	9.46	0.02384	-0.00954	-0.0248	0.05722	-0.02174	0.01564	-0.00248	-0.01564	0.05112		
477	0.02079	-0.09155		9.48	-0.0248	0.02079	-0.0248	0.033	0.05112	0.02689	0.06027	-0.00057	0.00858	-0.00954	9.48	-0.0248	0.04501	0.033	0.02689	-0.05512	-0.0309	0.03586	0.02995	-0.02785	0.02384	
478	-0.0309	-0.11292		9.5	-0.02785	0.03586	0.05417	-0.00648	0.05417	-0.01564	0.05417	0.03586	-0.06027	-0.06428	9.5	-0.06123	0.05112	0.05722	0.00858	-0.07648	-0.01259	0.05722	0.03586	-0.01259	-0.00954	
479	-0.00362	0.00858		9.52	0.00248	-0.01564	0.05722	-0.0309	0.02079	-0.0309	0.01469	0.02689	0.04807	-0.0946	9.52	-0.07648	0.02689	0.02995	-0.02174	-0.07038	0.01774	0.02689	-0.01259	0.01469	-0.0309	
480	0.02995	0.07534		9.54	0.03586	-0.06428	0.02384	-0.01869	-0.00362	-0.0248	-0.02174	0.01163	-0.05207	0.0248	9.54	0.05512	-0.00057	-0.00362	-0.037	-0.01259	0.033	0.00057	-0.05207	0.02689	-0.02174	
481	0.05722	0.03586		9.56	0.03586	-0.10071	-0.01869	0.02079	-0.0248	0.02995	-0.02785	-0.07343	-0.02174	0.03891	9.56	0.00858	-0.02785	-0.02174	-0.00362	0.04501	0.02995	-0.07648	-0.03362	0.03586	0.00553	
482	-0.02174	-0.02174		9.58	-0.02785	-0.04902	-0.037	0.04501	0.00553	0.04501	-0.00362	-0.10071	-0.0248	0.06638	9.58	0.04807	-0.0248	-0.02785	0.01774	0.05417	-0.0248	-0.0309	-0.07343	-0.01564	0.033	
483	-0.0885	-0.0248		9.6	-0.07038	0.04196	-0.00362	0.01163	0.03586	-0.00362	0.033	-0.05512	0.00858	0.02689	9.6	0.04807	0.0248	-0.00648	0.02689	0.02384	-0.06123	-0.00648	-0.01259	-0.04902	0.033	
484	-0.11292	0.00057		9.62	-0.10071	0.06638	0.03586	-0.04902	0.03586	-0.06123	0.02689	0.03586	0.03891	0.00648	9.62	0.02079	0.02079	0.01163	0.033	0.00057	-0.0248	0.00858	0.03586	-0.07954	-0.06428	
485	-0.0057	0.06638		9.64	0.00057	0.02384	-0.03891	-0.0885	-0.02174	-0.09766	-0.0309	0.06638	0.033	0.03948	9.64	-0.00954	0.03586	0.03586	-0.0248	-0.07648	-0.01259	0.02689	-0.05417	-0.07954	-0.05497	
486	0.06943	0.01163		9.66	0.03891	-0.00648	-0.01259	-0.0824	-0.07648	-0.06123	-0.07343	0.033	-0.0248	-0.01259	9.66	-0.0309	-0.00057	0.02079	-0.05817	-0.02174	0.00858	0.02384	-0.02689	-0.00362	-0.07648	
487	0.03891	-0.05152		9.68	0.06332	-0.0309	-0.06733	0.00858	-0.10071	-0.03062	-0.07038	0.02384	0.02995	9.68	-0.01869	-0.04005	-0.037	-0.07954	-0.0362	-0.0512	-0.0248	0.00248	-0.06428	-0.04807	-0.07648	
488	-0.02159	-0.11292		9.7	0.02079	-0.01259	-0.09766	0.06332	-0.037	-0.0362	-0.037	-0.02785	-0.09766	0.04501	9.7	0.00248	-0.07038	-0.06123	-0.05817	0.01469	0.04196	-0.05817	-0.02785	-0.05722	-0.0248	
489	-0.0785	-0.05152		9.72	-0.01259	0.01774	-0.05401	0.04501	0.04501	0.02995	0.04807	-0.00954	-0.04005	-0.00057	9.72	0.03586	-0.07954	-0.08545	0.01163	0.03586	0.02079	-0.07954	-0.02785	0.01774	0.03891	
490	-0.00057	0.05722		9.74	-0.04005	0.04501	0.033	0.00858	0.06027	-0.01259	0.06027	0.02079	0.03586	-0.06123	9.74	0.02995	-0.02785	-0.05207	0.05417	0.01469	-0.00648	-0.04902	-0.00557	-0.00362	-0.05417	
491	0.05722	0.06638		9.76	-0.00648	0.01469	0.06638	-0.02785	0.0384	-0.037	0.02384	0.04501	0.03632	-0.01273	9.76	-0.01259	0.03079	0.04501	-0.07648	-0.0309	0.01469	0.02384	-0.07648	0.02785	0.02995	
492	0.02079	0.01163		9.78	0.01774	-0.04002	0.02384	-0.01869	-0.00954	-0.00954	-0.01259	0.02079	0.02384	-0.07038	9.78	-0.04902	0.05722	0.05722	-0.02174	-0.06123	-0.01564	0.05417	0.03586	-0.01564	-0.00057	
493	-0.04902	-0.04005		9.8	0.04501	-0.0885	-0.00954	0.00858	-0.07648	-0.02995	-0.02785	-0.037	-0.01259	0.02384	9.8	-0.07343	0.033	0.03891	-0.01259	-0.0824	-0.0248	0.04501	0.00858	-0.0362	-0.02785	
494	-0.10986	-0.00362		9.82	-0.00362	-0.07954	-0.0309	0.04501	-0.00648	0.05112	-0.0362	-0.0824	-0.0309	0.06638	9.82	-0.07648	0.00858	0.01163	-0.02785	-0.04902	0.033	0.00858	-0.03586	0.02689	-0.02174	
495	-0.06428	0.03586		9.84	-0.04902	0.00858	-0.00648	0.02079	0.02079	0.0248	0.02995	-0.0885	-0.00954	0.03586	9.84	-0.01869	-0.02785	-0.01564	-0.01259	0.01469	0.02995	-0.01259	-0.07343	0.03586	-0.06638	
496	-0.0501	0.05112		9.86	-0.09766	0.06027	0.02384	-0.037	0.04501	-0.05207	0.04501	-0.00954	-0.02384	-0.00057	9.86	0.03586	-0.07648	-0.02785	-0.08545	0.01163	-0.01869	-0.0309	-0.07954	-0.00057	0.02689	
497	0.06638	-0.02174		9.88	-0.00362	0.04196	-0.0037	-0.0824	-0.0362	-0.04902	-0.06428	0.05722	0.0248	-0.00057	9.88	0.05722	-0.02689	-0.01564	-0.0885	0.02079	-0.04902	-0.02785	-0.07648	-0.04597	0.02995	
498	-0.0885	-0.0248		9.9	0.01774	-0.04902	-0.01259	-0.05401	0.04501	0.04501	0.02995	0.04807	-0.00954	-0.00057	9.9	0.02079	-0.07954	-0.0885	0.01163	0.03586	0.02079	-0.07954	-0.02785	0.01774	0.03891	
499	-0.0248	-0.10711		9.92	0.06332	-0.0248	-0.04597	-0.01564	-0.0946	-0.02079	-0.0946	-0.02384	-0.04597	-0.02079	9.92	-0.00057	0.033	0.03586	-0.01259	-0.01564	-0.04902	0.033	0.02995	-0.04501	-0.0824	-0.037
500	-0.0259	0.01469		9.94	0.04501	-0.0309	-0.0885	-0.0272	-0.06428	0.06638	-0.06123	0.02174	-0.0824	0.04807	9.94	-0.03395	0.02079	0.033	-0.04902	-0.07648	-0.01259	0.03586	0.01964	-0.02174	-0.07038	
501	0.02689	0.07248		9.96	0.00553	0.00858	-0.0824	-0.05417	0.02689	0.03586	-0.02995	-0.02785	-0.0824	-0.00858	9.96	-0.02785	-0.0309	-0.02174	-0.07954	-0.01259	0.03891	-0.01259	0.01163	0.03891	-0.0824	
502	-0.05417	0.02995		9.98	-0.02785	0.01964	0.00248	0.01469	0.06638	-0.00362	0.06638	-0.0362	-0.0248	-0.04902	9.98	-0.00648	-0.06428	-0.05207	-0.07648	0.00858	0.05722	-0.04005	-0.01564	0.05417	-0.04597	
503	-0.0885	-0.01869		10	-0.01869	0.033	-0.0272	-0.0248	0.02384	-0.02785	0.03586	-0.06027	-0.06027	-0.06027	10	0.02079	-0.0885	-0.0824	-0.0362	0.033	0.02689	-0.07954	-0.04005	0.033	0.02995	
504																										

	AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC
	9.4	0.02689	0.01774	0.02079	0.02689	-0.0309	0.00858	0.03586	-0.04005	-0.06123	-0.01564
	9.42	0.04196	0.02384	0.00248	0.01469	-0.04902	0.02079	0.02079	-0.05207	-0.04005	-0.0248
	9.44	0.02689	0.01163	-0.01259	-0.00362	-0.05817	0.02995	0.00858	-0.04902	0.00248	-0.04005
	9.46	0.01774	-0.00648	-0.0248	-0.01869	-0.04597	0.00553	-0.00954	-0.0248	0.02995	-0.05512
	9.48	-0.00362	-0.02785	-0.0248	-0.02785	-0.01259	-0.01259	-0.0248	0.00858	0.04196	-0.05207
	9.5	-0.01564	-0.04902	-0.01869	-0.0309	0.01774	-0.02785	-0.02785	0.02995	0.033	-0.02785
	9.52	-0.02785	-0.05207	-0.00057	-0.02174	0.03891	-0.04292	-0.02174	0.04196	0.02384	0.00858
	9.54	-0.0248	-0.05207	0.00858	-0.00057	0.03891	-0.05207	-0.00362	0.02995	0.00248	0.033
	9.56	-0.01259	-0.0309	0.02384	0.00858	0.02079	-0.04902	0.00858	0.01774	-0.01259	0.03586
	9.58	-0.00362	0.00248	0.02384	0.02689	0.00858	-0.01869	0.01469	-0.00057	-0.0248	0.02079
	9.6	0.00553	0.02995	-0.00057	0.02384	-0.00648	0.01469	0.02079	-0.01564	-0.02785	0.01774
	9.62	0.02384	0.03891	-0.01564	0.00248	-0.02174	0.03586	-0.00057	-0.03395	-0.01564	-0.00057
	9.64	0.02079	0.02689	-0.0309	-0.01564	-0.02785	0.03586	-0.01869	-0.02785	-0.00362	-0.01564
	9.66	-0.00362	0.02079	-0.04902	-0.0309	-0.01869	0.02689	-0.0309	-0.01564	0.00248	-0.0248
	9.68	-0.02174	-0.00057	-0.05817	-0.04902	-0.00362	0.01163	-0.04597	0.00248	0.02079	-0.02785
	9.7	-0.04005	-0.01564	-0.05207	-0.05207	0.00858	-0.00057	-0.05207	0.01774	0.02384	-0.01564
	9.72	-0.05207	-0.0309	-0.02174	-0.04902	0.01774	-0.01564	-0.04597	0.02689	0.00248	0.00248
	9.74	-0.05207	-0.0248	0.01469	-0.01564	0.02384	-0.02785	-0.01564	0.02079	-0.01259	0.00553
	9.76	-0.04005	-0.01564	0.03586	0.01469	0.01163	-0.0248	0.01774	-0.00057	-0.0309	0.02079
	9.78	-0.00954	-0.00057	0.03891	0.03586	-0.01259	-0.00648	0.03586	-0.0248	-0.04597	0.02384
	9.8	0.02079	0.00858	0.02079	0.03891	-0.02785	0.00553	0.03891	-0.037	-0.05512	-0.00362
	9.82	0.03586	0.02079	0.00553	0.02079	-0.04292	0.01774	0.02384	-0.05207	-0.04292	-0.02174
	9.84	0.03586	0.02689	-0.00954	0.00858	-0.05207	0.02689	0.01163	-0.05512	-0.01259	-0.04005
	9.86	0.02079	0.00248	-0.02174	-0.00362	-0.04902	0.01163	-0.00648	-0.03395	0.01774	-0.04902
	9.88	0.00553	-0.01259	-0.02785	-0.01869	-0.02785	-0.00648	-0.02174	-0.00057	0.03586	-0.05512
	9.9	-0.00648	-0.03395	-0.02174	-0.02785	0.00858	0.02785	-0.0309	0.02689	0.03891	-0.037
	9.92	-0.0248	-0.05207	-0.00648	-0.02174	0.033	-0.04005	-0.02174	0.03586	0.02079	-0.00362
	9.94	-0.02785	-0.05512	0.00858	-0.00362	0.04196	-0.05207	-0.00362	0.02689	-0.00362	0.02079
	9.96	-0.02174	-0.04597	0.01774	0.00858	0.02689	-0.05207	0.00858	0.00858	-0.01259	0.03891
	9.98	-0.00362	-0.01259	0.02689	0.01469	0.02079	-0.0309	0.01774	-0.00057	-0.0248	0.03586
	10	-0.00362	0.01774	0.01469	0.02384	-0.00057	0.00248	0.02689	-0.01259	-0.0248	0.02079

Data points for $k=4$

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	12V= 8.333 RPS												10V= 6.945 RPS									
2	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
3	0	-0.00362	-0.10071	-0.00648	-0.00648	-0.0248	-0.00362	-0.1585	-0.00362	-0.00362	-0.01259	0	-0.10376	0.13618	-0.00954	-0.08545	0.10586	-0.10986	-0.13409	0.06027	0.14229	-0.14629
4	0.02	-0.00648	-0.10376	-0.00362	-0.00362	-0.01564	-0.00648	-0.05512	-0.00057	-0.00648	-0.02174	0.02	-0.09766	0.06638	-0.00648	0.13313	0.08144	0.12093	-0.07648	0.12093	-0.0946	0.07534
5	0.04	-0.00648	0.06638	-0.00362	-0.00648	-0.01564	-0.00648	0.17872	-0.00648	-0.00362	-0.0309	0.04	0.03891	-0.1524	-0.00648	0.02079	-0.1585	0.05722	0.14534	-0.14324	-0.13714	0.09975
6	0.06	-0.00648	0.12093	-0.00362	-0.00648	-0.00362	-0.01259	-0.11597	-0.00057	-0.00648	-0.04597	0.06	0.11787	-0.06123	-0.00362	-0.17967	-0.02785	-0.17357	-0.00057	-0.07954	0.10586	-0.15545
7	0.08	-0.00954	-0.037	0.00553	-0.01259	0.00248	-0.00648	-0.0885	-0.00057	-0.00362	-0.04902	0.08	0.01469	0.14839	-0.00362	0.04807	0.1543	0.00858	-0.17357	0.1543	0.0845	-0.04597
8	0.1	-0.00362	-0.14324	0.01774	-0.00648	0.00858	-0.00648	0.16956	-0.00057	-0.00954	-0.04005	0.1	-0.11902	-0.00362	-0.00362	0.14534	-0.05817	0.1543	0.06638	0.02384	-0.14935	0.15736
9	0.12	-0.01259	-0.00648	0.02079	-0.00648	0.00858	-0.00648	-0.0885	-0.00057	-0.01259	-0.01869	0.12	-0.08545	-0.15545	-0.00362	-0.0885	-0.14629	-0.05207	0.13313	-0.17052	-0.0309	-0.00057
10	0.14	-0.00648	0.13313	0.01163	-0.00648	0.00858	-0.00362	-0.11292	0.00553	-0.01564	0.00858	0.14	0.06332	0.033	-0.00362	-0.13714	0.10891	-0.1585	-0.10376	0.00248	-0.1543	-0.17052
11	0.16	-0.00648	0.00248	0.00553	-0.01259	0.01163	-0.00954	0.18787	0.00553	-0.0309	0.04196	0.16	0.10586	0.13618	-0.00362	0.10891	0.09975	0.06332	-0.12798	0.14534	-0.05817	0.02689
12	0.18	-0.00362	-0.14935	-0.01259	-0.00648	0.00858	-0.00648	-0.05207	-0.00362	-0.037	0.05722	0.18	-0.03395	-0.10376	-0.00362	0.0845	-0.14019	0.12093	0.11196	-0.07343	-0.15545	0.14534
13	0.2	-0.00954	-0.0309	-0.04005	-0.00648	0.01163	-0.00648	-0.16136	0.02079	-0.0309	0.04501	0.2	-0.13409	-0.11292	-0.00648	-0.1585	-0.0885	-0.12798	0.08144	-0.12798	0.10891	-0.10071
14	0.22	-0.00648	0.13618	-0.06123	-0.00648	0.01163	-0.00648	0.1543	0.02689	-0.0248	-0.00362	0.22	-0.0248	0.12398	-0.00362	-0.03395	0.14534	0.10574	-0.1585	0.10891	0.10281	-0.11597
15	0.24	-0.00648	0.01469	-0.06428	-0.00954	0.01163	-0.00362	0.00553	0.02689	-0.00362	-0.05817	0.24	0.11482	0.08144	-0.00362	0.14839	0.02384	0.14229	-0.0248	0.07248	-0.13714	0.12093
16	0.26	-0.01259	-0.1585	-0.04005	-0.00362	0.00858	-0.00648	-0.17357	0.01163	0.02079	-0.08545	0.26	0.07248	-0.14324	-0.00362	-0.05207	-0.16136	0.00248	0.1543	-0.17662	-0.0885	0.04807
17	0.28	-0.00648	-0.00362	0.00858	-0.00648	0.00858	-0.00362	0.12093	-0.01259	0.04501	-0.06123	0.28	-0.09766	-0.08545	-0.00648	-0.1524	0.02995	-0.17662	-0.05817	-0.00648	0.13924	-0.17662
18	0.3	-0.00648	0.14839	0.05112	-0.00648	0.01163	-0.01259	0.03891	-0.04597	0.04807	0.01163	0.3	-0.11292	0.14534	-0.00648	0.10891	0.13313	0.06027	-0.15545	0.15736	0.02079	0.02079
19	0.32	-0.00362	-0.01564	0.06332	-0.00648	-0.00954	-0.01259	-0.19493	-0.07038	0.02384	0.07534	0.32	0.05417	0.02384	-0.00648	0.10586	-0.11292	0.13924	0.10891	-0.04292	-0.1585	0.1543
20	0.34	-0.00648	-0.15545	-0.02995	-0.00648	-0.02785	-0.01564	0.13313	-0.06733	-0.0248	0.06943	0.34	0.11482	-0.15545	-0.00362	-0.14019	-0.10071	0.10376	0.10281	-0.16136	-0.10376	-0.07038
21	0.36	-0.00648	0.01163	-0.037	-0.00648	-0.05207	-0.02785	0.06943	-0.0309	-0.06733	0.01259	0.36	-0.04902	0.01163	-0.00648	-0.08545	0.13924	-0.12798	0.13714	0.06027	0.14229	-0.14935
22	0.38	-0.00648	0.14534	-0.0885	-0.01259	-0.06428	-0.037	-0.19493	0.01774	-0.07648	-0.10071	0.38	-0.13714	0.13924	-0.00362	0.14229	0.05722	0.11787	-0.09766	0.12398	-0.10986	0.09365
23	0.4	-0.01259	-0.06123	-0.07648	-0.00648	-0.05207	-0.04005	0.05722	0.06332	-0.04005	-0.08545	0.4	-0.00057	-0.0824	-0.00648	0.02079	-0.16136	0.07248	-0.13924	-0.12493	-0.10986	0.09975
24	0.42	-0.00362	-0.14324	-0.00057	-0.00648	-0.02174	-0.02785	0.17872	0.06638	0.02384	0.01774	0.42	0.12703	-0.13714	-0.00362	-0.16441	-0.04005	-0.1524	0.0332	0.09766	0.13313	-0.14629
25	0.44	-0.00954	0.07534	0.07534	-0.00648	0.03586	-0.00362	-0.17357	0.01774	0.07534	0.0967	0.44	0.01163	0.10891	-0.00648	0.02995	0.15736	-0.02174	-0.16441	0.14229	0.06027	-0.05207
26	0.46	-0.00648	0.10586	0.07534	-0.00954	0.06638	0.02384	0.02384	-0.06733	0.06027	0.06943	0.46	-0.14629	0.09975	-0.00362	0.13924	-0.04005	0.1543	0.01774	0.00248	-0.16136	0.14839
27	0.48	-0.00954	-0.13714	-0.01564	-0.00057	0.05722	0.04501	0.13313	-0.10376	-0.0248	-0.05512	0.48	-0.03395	-0.13409	-0.00954	-0.12188	-0.14935	-0.06428	0.14229	-0.17357	-0.04005	-0.02785
28	0.5	-0.00954	-0.07038	-0.11292	-0.00648	-0.00362	0.05112	-0.18578	-0.04902	-0.10071	-0.11902	0.5	0.13008	-0.0946	-0.00648	-0.09766	0.07534	0.14629	-0.10071	0.06332	0.15736	-0.16136
29	0.52	-0.01564	0.14534	-0.0824	0.00248	-0.07343	0.033	0.02689	0.04501	-0.07648	-0.04005	0.52	0.05722	0.13313	-0.01259	0.14839	0.10891	0.11787	-0.11902	0.13924	-0.0309	0.08755

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	
473	9.4	-0.07954	-0.14935	0.12398	-0.11597	-0.14324	0.17262	0.14534	0.16346	-0.1524	-0.1524	9.4	0.09975	0.09365	-0.14324	-0.16747	-0.07648	-0.15545	-0.07343	-0.0824	0.14534	-0.0248	
474	9.42	-0.12493	-0.07954	0.08755	0.18787	-0.0824	-0.11292	0.02079	-0.12798	-0.04902	0.13924	9.42	-0.12493	-0.15545	-0.07343	-0.01564	0.14534	-0.05207	0.1543	0.1543	0.00858	0.15736	
475	9.44	0.1543	0.16041	-0.18578	-0.04005	0.18177	-0.0885	-0.17357	-0.07648	0.17872	0.01163	9.44	-0.10071	-0.04597	0.1543	0.1543	0.03586	0.15736	0.02384	0.02079	-0.17662	-0.05207	
476	9.46	-0.0309	-0.02785	0.02079	-0.17052	-0.09155	0.18787	0.10891	0.18177	-0.11292	-0.18883	9.46	0.13008	0.1543	0.033	-0.05817	-0.16441	0.00858	-0.16747	-0.17052	0.05722	-0.13714	
477	9.48	-0.16136	-0.1524	0.14534	0.14534	-0.11292	-0.07648	0.06027	-0.05512	-0.11292	0.15144	9.48	0.05722	0.01163	-0.16747	-0.14324	-0.0248	-0.17662	-0.01564	-0.00057	0.13313	0.08755	
478	9.5	0.16651	0.07839	-0.15545	0.02079	0.16651	-0.14324	-0.19798	-0.1585	0.17872	0.04501	9.5	-0.1585	-0.16747	-0.01259	0.09975	0.1543	0.01469	0.1543	0.15144	-0.09766	0.08755	
479	9.52	0.02384	0.10586	-0.0309	-0.18272	-0.06428	0.16041	0.12093	0.13313	-0.04902	-0.19798	9.52	-0.01564	0.00248	0.14839	0.08755	-0.05207	0.14839	-0.06123	-0.07648	-0.13103	-0.17357	
480	9.54	-0.18578	-0.13409	0.16651	0.11196	-0.12798	-0.00648	0.07534	0.04501	-0.13714	0.0845	9.54	0.14534	0.1543	-0.05817	-0.17052	-0.14324	-0.0885	-0.13409	-0.12493	0.11196	-0.0309	
481	9.56	0.08144	-0.07954	-0.14935	0.05417	0.18177	-0.16136	-0.19188	-0.17357	0.14534	0.09975	9.56	-0.05512	-0.07038	-0.14324	-0.01259	0.0906	-0.11902	0.0967	0.10586	0.07839	0.16041	
482	9.58	0.0967	0.14229	-0.05512	-0.19798	-0.0248	0.12398	0.03586	0.07534	-0.01564	-0.18272	9.58	-0.14935	-0.12798	0.09365	0.1543	0.0967	0.11787	0.08755	0.07534	-0.1524	-0.0309	
483	9.6	-0.17967	0.05417	0.18177	0.12093	-0.16747	0.02995	0.13618	0.09975	-0.17052	0.04196	9.6	0.09365	0.0967	0.0906	-0.03395	-0.16136	0.06638	-0.16747	-0.17052	-0.0248	-0.16441	
484	9.62	0.03586	-0.14935	-0.0824	0.08144	0.12398	-0.19188	-0.17052	-0.19493	0.16346	0.13008	9.62	-0.11787	0.08755	-0.16136	-0.16136	-0.04005	-0.17967	-0.02174	-0.01259	0.14839	0.06027	
485	9.64	0.13313	-0.06123	-0.14935	-0.19188	0.04196	0.13924	0.00248	0.05417	0.02079	-0.19493	9.64	-0.12188	-0.16136	-0.02785	0.05417	0.15736	0.00248	0.15736	0.1543	-0.06428	0.12398	
486	9.66	-0.19188	0.13924	0.15736	0.033	-0.18272	0.06638	0.14534	0.14534	-0.18883	0.04196	9.66	-0.11902	-0.037	0.15736	0.13313	-0.01259	0.15736	-0.0248	-0.04292	-0.1524	-0.12188	
487	9.68	0.01774	0.02079	0.00553	0.14534	0.09975	-0.19188	-0.17662	-0.1524	0.08755	0.14839	9.68	0.11482	0.15144	-0.01259	-0.12188	-0.17052	-0.04902	-0.16747	-0.1585	0.11196	-0.07954	
488	9.7	0.16041	-0.14935	-0.17052	-0.16441	0.07248	0.05112	-0.00057	-0.06123	0.0845	-0.17052	9.7	0.07839	0.00248	-0.16747	-0.0885	0.02384	-0.1585	0.03586	0.06638	0.09975	0.14839	
489	9.72	-0.14935	-0.02174	0.11787	-0.01564	-0.19798	0.12093	0.16956	0.17872	-0.18272	-0.037	9.72	-0.14935	-0.17357	0.033	0.14229	0.14229	0.06638	0.14534	0.12703	-0.14019	0.00858	
490	9.74	0.07448	0.13313	0.04807	0.16041	0.0586	-0.17357	-0.13714	-0.09155	0.05722	-0.1762	9.74	-0.037	0.01774	0.13924	0.02553	-0.0946	-0.12398	-0.06028	-0.12493	-0.09754	0.17357	
491	9.76	0.17872	0.01163	-0.19798	-0.16747	0.02081	0.08588	-0.07954	-0.10986	0.12398	-0.1303	9.76	0.14839	0.14534	-0.10071	-0.17662	-0.11292	-0.12798	-0.10376	-0.0885	0.14839	0.06943	
492	9.78	-0.0824	-0.14935	0.11196	-0.18169	-0.18578	0.14534	0.13567	0.16346	-0.19798	-0.07633	9.78	-0.037	-0.08545	-0.10986	0.06027	0.11787	-0.07648	0.13008	0.14229	0.01469	0.13313	
493	9.8	-0.12798	-0.04005	0.10281	-0.17872	0.14549	-0.17357	-0.0885	-0.04005	0.05112	0.16346	9.8	-0.15545	-0.12188	0.12398	0.13618	0.05417	0.14534	0.03001	0.00858	-0.1585	-0.10986	
494	9.82	0.1543	0.12093	-0.19188	-0.12798	0.1543	-0.00057	-0.11587	-0.1524	0.13924	-0.19192	9.82	0.07534	0.11196	0.04807	-0.09646	-0.17357	0.00553	0.14757	-0.17357	0.033	-0.12188	
495	9.84	-0.037	0.07248	0.08588	-0.10071	-0.16136	0.16956	0.16346	0.16651	-0.17357	-0.0827	9.84	-0.13313	0.07248	-0.17357	-0.13714	0.00248	-0.17662	0.07174	0.06332	0.13313	0.11787	
496	9.86	-0.1585	-0.16861	0.16041	0.17872	-0.02174	-0.14019	-0.05817	0.01774	-0.037	-0.18824	9.86	-0.09367	-0.16747	0.07248	0.10734	0.0586	0.1543	0.06027	0.15144	0.13313	-0.11902	-0.05722
497	9.88	0.16956	-0.11292	-0.14935	-0.07308	0.15736	-0.0885	-0.13409	-0.18578	0.17262	-0.08545	9.88	-0.17314	0.01564	0.1543	0.0845	-0.05207	0.13924	-0.06123	-0.10376	-0.1071	-0.1585	
498	9.9	0.05503	0.04807	-0.05207	-0.17409	-0.15545	0.18177	-0.17567	0.0906	-0.13103	-0.04019	9.9	0.10281	0.15736	-0.06123	-0.1524	-0.16136	-0.10071	-0.14935	-0.13103	0.14839	-0.00062	
499	9.92	-0.18578	0.11482	0.16956	0.1543	-0.04597	-0.09155	-0.02174	0.09975	-0.07343	0.16651	9.92	0.0906	-0.0248	-0.1524	-0.02785	0.07248	-0.13409	0.07248	0.10891	0.04501	0.15144	
500	9.94	0.00646	-0.037	-0.11292	-0.02785	0.18177	-0.11902	-0.13714	-0.17357	0.16956	-0.0309	9.94	-0.14324	-0.16136	0.08144	-0.04839	0.11787	0.10586	0.14288	0.0845	-0.16747	-0.0885	
501	9.96	0.0967	-0.12798	-0.0885	-0.1585	-0.11902	0.16346	0.12093	0.033	-0.11902	-0.1524	9.96	-0.04902	0.04501	0.13914	-0.05817	-0.13714	0.08755	-0.13409	-0.1524	-0.01869	-0.12798	
502	9.98	-0.18272	-0.0309	0.18177	0.16651	-0.11597	-0.04902	0.05417	0.13924	-0.0885	0.14229	9.98	0.14229	0.13924	0.10871	-0.14935	-0.07308	-0.1524	-0.06123	-0.03395	0.1543	0.00858	
503	10	0.033	0.09975	-0.05817	0.01774	0.17872	-0.14629	-0.17357	-0.17357	0.18787	0.00248	10	-0.01259	-0.10681	-0.06428	0.10891	0.14534	0.03395	0.14534	0.14839	-0.06428	0.07534	
504																							

#	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	
1	8V= 5.556 RPS												6V= 4.167 RPS												
2	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	
3	0.14229	-0.14629		0	-0.11902	-0.00648	-0.04902	-0.00648	-0.0946	-0.07648	-0.11292	-0.10986	0.12703	-0.14019	0	-0.00362	0.10891	-0.00362	-0.12188	-0.12493	-0.08545	0.07248	-0.12188	-0.05207	-0.037
4	-0.0946	0.07534		0.02	-0.07038	-0.00648	-0.14324	-0.00648	0.0845	-0.13714	-0.10681	-0.10986	-0.0248	0.07648	0.02	-0.00648	0.06332	0.11196	-0.00648	-0.07038	-0.10986	-0.07648	-0.07648	-0.13409	0.09975
5	-0.13714	0.09975		0.04	0.06943	-0.00362	0.02384	-0.00648	0.0967	0.02689	0.07839	0.06943	-0.14935	0.11482	0.04	-0.00362	-0.07343	0.05722	0.10586	0.07248	0.0248	-0.12493	0.05722	-0.02174	0.0845
6	0.10586	-0.15545		0.06	0.09975	-0.00362	0.13924	-0.00648	-0.0946	0.13313	0.10586	0.10281	-0.02174	0.07534	0.06	-0.00362	-0.12188	-0.0885	0.07534	0.10281	0.10891	0.00553	0.11196	0.09975	-0.04902
7	0.0845	-0.04597		0.08	-0.03395	-0.01259	-0.00362	-0.00648	-0.11902	-0.03395	-0.0885	-0.07954	0.13008	-0.12188	0.08	-0.00648	-0.02174	-0.10986	-0.06123	-0.0248	0.04501	0.11196	-0.00648	0.08755	-0.12798
8	-0.14935	0.15736		0.1	-0.13409	-0.00954	-0.14629	-0.00362	0.05722	-0.13714	-0.12188	-0.12798	0.02384	-0.0885	0.1	-0.00648	0.09975	-0.00362	-0.13409	-0.11902	-0.10071	0.06638	-0.11292	-0.05207	-0.04597
9	-0.0309	-0.00057		0.12	-0.0309	-0.00954	-0.037	-0.00954	0.12398	-0.00057	0.06943	0.05112	-0.13714	0.09365	0.12	-0.00648	0.07839	0.09975	-0.04005	-0.05512	-0.10376	-0.07648	-0.07038	-0.12493	0.0967
10	0.1543	-0.17052		0.14	0.10586	-0.01259	0.12398	-0.00648	-0.0309	0.11787	0.12398	0.13008	-0.04902	0.0906	0.14	-0.00648	-0.06428	0.04807	0.0967	0.07248	0.03586	-0.12798	0.06638	-0.04005	0.08755
11	-0.05817	0.02689		0.16	0.08144	0.00248	0.04501	-0.01259	-0.14629	0.00553	-0.04005	-0.02785	0.11787	-0.10986	0.16	-0.00648	-0.12188	-0.0885	0.0845	0.0906	0.11482	-0.00954	0.0967	0.09365	-0.06733
12	-0.15545	0.14534		0.18	-0.07343	0.00858	-0.13103	-0.01259	-0.0248	-0.14629	-0.14629	-0.1524	0.04807	-0.11292	0.18	-0.00648	-0.02174	-0.10986	-0.06733	-0.037	0.02995	0.11482	-0.0248	0.0845	-0.12188
13	0.10891	-0.10071		0.2	-0.12493	0.01163	-0.06123	-0.0248	0.13313	-0.0309	-0.01869	-0.01564	-0.13714	0.08755	0.2	-0.00648	0.0906	0.01469	-0.11902	-0.12493	-0.0986	0.05722	-0.12493	-0.07343	-0.01259
14	0.10281	-0.11597		0.22	0.00248	0.033	0.10891	-0.0309	0.02995	0.13008	0.12703	0.13924	-0.06733	0.11196	0.22	-0.00648	0.05722	0.11196	-0.00954	-0.04005	-0.10071	-0.10071	-0.05207	-0.11597	0.0967
15	-0.13714	0.12093		0.24	0.12093	0.03891	0.05722	-0.04292	-0.13714	0.05417	0.02689	0.01774	0.11787	-0.06123	0.24	-0.00648	-0.0824	0.06027	0.10281	0.09365	0.04196	-0.10376	0.07839	-0.00057	0.05417
16	-0.0885	0.04807		0.26	0.02689	0.02689	-0.12798	-0.04902	-0.05207	-0.12188	-0.13714	-0.13714	0.0845	-0.14935	0.26	-0.00362	-0.12188	-0.07648	0.05722	0.0967	0.10891	0.02689	0.10281	0.10586	-0.10071
17	0.13924	-0.17662		0.28	-0.11902	0.00858	-0.07648	-0.04597	0.11196	-0.09766	-0.04902	-0.04597	-0.09766	0.02689	0.28	-0.00648	-0.00057	-0.12188	-0.0946	-0.0309	0.00553	0.10891	-0.00954	0.05112	-0.11292
18	0.02079	0.02079		0.3	-0.0824	-0.0248	0.11482	-0.02785	0.06027	0.09365	0.11787	0.11787	-0.12798	0.13618	0.3	-0.00362	0.09975	-0.00954	-0.11597	-0.13103	-0.11292	0.02384	-0.11902	-0.09766	0.03586
19	0.1585	0.1543		0.32	0.07534	-0.05817	0.09365	-0.00954	-0.13103	0.10281	0.04807	0.04807	0.05417	-0.0309	0.32	-0.00648	0.07248	0.10586	0.02079	-0.05817	-0.07648	-0.11902	-0.08545	-0.10891	0.11787
20	0.02079	-0.07038		0.34	0.09975	-0.07038	-0.10071	0.02689	-0.0824	-0.10071	0.13409	0.13714	0.12398	0.14324	0.34	-0.00362	-0.05817	0.06638	0.11196	0.07839	0.06027	-0.0885	0.05722	0.02689	0.03586
21	0.14229	-0.14935		0.36	-0.05512	-0.05207	-0.12493	0.04501	0.09365	-0.11292	-0.07343	0.06428	-0.05207	0.00248	0.36	-0.00648	-0.12188	-0.07648	0.05112	0.09975	0.09975	0.05722	0.11482	0.11482	-0.09766
22	-0.10986	0.09365		0.38	-0.13409	-0.00648	0.05417	0.05112	0.10281	0.05722	0.10281	0.11196	-0.14019	0.13008	0.38	-0.00362	-0.04005	-0.11292	-0.08545	-0.03395	-0.0248	0.11482	-0.00362	0.04501	-0.10681
23	-0.10986	0.09975		0.4	0.00858	0.04501	0.13008	0.02079	-0.07648	0.11482	0.09365	0.0845	0.02079	-0.00954	0.4	-0.00362	0.08755	-0.01564	-0.11902	-0.12188	-0.12798	0.02079	-0.11902	-0.0946	0.02079
24	0.13313	-0.14629		0.42	0.12093	0.06638	-0.05512	-0.01564	-0.13714	-0.06428	-0.0845	-0.10376	0.12703	-0.14935	0.42	-0.00362	0.0906	0.0906	0.00248	-0.04597	-0.04292	-0.11902	-0.07038	-0.11292	0.11787
25	0.06027	-0.05207		0.44	0.04196	0.05722	-0.14324	-0.07038	0.033	-0.13714	-0.13409	-0.12493	-0.0248	-0.01259	0.44	-0.00362	-0.04597	0.06027	0.11482	0.07534	0.0967	-0.0946	0.06027	0.01163	0.02995
26	-0.16136	0.14839		0.46	-0.12188	-0.00057	0.02689	-0.07648	0.13313	0.04807	0.04501	0.06943	-0.14935	0.13618	0.46	-0.00362	-0.11902	-0.07954	-0.04807	0.0845	0.0967	0.04807	0.09975	0.11787	-0.10986
27	-0.04005	-0.02785		0.48	-0.0946	-0.07343	0.12703	-0.06123	-0.0309	0.13008	0.13313	0.11787	0.00248	0.02689	0.48	-0.00648	-0.037	-0.11902	-0.09766	-0.05512	-0.04597	0.11787	-0.01869	0.02995	-0.0885
28	0.15736	-0.16136		0.5	0.07839	-0.0885	-0.03395	0.00248	-0.13714	-0.0248	-0.04902	-0.07648	0.13313	-0.14324	0.5	-0.00648	0.08755	0.00858	-0.10376	-0.12493	-0.12798	-0.01259	-0.12188	-0.10986	0.05112
29	-0.0309	0.08755		0.52	0.10586	-0.04902	-0.14935	0.04807	-0.00648	-0.14629	-0.13714	-0.12798	0.02995	-0.07038	0.52	-0.00648	0.0845	0.11196	0.02689	-0.01259	-0.04292	-0.11902	-0.06123	-0.09155	0.10586

#	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR		
473	0.14534	-0.0248		9.4	0.12093	-0.06123	0.13313	0.04501	0.0906	-0.05207	-0.12798	-0.037	0.00553	0.10586	9.4	0.01774	-0.12493	-0.04902	0.0845	-0.00954	-0.11292	-0.12798	-0.07038	-0.12188	0.10891	
474	0.00058	0.15736		9.42	0.05722	0.10586	0.02079	-0.12493	0.09365	-0.13714	0.06027	-0.1524	-0.13714	0.06332	9.42	0.11196	-0.037	-0.12188	0.10586	-0.11597	0.00858	-0.05207	-0.12798	-0.00057	0.06027	
475	-0.17662	-0.05207		9.44	-0.12188	0.06943	-0.13714	-0.0885	-0.10071	0.01469	0.13008	-0.00648	-0.0309	-0.12798	9.44	0.02689	0.0906	-0.02785	-0.02174	-0.0885	0.10891	0.09365	-0.01259	0.10586	-0.07343	
476	0.05722	-0.13714		9.46	-0.07648	-0.13103	-0.07648	0.0967	-0.11292	0.13008	-0.0309	0.13924	0.11482	-0.07954	9.46	-0.11292	0.0906	0.09365	-0.12798	0.05722	0.02689	0.0967	0.11196	0.06943	-0.12493	
477	0.13313	0.08755		9.48	0.0967	-0.0885	0.10891	0.09975	0.06943	-0.0309	-0.14935	0.00553	0.04196	0.10891	9.48	-0.0885	-0.04005	0.06638	-0.06123	0.11196	-0.0986	-0.0309	0.06027	-0.07648	0.01259	
478	-0.09766	0.08755		9.5	0.08755	0.09975	0.07534	-0.10376	0.12093	-0.14935	-0.01564	-0.14324	-0.13714	0.09365	9.5	0.06027	-0.13103	-0.08545	0.0845	-0.00057	0.09155	-0.13103	-0.08545	-0.12493	0.11482	
479	-0.13103	-0.17357		9.52	-0.11902	0.10281	-0.11292	-0.10986	-0.04902	0.02079	0.13313	-0.02785	-0.06428	-0.09155	9.52	0.11482	0.06123	-0.11292	0.09365	-0.11292	0.04807	-0.05512	-0.10681	-0.00954	0.05722	
480	0.11196	-0.0309		9.54	-0.10986	-0.08545	-0.0946	0.06332	-0.14629	0.13618	0.01774	0.13008	0.10586	-0.12798	9.54	0.01163	0.0845	0.02079	-0.04902	-0.07038	0.10891	0.0845	0.01163	0.10891	-0.0946	
481	0.07839	0.16041		9.56	0.08755	-0.13714	0.08755	0.10891	0.01469	0.00858	-0.13714	0.01163	0.09365	0.05417	9.56	-0.11902	0.09975	0.11787	-0.11902	0.05112	0.02384	0.09975	0.09975	0.06332	-0.10376	
482	-0.1524	-0.0309		9.58	0.11482	0.04501	0.09975	-0.0824	0.13313	-0.14629	-0.04005	-0.14935	-0.08545	0.13008	9.58	-0.0885	-0.04005	0.05417	-0.0309	0.10586	-0.10376	-0.05207	0.04196	-0.08545	0.02079	
483	-0.0248	-0.16441		9.6	-0.06428	0.13313	-0.09766	-0.12798	-0.01259	-0.05207	0.11787	-0.02785	-0.13103	-0.05817	9.6	0.04501	-0.11902	-0.0885	0.09365	-0.00648	-0.10071	-0.12188	-0.10071	-0.10986	0.10586	
484	0.14839	0.06027		9.62	-0.14935	-0.04292	-0.11597	0.05417	-0.14629	0.12398	0.04501	0.13313	0.03586	-0.13714	9.62	0.10891	-0.037	-0.11902	0.07534	-0.11902	0.03586	-0.0309	-0.10071	0.01774	0.033	
485	-0.06428	0.12398		9.64	0.01774	-0.13714	0.07248	0.13008	-0.01564	0.05417	-0.13714	0.05112	0.12703	0.2079	9.64	0.00057	0.0845	0.00553	0.03743	-0.1743	0.11482	0.09365	0.03586	0.10586	-0.10681	
486	-0.1524	-0.12188		9.66	0.13313	0.00553	0.12093	-0.01689	0.12703	-0.13103	-0.06123	-0.12798	-0.0309	0.2008	9.66	-0.11902	0.0845	0.11196	-0.12188	0.06027	0.01163	0.06943	0.11787	0.02995	-0.09766	
487	-0.11196	-0.09754		9.68	-0.01564	0.13008	-0.05207	-0.14629	0.02079	-0.07308	0.12093	-0.09155	-0.13714	0.13878	9.68	-0.06428	-0.04902	0.03586	0.0248	0.10586	-0.11597	0.03743	0.033	-0.11292	0.04807	
488	0.09975	0.14839		9.7	-0.13714	-0.1564	-0.14324	-0.12785	-0.14629	0.10891	0.0845	0.09365	-0.00362	-0.1585	9.7	0.07248	-0.12493	-0.10376	0.11482	0.02079	-0.07343	-0.11902	-0.10986	-0.09766	0.11787	
489	-0.14019	0.00858		9.72	-0.02174	-0.01524	-0.0648	0.03303	0.00543	0.10986	0.0967	0.11787	0.01163	0.3618	9.72	0.09975	-0.02785	-0.0496	0.05722	0.10071	0.0638	0.0248	-0.10376	0.05722	0.02079	
490	-0.07954	-0.17357		9.74	0.12703	-0.00057	0.13618	0.033	0.13313	-0.13103	-0.11597	-0.10071	0.01163	0.1618	9.74	-0.00543	0.0967	0.04501	-0.0885	-0.09766	0.09365	0.10891	0.03586	0.11482	-0.11292	
491	0.14839	0.06943		9.76	0.02079	0.13313	0.00858	-0.13714	0.05722	-0.0824	0.07248	-0.10681	-0.14324	0.13618	9.76	-0.12798	0.09365	0.10891	-0.11597	0.02995	-0.02174	0.0638	0.11787	0.01163	-0.09766	
492	0.01469	0.13313		9.78	-0.14935	0.02689	-0.14324	-0.01297	-0.12493	0.11196	0.11196	0.06943	-0.04005	-0.14324	9.78	-0.02785	-0.04005	-0.02007	0.00058	0.11787	-0.12493	-0.07648	0.01163	-0.10986	0.04807	
493	-0.1585	-0.0986		9.8	-0.037	-0.14019	-0.03395	-0.1596	0.046	0.09365	-0.07648	0.10586	0.11787	-0.06428	9.8	0.09975	-0.13103	-0.12493	0.10891	0.02079	-0.04902	-0.12188	-0.11292	-0.09766	0.11482	
494	0.0333	-0.12188		9.82	0.11787	-0.07343	0.12093	0.06027	0.0906	-0.07648	-0.12493	-0.0885	0.07534	0.12093	9.82	0.08755	-0.05512	-0.07308	0.04196	-0.10071	0.08755	-0.01259	-0.07544	0.04196	-0.00362	
495	0.13313	0.11787		9.84	0.06638	0.10891	0.033	-0.13714	0.09975	-0.12798	0.03586	-0.12188	-0.10376	0.06332	9.84	-0.04902	0.08144	-0.07534	-0.10071	-0.0885	0.09365	0.10891	0.05722	0.11787	-0.11902	
496	-0.11902	0.05722		9.86	-0.11292	-0.13734	-0.14324	-0.07648	-0.09616	0.05112	0.12093	0.07248	-0.11292	-0.12188	9.86	-0.13103	0.0906	-0.11586	-0.09766	0.063	-0.03174	0.03586	0.10586	-0.0858	-0.07343	
497	0.00058	0.15736		9.88	-0.10986	-0.02785	-0.05207	0.09975	-0.10986	-0.04902	0.02079	0.13313	0.06027	-0.07648	9.88	-0.04902	-0.04902	-0.01298	-0.0309	0.03586	-0.05512	-0.12493	-0.09155	-0.01259	-0.11597	0.05722
498	-0.14839	-0.00362		9.9	0.07248	-0.0946	0.09975	0.07248	-0.0946	-0.14324	-0.04005	-0.12093	0.09365	0.05417	9.9	0.09975	-0.11902	-0.11586	-0.0986	0.010891	0.02384	-0.06428	-0.10986	-0.12798	0.07648	0.10586
499	0.04501	0.15144		9.92	0.10586	0.0845	0.0845	-0.09155	0.10891	-0.14019	0.02384	-0.14935	-0.06123	0.18144	9.92	0.0906	0.00543	0.00543	-0.06428	-0.0986	-0.11292	0.0845	-0.04196	-0.05512	0.06943	-0.0248
500	-0.16747	-0.0845		9.94	-0.06428	0.0967	-0.09766	0.13313	-0.08545	-0.02394	-0.13924	-0.06428	-0.13103	0.18144	9.94	-0.06123	0.08144	-0.0748	-0.12493	-0.0946	0.09365	0.11196	0.0906	0.0995	0.13714	
501	-0.18689	-0.12798		9.96	-0.12493	-0.10681	-0.11902	-0.04807	-0.12798	0.13008	-0.06428	-0.13618	0.02384	-0.0515	9.96	-0.11902	0.0906	0.10891	-0.07343	0.03981	-0.04597	0.02995	0.09975	0.09975	-0.04005	-0.04292
502	0.01543	0.0309		9.98	0.03891	-0.10986	0.06027	0.13008	-0.07248	-0.04005	-0.14629	0.0858	0.11787	0.01281	9.98	-0.00954	-0.04597	-0.0309	0.06943	0.11196	-0.12188	-0.10986	-0.0309	-0.12493	0.09365	
503	-0.06428	0.17355		10	0.12398	0.08144	0.12079	-0.05207	0.12798	-0.14629	-0.04902	-0.14324	-0.05644	0.09975	10	0.10586	-0.04597	-0.12188	0.10586	0.06027	-0.00954	-0.02785	-0.09155	-0.12798	-0.04005	0.09365
504																										

AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC
4V= 2.778 RPS										
Time	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
0	-0.01259	0.04807	-0.02174	0.06943	-0.01564	-0.06123	0.07248	0.07248	0.05722	0.06638
0.02	-0.00362	0.07534	-0.07648	0.00858	-0.00648	-0.0885	0.02995	0.00858	-0.00362	0.00858
0.04	-0.01259	0.04501	-0.08545	-0.06123	-0.00362	-0.05207	-0.04292	-0.05512	-0.06428	-0.06123
0.06	-0.00362	-0.02174	-0.037	-0.0885	-0.01259	0.01774	-0.0946	-0.0885	-0.0885	-0.0885
0.08	-0.00648	-0.07954	0.03586	-0.06123	-0.00648	0.06943	-0.06733	-0.06428	-0.04902	-0.05512
0.1	-0.00648	-0.0824	0.07534	0.01163	-0.00648	0.07248	-0.00057	-0.00057	0.02384	0.00858
0.12	-0.00648	-0.0309	0.05417	0.06943	-0.01259	0.02384	0.06027	0.06332	0.07248	0.06943
0.14	-0.00362	0.03586	-0.02174	0.06943	-0.00648	-0.05512	0.06943	0.06943	0.06943	0.07248
0.16	-0.00648	0.08144	-0.07954	0.00858	-0.00362	-0.0885	0.04196	0.02384	0.01774	0.02689
0.18	-0.00648	0.05112	-0.0885	-0.07038	-0.00954	-0.06428	-0.0309	-0.04902	-0.05512	-0.04597
0.2	-0.00648	-0.02174	-0.0248	-0.09155	-0.00648	0.00248	-0.0824	-0.08545	-0.0885	-0.0885
0.22	-0.00648	-0.0824	0.04501	-0.04902	-0.00362	0.06332	-0.08545	-0.06123	-0.06733	-0.07648
0.24	-0.00362	-0.07954	0.07839	0.02384	-0.01259	0.07248	-0.0248	0.00248	-0.00057	-0.01259
0.26	-0.00362	-0.02785	0.05722	0.07534	-0.00648	0.02384	0.04807	0.05112	0.05722	0.06027
0.28	-0.00648	0.04196	-0.01259	0.06638	-0.00362	-0.05817	0.08144	0.07248	0.07534	0.07839
0.3	-0.00362	0.07534	-0.07648	0.00858	-0.01259	-0.0885	0.03586	0.02384	0.02384	0.033
0.32	-0.00362	0.04807	-0.0885	-0.06428	-0.00648	-0.06428	-0.0309	-0.04902	-0.05207	-0.04292
0.34	-0.00648	-0.0248	-0.04005	-0.09155	-0.00648	-0.00362	-0.0824	-0.0885	-0.0885	-0.08545
0.36	-0.00648	-0.07954	0.02384	-0.05512	-0.00954	0.05722	-0.07648	-0.06733	-0.06123	-0.07648
0.38	-0.00954	-0.0824	0.07839	0.01469	-0.00648	0.07248	-0.02785	-0.00362	0.00553	-0.00954
0.4	-0.00648	-0.02174	0.06027	0.07248	-0.00362	0.02689	0.04807	0.06638	0.06332	0.05112
0.42	-0.00954	0.03891	-0.01259	0.06943	-0.00954	-0.05207	0.07534	0.07534	0.07248	0.07248
0.44	-0.00362	0.08144	-0.07343	0.00858	-0.00362	-0.0885	0.04501	0.03586	0.02079	0.033
0.46	-0.01259	0.04501	-0.08545	-0.06123	-0.00648	-0.06428	-0.037	-0.037	-0.06123	-0.04005
0.48	-0.00648	-0.01259	-0.037	-0.0885	-0.01564	-0.00057	-0.08545	-0.0885	-0.0946	-0.0885
0.5	-0.01564	-0.07954	0.02995	-0.05207	-0.01869	0.06332	-0.07648	-0.07648	-0.06428	-0.07648
0.52	-0.00362	-0.0885	0.07534	0.00858	-0.0248	0.07534	-0.01259	-0.02174	0.00858	-0.00648
0.54	-0.01259	-0.04597	-0.05112	-0.06943	-0.037	-0.033	-0.04501	-0.04807	-0.06943	-0.05722

AS	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC
9.4	-0.0885	-0.0885	-0.05207	0.07248	-0.01869	0.06943	-0.07954	0.01774	-0.00648	-0.04902
9.42	-0.04292	-0.04005	-0.09155	0.01469	-0.07954	0.06943	-0.08545	-0.04902	-0.00362	-0.0946
9.44	0.02689	0.033	-0.06123	-0.05207	-0.0824	0.02689	-0.03395	-0.0885	-0.00648	-0.07038
9.46	0.07248	0.08144	0.01163	-0.09155	-0.0248	-0.04597	0.03586	-0.07648	-0.00362	-0.00057
9.48	0.06638	0.05722	0.06943	-0.06428	0.04501	-0.0885	0.07534	-0.00362	-0.00648	0.06027
9.5	0.01163	-0.01259	0.07534	0.01163	0.07248	-0.07343	0.06027	0.06027	-0.00362	0.07839
9.52	-0.06123	-0.07954	0.02689	0.07248	0.05417	-0.01259	0.00248	0.07839	-0.00362	0.03586
9.54	-0.0885	-0.0824	-0.05207	0.06943	-0.00954	0.05722	-0.06123	0.033	-0.00362	-0.03395
9.56	-0.05512	-0.04005	-0.0885	0.02689	-0.07648	0.07534	-0.09155	-0.05207	-0.00648	-0.08545
9.58	0.01774	0.03586	-0.07038	-0.04597	-0.08545	0.033	-0.05512	-0.08545	-0.00362	-0.0824
9.6	0.06943	0.07248	-0.00648	-0.09155	-0.04292	-0.04902	0.01774	-0.06428	-0.00648	-0.0248
9.62	0.06943	0.05417	0.06027	-0.07038	0.02689	-0.08545	0.06943	-0.00362	-0.00362	0.05112
9.64	-0.00362	-0.01259	0.07248	-0.00648	0.07248	-0.07038	0.06943	0.05722	-0.00954	0.08144
9.66	-0.06428	-0.07648	0.02689	0.05722	0.06332	-0.01259	0.00858	0.06638	-0.00954	0.04501
9.68	-0.09766	-0.0824	-0.04902	0.07839	-0.00362	0.04807	-0.06123	0.02995	-0.01259	-0.0309
9.7	-0.04902	-0.02785	-0.0885	0.03586	-0.07648	0.07248	-0.08545	-0.04005	-0.01259	-0.07954
9.72	0.01163	0.03891	-0.06428	-0.04597	-0.08545	0.03586	-0.05207	-0.0885	-0.00648	-0.07648
9.74	0.06943	0.06943	-0.00362	-0.08545	-0.04292	-0.037	0.00248	-0.07038	-0.00362	-0.02785
9.76	0.06027	0.06027	0.06027	-0.06733	0.02384	-0.09155	0.06638	-0.00954	-0.00648	0.04501
9.78	0.00858	-0.00954	0.06943	-0.00648	0.07248	-0.07343	0.06943	0.05722	-0.00648	0.07248
9.8	-0.06733	-0.07038	0.01774	0.05722	0.06027	-0.00648	0.01163	0.07534	-0.00648	0.04196
9.82	-0.0885	-0.0885	-0.05207	0.07248	-0.00362	0.06027	-0.05817	0.04501	-0.00648	-0.02785
9.84	-0.04902	-0.04902	-0.09155	0.02689	-0.07954	0.08144	-0.0946	-0.02174	-0.00362	-0.08545
9.86	0.02384	0.02689	-0.05512	-0.04597	-0.0885	0.033	-0.05817	-0.0824	-0.00648	-0.07954
9.88	0.07248	0.07839	0.00858	-0.0885	-0.04005	-0.0309	0.01774	-0.0885	-0.00362	-0.01869
9.9	0.06943	0.06332	0.06943	-0.06428	0.02995	-0.0824	0.07248	-0.037	-0.00362	0.04501
9.92	0.01163	-0.00954	0.07534	0.00553	0.07534	-0.07954	0.07248	0.03891	-0.00362	0.08144
9.94	-0.05817	-0.07343	0.02384	0.06638	0.06332	-0.02785	0.02384	0.08144	-0.00362	0.04807
9.96	-0.0885	-0.0946	-0.04902	0.08144	0.00248	0.04501	-0.05207	0.05722	-0.00362	-0.01869
9.98	-0.06123	-0.04292	-0.0946	0.033	-0.06123	0.07839	-0.0885	-0.0248	-0.00362	-0.07954
10	0.01469	0.02384	-0.07038	-0.04005	-0.0885	0.04807	-0.07038	-0.07648	-0.00954	-0.0885